

UNIVERSITY OF CALGARY

Maneuvering Mechanisms for Omnidirectional Robots using Semi-circular Mecanum Wheels

by

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Abstract

The recently designed Semi-circular Mecanum Double Wheel (SMDW) is a type of wheel that provides vehicles with enhanced omnidirectional motion on rough terrain and obstacle overcoming capabilities. This thesis proposes a new maneuvering methodology that allows robots with SMDWs to overcome obstacles in diverse approach configurations allowing robots to select their position and orientation according to their environment. The techniques developed prior to this work considered only the obstacle overcoming in a lateral motion, significantly restricting the holonomic abilities of the device. The methodology proposed in this thesis considers the generation of cubic spline trajectories that connect the initial position and orientation of the robot with the target configurations and the SMDWs states necessary to perform the obstacle overcoming task. For the generation of those trajectories a Particle Swarm Optimization (PSO) search technique explores the navigation space of the robot in order to find suitable control point locations to achieve the required robot's position and wheels states. The obtained trajectories were tested using a novel physical simulator that incorporates the effects of the peripheral rollers in order to evaluate the performance over a wide range of obstacles and terrain configurations. The new methodology provides an effective tool to enable robots using SMDWs to move and navigate more effectively in complex outdoor and indoor rough terrains.

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List of Symbols, Abbreviations and Nomenclature

AGV	Autonomous Ground Vehicle
AMR	Autonomous Mobile Robots
AR2S	Autonomous Reconfigurable Robotics Systems Laboratory
EMDW	Elliptical Mecanum Double Wheel
MW	Mecanum Wheel
ODE	Open Dynamics Engine
ODV	Omni-Directional Vehicle
OGV	Omnidirectional Ground Vehicle
PSO	Particle Swarm Optimization
ROS	Robot Operating System
SMDW	Semicircular Mecanum Double Wheel
UAV	Unmanned Air Vehicle
UGV	Unmanned Ground Vehicle
V-REP	Virtual Robot Experimentation Platform
α	Angle of approach between the robot and the obstacle
β	Vector of penalty coefficients for wheels state deviation
γ	Vector of penalty coefficients for position deviation
Δt	Spline trajectory time interval
ψ	Robot centre angular orientation
$\psi(s)$	Robot spline trajectory for angular orientation
ζ_m	Angle of the tangent at the point of contact
Γ_0	Semi-circular Mecanum Double Wheel wire frame model matrix
$v(\theta)$	Wheel hub-obstacle point of contact
$\rho(\theta)$	Radius between the wheel centre and the point of contact
θ	Angular position of the vehicle wheel
θ_i	Angular position of the vehicle wheel “i”
Θ	Vector of robot’s wheels angular position
Θ_f	Vector of robot’s target final wheels angular position
Θ_d	Vector of robot’s current final wheels angular position

Θ_0	Vector of robot's initial wheels angular position
$\dot{\Theta}$	Vector of robot's wheels angular velocity
c_1	Cognitive knowledge acceleration constant
c_2	Social knowledge acceleration constant
h	Obstacle height
k	PSO Iteration index
$l(\alpha)$	Horizontal projection of the collision line
P_0	Vector of robot's initial position
P_1, P_2	Spline control points
p_{bij}	Personal best position for particle "i" on "j" dimension
$\hat{p}_{bij}$	Global best position for particle "i" on "j" dimension
P_d	Vector of robot's target final position
P_f	Vector of robot's current final position
$P_{ij}(k)$	Position equation for particle "i" of "j" dimension
R_c	Semi-circular Mecanum Double Wheel hub radius
R_e	Semi-circular Mecanum Double Wheel hub semi-minor axis
r_{1ij}	Cognitive knowledge stochastic coefficient
r_{2ij}	Social knowledge stochastic coefficient
R_y	Transformation matrix for rotation about y-axis
R_z	Transformation matrix for rotation about z-axis
$s(t)$	Auxiliary variable for spline trajectory time parameterization
$S(X, \Theta)$	Fitness cost of a spline trajectory
x	Robot centre position x-axis component
$x(s)$	Robot spline trajectory for x-axis component
X	Vector of the robot centre position
$\dot{X}$	Vector of the robot centre velocity
x_{max}	x-axis upper boundary of the navigation space
y_{max}	y-axis upper boundary of the navigation space
x_{min}	x-axis lower boundary of the navigation space
t	Time

$v_{ij}(k)$	Update velocity equation for particle “i” of “j” dimension
v_{max}	Maximum update velocity allowed
y	Robot centre position y-axis component
$y(s)$	Robot spline trajectory for y-axis component
y_{min}	y-axis lower boundary of the navigation space
w	Inertia weigh coefficient
w_x	Wheel frame of reference x-axis
w_y	Wheel frame of reference y-axis
w_z	Wheel frame of reference z-axis
z'	Slope of the tangent at the point of contact

Chapter 1: Introduction

1.1 Ground Vehicles

The use of autonomous mobile robots (AMR) in different applications, ranging from industrial facilities surveillance tasks to search and rescue applications is becoming increasingly popular in over the last years. An AMR is defined as “a system which operates with a semi of full degree of autonomy in order to perform a goal oriented task excluding manufacturing operations” [1]. Autonomous mobile robots are very versatile devices with the ability to make decisions by themselves in order to complete the assigned tasks. In addition AMRs possess the necessary resources such as energy, perception, and computation capabilities. These basic features that define an AMR [1] are fundamental factors for today and future usefulness and flexibility.

The complexity of the task being executed affects directly the autonomy capability of an AMR. The degree of unstructuredness of its operational environment has a direct impact to the efficiency of the robot. In several cases, an increase in the level of human intervention during the robot's decision making progress is required to overcome a non-structured environment. Moreover in order to improve the performance on an AMR it is required to expand its capabilities to enable them to cope with environment disturbances.

One of the main strengths of autonomous mobile robots is that they can be built and designed to operate on a wide range of different environments. The flexibility of these devices has allowed the development of: unmanned ground vehicles (UGVs), unmanned water vehicles (UWVs), autonomous underwater vehicles (AUVs) and unmanned aerial vehicles (UAVs). The work developed in this thesis concerns an unmanned ground vehicle application.

The UGVs have been and are being used in a wide range of fields. The applications are classified in the following fields: transportation, surveillance, exploration, inspection/maintenance, harvesting, housekeeping and education/entertainment [1]. The demand of ground robots capable to maneuver on difficult to access and constrained spaces is increasing, particularly in the fields of exploration, surveillance, search and rescue, military and inspection/maintenance. These applications imply a considerable number of tasks performed at unstructured and/or unknown environment. However, most of most UGVs and their navigation

tools have been developed for engineered or structured terrains. Therefore, an improvement in the autonomy and adaptability capabilities of ground robots is essential to perform missions under those conditions.

The most common ground vehicles implementation considers wheels, tracks, legs and their combination (hybrid) as locomotion ways [2]. The selected locomotion mechanism to be used in a given UGV affects directly the following aspects of the system: maneuverability capabilities, motion control complexity, device scalability, energy consumption, mechanical complexity and the type of terrain that the system can deal with. The selection of the appropriate locomotion system is fundamental to meet the demands of the intended application and provide effective robot performance.

The most common locomotion type used in ground robots are the wheeled systems and the legged systems. The wheeled vehicles (including wheeled and tracked vehicles) are fairly the most popular implementation of ground vehicles [3]. The main characteristic of a wheeled vehicle is that the wheels or tracks are in contact with the ground at all times during the vehicle operation. A wheeled vehicle has the advantage of possessing a simple and robust design which relies on a large accumulated knowledge base and years of design experience.

Wheeled locomotion designs allow a simple and inexpensive construction, manufacturing and maintenance [4]. Additionally a wheeled platform exhibits good performance providing fast and reliable locomotion, with efficient-low energy consumption; and a relatively simple motion control. In general terms the motion control of a wheeled vehicle is based on the vehicle kinematics and the type of wheel being used [5][3].

Contrasting with all the advantages that a wheeled UGV possesses there are a series of shortcomings and challenges that these type of vehicle face when traversing non-flat unstructured terrains. The main reason for these limitations is the discontinuity of the contact between the wheels and the surface. These limitations restrict heavily the ability to overcome obstacles. Finally the typical wheels mechanism (not considering tracked vehicles) can be affected and obstructed by loose soil while traversing on non-structured terrain.

The legged systems, on the other hand, are characterized by a series of point contacts between the robot and the ground [3]. This locomotion system is inspired by nature and tries to mimic or replicate the limbs of different animals in order to provide locomotion and perform the

motion control. In contrast to wheeled vehicles, legged UGV's possess increased mobility in difficult unstructured terrains which account for approximately 50% of the earth's surface [5] .

Despite their improved mobility, current legged systems have considerably more complex designs, are less energy efficient and slower. A legged robot also has higher requirements of power and torque peaks than a wheeled robot [3][4][5]. Also the motion control a legged robot is more complex since dynamic stability and gait analysis must be taken in addition to the vehicle kinematics.

Wheeled systems possess several advantages when it comes to their use in autonomous robots such as less complex the decision making process due to its simpler motion control mechanism. In summary additional efforts are required in order to improve the performance of wheeled vehicles for handling obstacles when operating in non-structured terrains. The work presented in this thesis addresses such challenges. The research addressed the obstacle overcoming capabilities of a wheeled UGV in order to improve its mobility and bring it closer to the mobility seen in legged vehicles.

1.2 Omnidirectional Ground Vehicles

In general terms the motion of wheeled robots is governed almost entirely by its kinematics[1][3]. The kinematic model of such vehicles is affected mainly by the following factors: type of wheel, the geometry of the vehicle frame, and the actuation system location. Among these factors the type of wheel can be considered as the main parameter for vehicle design since the vehicle frame design and the actuation layout selection are directly related with it. The kinematics of a vehicle will establish the configuration and workspace constraints of it. The most critical constraints of a wheeled platform are the contact points between the robot and the ground [2].

Ground robots are classified as holonomic and non-holonomic based on the limits of their kinematic constraints. A Non-holonomic vehicle observes restrictions on the motions and maneuvers that it can perform based on its initial position. Due to the degrees of freedom of its wheels, such type of vehicle cannot modify its position state arbitrarily. A good example of the restrictions faced by a non-holonomic platform is the inability to perform sideways motion and in-situ rotation [2] .

Conversely a holonomic vehicle also known as an omnidirectional vehicle possesses the ability to move arbitrarily in any planar direction regardless its current position [6]. A holonomic ground vehicle has three degrees of freedom in planar motion. These degrees of freedom confer to the vehicle the ability to change direction without the need to stop for re-configuration and allow it to move in any direction with the same degree of mobility[7].

Omnidirectional ground vehicles (OGV) have several key features which allow them to maneuver effectively in cluttered and reduced space environments. They also have multiple advantages for applications where high maneuverability is desired or required such as those encountered in confined spaces. The conditions mentioned above are common in several surveillance, exploration, maintenance, military and mining applications[5][2]. Omnidirectional ground robots also exhibit a high degree of agility due to their ability to change direction smoothly without affecting significantly its velocity[6].

A considerable amount of work has been developed for OGVs applications. An extensive knowledge and variety of applications have been developed for hard flat surfaces and indoors conditions. However, the amount of outdoors applications in uneven terrain with obstacles has been increasing recently [8][9][10][11][12][13] (details of some of these studies are provided in Chapter 2). Therefore, a clear need to continue expanding the OGV capabilities to handle obstacles and non-structured surfaces is present.

1.3 Wheeled Ground Vehicles

The use of particular types of wheels is one of the approaches to implement ground robots with holonomic capabilities. The types of wheels that can provide omnidirectional capabilities include castor/caster wheels, spherical wheels and omnidirectional wheels [3]. Some other wheel designs [14][15] which do not fit on a single of the above categories can provide holonomic capabilities; however, those particular implementations are not discussed here.

The three types of wheels suitable for omnidirectional applications are classified into two more categories: conventional and special wheels. The main difference between these categories is that for special wheels the traction is achieved by applying constraints in one direction and removing this constraints later by allowing passive motion in another direction[2]. The spherical and caster wheels are fit into the conventional category while omnidirectional wheels belong to

the special category. The most important characteristics and features of these wheels are provided below.

1.3.1 Castor Wheel

A castor wheel is a type of wheel which consists basically in a circular wheel with the ability to steer around its vertical axis (Figure 1.1). As a result the caster wheels possess the ability to control the spinning of the wheel (θ) and the steering angle (β). While in rolling motion the caster wheels have the same constraints as a conventional wheel the offset steering axis modifies the sliding constraints of the wheel. By controlling the steering motion of the wheel (β) without altering the position of the vehicle frame a caster wheel gives to it the possibility of move laterally in any direction [3], however this come at the expense of potentially undesired motion of the UGV.

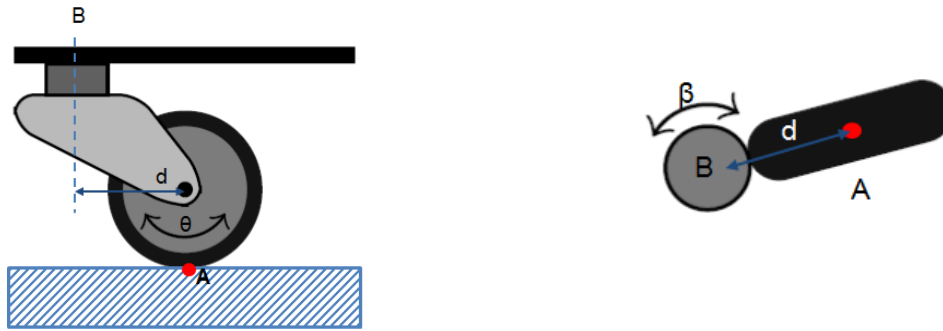


Figure 1.1 Castor Wheel [1]

1.3.2 Spherical Wheel

The spherical wheel is simply a sphere shape attached to the vehicle frame trough a ball joint. As can be seen in Figure 1.2 , the geometry of a sphere does not have a principal axis of rotation. In consequence constraints in rolling and sliding are not present in vehicles equipped with this locomotion system.

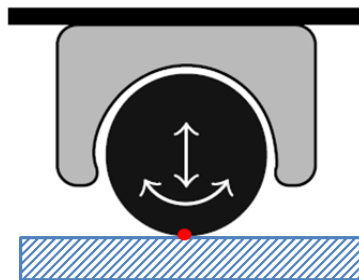


Figure 1.2 Spherical Wheel [1]

A brief comparison of advantages and disadvantages in the use of spherical wheels is shown in the following table.

Table 1.1 Spherical Wheel Features Summary

Advantages	Shortcomings
- Inherited omnidirectional geometrical capabilities offering full holonomic constraints with minimal restrictions -Low wheel weight and simple construction -Smooth continues motion on flat hard surfaces	- Requires two actuators with individual control and complex drive mechanisms to achieve holonomic motion [7] -Very limited ability to traverse uneven terrain and particular sensitivity to loose terrain -Sensitive to vehicle vibrations and terrain discontinuities [3]

1.3.3 Omnidirectional Wheel

The omnidirectional wheel consists of a mechanical assembly of a rotating hub connected to the frame of the vehicle and a series of peripheral free rotating rollers attached to it [3]. There are several different designs of omnidirectional wheels as show in Figure 1.3. The term omnidirectional wheel is commonly used as synonym of the Mecanum wheel, however the Mecanum wheel is a particular design of omnidirectional wheel. Despite the variety of forms and designs all omnidirectional wheels have the same principle of operation.

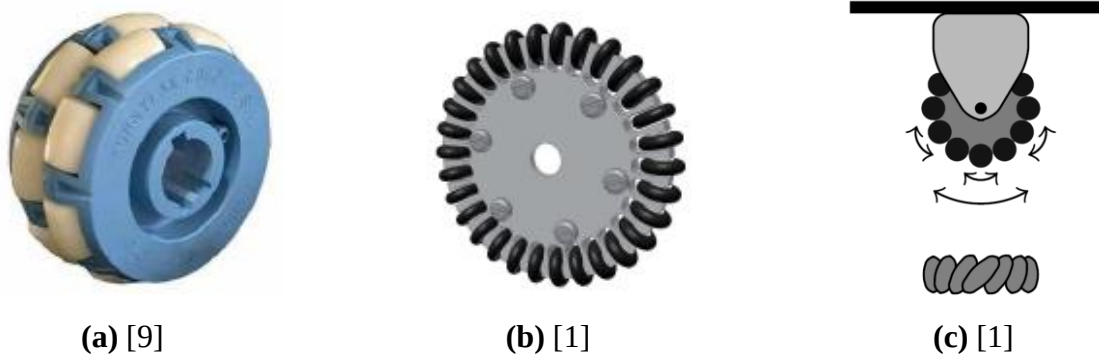


Figure 1.3 Omnidirectional Wheels (a) double (b) segmented (c) Mecanum

An omnidirectional wheel operates by rotating the wheel hub providing traction and rolling constrained motion to the assembly (the rotation axis is connected to the OGV frame) while the contact between the assembly and the ground is made by one of the peripheral rollers. The lateral force at the point of contact induces motion on this free rotating roller. The passive

motion induced on the roller allows a slipping motion and exerts a force perpendicular to the rotation axis of the roller.

The induced force on the roller is constrained by the angle of its rotation axis with respect to the wheels hub. In a multiple wheel vehicle the combination of the forces exerted in each wheel produces a resultant force on the vehicle centre of mass which result in a holonomic planar motion to the overall vehicle. This resultant force can be modified by changing the angular velocity and direction of the wheel hub.

The kinematics that governs the relation between the resultant force and the wheel's hub rotation is given by the number of wheels and their arrangement with respect to the vehicle's center of mass. A summary with the main features for the omnidirectional wheel is provided in Table 1.2

Table 1.2 Omnidirectional Wheel Features Summary

Advantages	Shortcomings
-Capability to be mounted on a car-like platform -Requires only one actuator with individual control per wheel to achieve omnidirectional motion -High payload capability [3] [8]	-Limited mobility on uneven terrain particularly in lateral motion - More complex mechanical assembly than caster and spherical wheels -Its applications are constrained by a limited set of allowable wheel position configurations[16]

1.4 Mobile Robots with Mecanum Wheels

The research presented in this thesis has the goal to improve the obstacle handling capabilities of an OGV equipped with a novel design of Mecanum wheels, the specific problem statement is provided in Chapter 3. Therefore, a detailed introduction to the Mecanum wheel principles of operation and features is provided here. The Mecanum wheel (MW) is a device designed and developed by the Swedish engineer Bengt Ilon in 1975. This wheel is built by a central circular wheel frame with a number of free moving angled rollers on the periphery of the wheel (Figure 1.4). The profile of the rollers designed to provide a perfect lateral circular silhouette to the wheel assembly (Figure 1.4) [8].

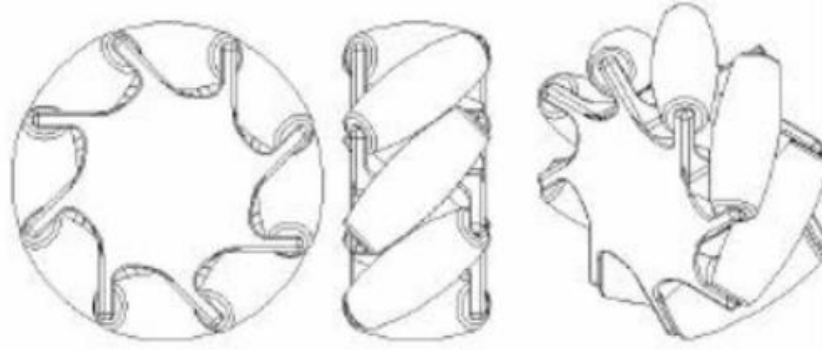


Figure 1.4 Traditional Mecanum Wheel Design [8]

The Mecanum wheel design sets the rotation axis of the rollers at a relative angle of 45° or 135° with respect to the rotational axis of the wheel frame. The holonomic movement is achieved by the free moving rollers which transform a portion of the wheel propelling force into a force normal to the wheel direction [17] (Figure 1.5). Under the same principle of operation as with any other omnidirectional wheel design, a resultant vector from the combination of the forces produced by each one of the individual determines the direction of the vehicle. The control scheme for a vehicle with Mecanum wheels is based on the kinematic model and geometry of the vehicle.

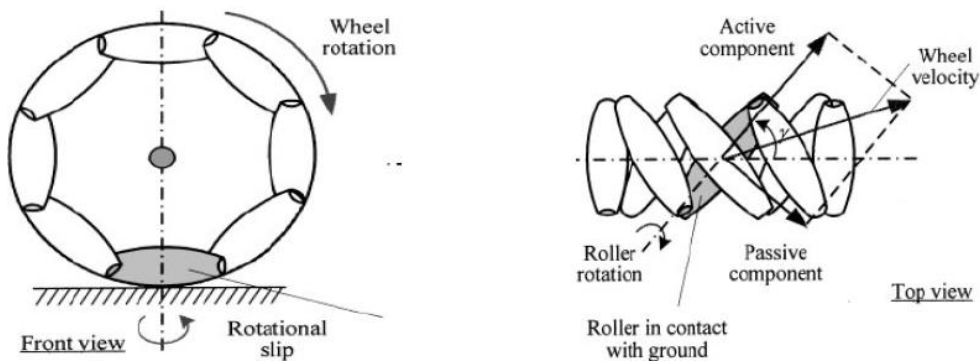


Figure 1.5 Mecanum Wheel Principles of Operation[17]

In order to achieve full holonomic capabilities a robot equipped with Mecanum wheels must have at least three wheels with non-parallel axes [16]. The kinematic model required for the motion control of a 4 wheel OGV with Mecanum Wheels has been analyzed extensively by several researchers[16][18][17]. Due to its popularity the robot layout shown in Figure 1.6 was selected as the basic Mecanum wheeled OGV for analysis and comparison along this thesis work. The kinematic model for such vehicle is given by the following mathematical expression:

$$\begin{bmatrix} v_x \\ v_y \\ \omega_z \end{bmatrix} = \frac{R}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 1 \\ -1 & 1 & -1 & 1 \\ \frac{1}{l_1 + l_2} & \frac{1}{l_1 + l_2} & \frac{1}{l_1 + l_2} & \frac{1}{l_1 + l_2} \end{bmatrix} \cdot \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \\ \omega_4 \end{bmatrix}$$

where:

R : is the wheels radius

v_x : is the vehicle's translational velocity in the x-axis

v_y : is the vehicle's translational velocity in the y-axis

ω_z : is the vehicle's rotational velocity along the z-axis

$\omega_1, \omega_2, \omega_3, \omega_4$: are the Mecanum wheels rotational velocities

l_1 : is the y-axis distance between the vehicle frame of reference and wheel contact point

l_2 : is the x-axis distance between the vehicle frame of reference and wheel contact point

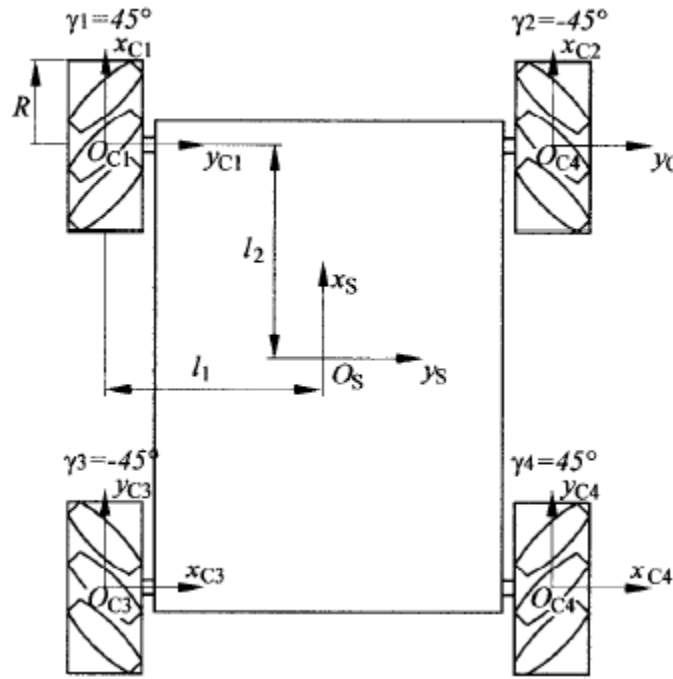


Figure 1.6 Ground Vehicle Configuration for 4 Mecanum Wheels [17]

Utilizing the mathematical model presented above it is possible to use inverse kinematics to determine the required wheel velocities in order to achieve a desired direction and velocity of the vehicle. As a result the angular velocities for each wheel can be obtained using the equation

below, provided that the desired translation and rotation velocities, and physical characteristics for the robot are given.

$$\begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \\ \omega_4 \end{bmatrix} = \frac{1}{R} \begin{bmatrix} 1 & 1 & -(l_1 + l_2) \\ 1 & -1 & l_1 + l_2 \\ 1 & -1 & -(l_1 + l_2) \\ 1 & 1 & l_1 + l_2 \end{bmatrix} \cdot \begin{bmatrix} v_x \\ v_y \\ \omega_z \end{bmatrix}$$

The calculation of the rotational velocity of the wheels from the inverse kinematics model allow the use of different schema to perform trajectory, navigation and position control [17][18][19][20]. In general terms the navigation of an OGV with Mecanum wheels requires individual control of the wheel's actuators, rotational velocity measurements from each wheel and in some cases the need to incorporate odometry measurements of the vehicle's frame of reference.

The use of OGVs with 4 traditional Mecanum wheels has been proven for many years in applications when the OGV is required to move on hard flat surfaces Vehicles of this type have also the potential to be used “in numerous outdoors applications, such as search and rescue missions, military activities, planetary explorations and mine operations”[8]. In recent years research has been carried out to develop robots for outdoor and unstructured terrain conditions [12][21][8][22]. Among this research the main approaches are the optimization of current vehicle designs and the development of new constructive solutions [8]. This thesis is based on the use of a constructive solution; a novel design, the semi-circular Mecanum double wheel (SMDW). As will be described in Section 2.3.2 this wheel allows to address the obstacle climbing and rough terrain navigation challenges not possible before by OGVs.

1.5 Thesis Outline

This thesis is organized as follow: Chapter 1 provides an introduction and overview to the field of omnidirectional ground vehicles. It also gives and insight to the concepts, features and applications of Mecanum wheels. Chapter 2 provides a review of the current research in obstacle overcoming and rough terrain applications with Mecanum wheels. The semi-circular Mecanum double wheel (SMDW) design and a previous rough terrain maneuvering scheme using these wheels are also covered in this review.

Chapter3 provides the research problem statement, the envisioned contributions and the needed definitions of the current work. Chapter 4 covers in detail the core topics of the proposed solution of the obstacle climbing with SMDW problem. The topics covered include: geometrical modeling of the SMDW, analysis of non-sideways obstacle climbing, the modified maneuvering scheme and the trajectory generation algorithm.

Subsequently Chapter 5 presents the tools and software selected to perform and evaluate the proposed solution. The process to develop a realistic model of the SMDW is detailed. Chapter 6 contains the experimental results for different obstacle climbing scenarios by an OGV with SMDWs. Chapter 7 discusses the proposed solution with respect of the experimental results. In addition the chapter describes the potential improvements, the future work and the goals achieved in the research work.

Chapter 2: Literature Review

In recent years, there has been a number of advances to develop robots equipped with Mecanum or omnidirectional wheels for outdoor applications. Several solutions have been developed to allow such vehicles to handle unstructured terrain and some obstacles. The semi-circular Mecanum double (SMDW) wheel is one of the solutions presented. The background provided for this device is fundamental for this research work. This chapter provides further details of different solutions and their approach to address the rough terrain and obstacle handling challenges. Also, previous work regarding the concept of SMDW is analyzed in detail. An existing maneuvering technique for vehicles with SMDW is presented along with important details related to it. The background information concludes with the review of trajectory generation and optimization that are key factors for the existing maneuvering algorithm.

2.1 All Terrain OGVs with Mecanum Wheels

Some outdoor solutions have been developed considering the use of conventional Mecanum wheels [14][8][23][12][24]. The results of these solutions is limited in terms of the maximum height of the obstacles analyzed and the level of unstructuredness of the environment. However, such research works have provided important information regarding the performance and the abilities of vehicles using Mecanum wheels to operate on such conditions. Due to its depth, detail and the information provided, there are two studies that outstand [23] [12] . A review with the main features of those studies is provided below.

The first study is a research project developed in 2000 by the US Navy for its Omni Directional Vehicles (ODV) cargo handling fleet [23]. In this study a 39” diameter Mecanum wheel (Figure 2.1) was analyzed. The construction of this Mecanum wheel intended to avoid the traditional undesired contact between the wheel and its rim, by attaching the rollers to an internal middle frame.

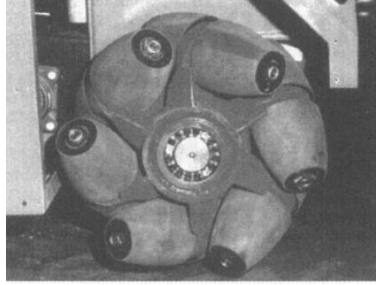


Figure 2.1 Navy ODV fleet Mecanum Wheel [23]

The results proved that the traction and controllability capabilities of the vehicle while operating on ice, inclined surfaces and wet surfaces were good. The study reported “an excellent capability to negotiate ramps, obstacles, mud and sand”[23]. Tests were performed to assess the capability to handle small obstacles. The test considered terrain sections having the following obstacles: chains, board and ropes with a height of 3 inches. The results showed that Mecanum wheeled vehicles negotiated successfully those obstacles and proved the feasibility to operate omnidirectional vehicles on hybrid environments (generally structured terrain with some obstacles, slopes and uneven surfaces).

However, in order to achieve the performance described above, the diameter of the wheels is relatively larger than the height of the obstacles. The height of the obstacles analyzed was just 7.5% of the wheel’s diameter. Such results proved that a vehicle with Mecanum wheel can handled obstacles properly, but the obstacle overcoming capability of such vehicles is very limited.

The second significant study was performed by the EADS Astrium [12] during the development of the Martian exploration platform Mars Cruiser One (Figure 2.2). Such study assessed the capabilities of Mecanum wheels on rough terrain by a series of experimental tests to analyze sinkage, maneuverability and traction. In order to mimic the potential conditions encountered on the Martian soil a mixture of sand, loam and stones was utilized during the tests which also considered flat, inclined and uneven environments.

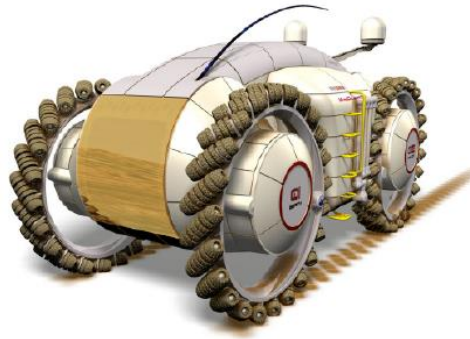


Figure 2.2 Mars Cruiser One Platform [12]

The results of the study confirmed that a Mecanum wheeled vehicle can traverse successfully hard surfaces of gravel and dirt furrows in frontal and lateral motion. However, while traversing on dirt, the vehicle had to maneuver slowly because high torques were required in order to keep control and stability on the vehicle's trajectory. The study also found that loose terrain such as sand, causes a considerable amount of sinkage over the time. Moreover while traversing laterally a significant level of slippage and trajectory deviation was observed on sand.

During the tests, the overcoming of different obstacles with a height up to the peripheral rollers diameter was performed successfully in different motion directions (frontal, lateral and angled). However, when taller obstacles were tested, up to 40% of the wheel diameter, the vehicle was unable to climb the obstacle. For most of the analyzed cases just one of the wheels of the vehicle was able to climb the obstacle, while the other slipped and took the vehicle out of its targeted trajectory. The best results for obstacle climbing were obtained when the vehicle approach the obstacle at an angle of 45° with respect to its longitudinal axis[12].

Both of the studies detailed above section confirmed the expected limited capabilities and reduced performance of Mecanum wheels while traversing on rough terrain, being the most critical of the shortcomings its ability to climb obstacle higher than the diameter of the peripheral rollers. Other less detailed studies [14][8][24] have showed similar results, thus supporting these conclusions.

2.2 All-Terrain Vehicles Hybrid Implementations with Omnidirectional Wheels

Other solutions consider an approach based on modifications into the construction and design of the vehicle. The modifications performed provide to the vehicles some capabilities for

obstacle climbing and navigation on uneven terrain. Some solutions combine Mecanum wheels with auxiliary systems. Such auxiliary systems are either passive climbing mechanisms or conventional non-holonomic wheels. Due to the use of such auxiliary systems these solutions can be considered hybrid devices that combine holonomic and non-holonomic features. Two of such relevant cases will be discussed [21] as they provide relevant information about its limitations and advantages of the hybrid systems. Other solutions exist but will not be discussed here as they provide similar conclusions.

The first solution to be reviewed is the MEGAN OGV platform designed at the Massey University in New Zealand [10][21]. MEGAN considers the use of four regular wheels combined with four Mecanum wheels on the same vehicle with the layout displayed in Figure 2.3. The platform incorporates a hydraulic system that allows the robot navigation system to choose between the two different set of wheels depending on the terrain condition. The principle of operation of this robot comprises switching between conventional outdoor wheels and Mecanum wheels based on the roughness of the terrain.

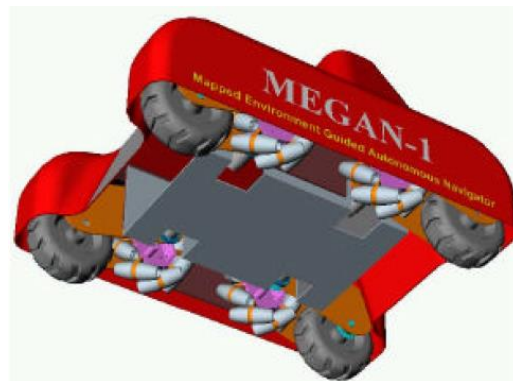


Figure 2.3 MEGAN-1 platform bottom view [21]

This solution attempted to provide a simple concept in order to address the rough terrain navigation limitations. However, there are two key issues that limit the usability of this robot. The first factor is the complexity added by having parallel and mutually exclusive traction systems. The complexity of the control increases too, because it is required to incorporate an algorithm for the wheel switchover with a corresponding terrain evaluation algorithm that dictates the switch. As a result additional efforts for the sensing and determination of the terrain roughness are required.

The second key factor is the loss of the holonomic capabilities by the vehicle while operating on rough terrain mode. Once the vehicle has switched to the conventional wheels the

vehicle is subjected to non-holonomic constraints, which requires to have two motion planners for the different kinematics of the vehicle. Finally, this solution still limits the use of omnidirectional wheels to relatively structured and flat surfaces.

The next relevant solution was developed by a research group in Japan comprising the Saitama University, The Institute of Physical and Chemical Research and The University of Tokyo[10]. The vehicle designed uses seven special omnidirectional wheels equipped with 12 free moving rollers oriented at 90° and an obstacle climbing link mechanism (Figure 2.4). Because of the wheel independent actuation requirement the main challenge for its control is the coordination of the rotational velocity of each wheel [10] [25].

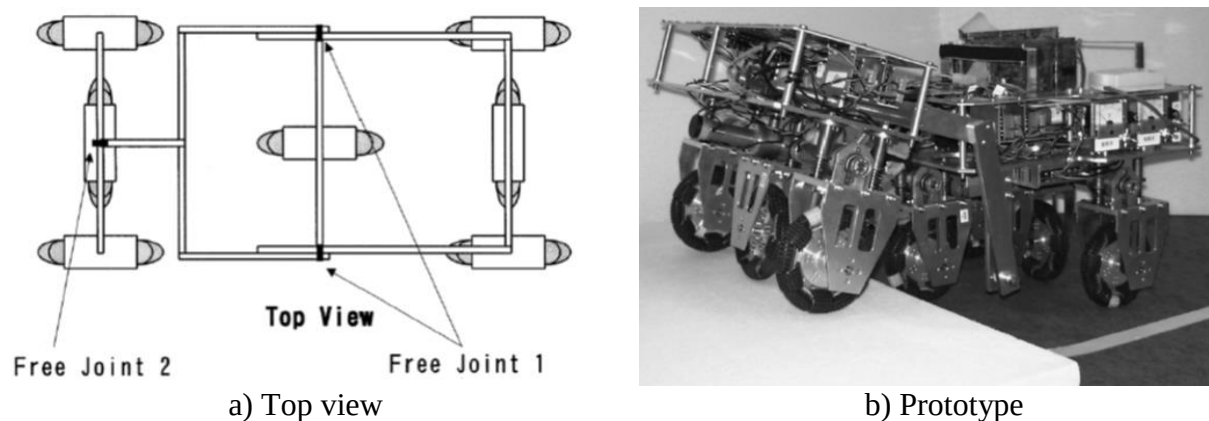


Figure 2.4 Japan research group omnidirectional vehicle with step-climbing ability [25]

While navigating on rough terrain the passive suspension system had the ability to adapt the body configuration of the vehicle depending on the terrain profile. This reconfiguration of the robot's frame allows it to climb relatively high obstacles and smoothly traverse on unstructured terrain.

The main feature of this vehicle is that, it can estimate the terrain conditions without the need of additional sensors or actuators [25]. Additionally the results presented in [25] indicate an improvement in energy efficiency compared with legged robots and a reduced slippage when the wheel actuation is coordinated appropriately. The ability to climb obstacle without prior knowledge of the terrain is the main strength of this solution. The vehicle was designed to operate in narrow spaces and floors with steps and slopes with unknown location. However, the control method only considered motion on flat surfaces and climb over steps in one direction

[10]. Therefore, this solution does not address the challenges of providing diagonal or lateral motion while navigating on rough terrain, neither at obstacle climbing.

All of the available solutions [8][10][14][24][25] provide creative solution that expands the capabilities on rough terrain for vehicles equipped with omnidirectional wheels. However, those OGVs don't preserve effectively the holonomic capabilities of the vehicle when operating on challenging environments.

2.3 All Terrain OGVs with Modified Omnidirectional Wheels

The latter group of solutions address the challenges of unstructured environments using particular wheel designs. This approach also known as constructive solutions have attracted an important stream of research efforts over the last years [8]. The approach is based on either modifications to the design of typical Mecanum wheels or to develop new wheel designs. Modifications include changes in the shape of the rollers, orientation of the rollers, and profile of the hub among others. Several alternative designs have been developed, built and tested successfully (e.g. [24][21][22][11][9]). In what follows we describe the wheel designs that can be considered game changers as they have provided significant contributions.

The first constructive solution was developed by a research group at the University of Massey in New Zealand [21]. This design targeted a Mecanum wheel with dynamically adjustable rollers. This device has the ability to adjust as needed the angle at which the peripheral rollers are mounted with respect to the central hub. It also incorporates a system to lock the rollers in place in order to inhibit their passive motion if required. As shown in

a) rollers at 45°

b) rollers at 90°

Figure 2.5 in order to perform the roller adjustments each wheel requires an additional actuation system (a disc controlled by a linear actuator). These features allow to the wheel an increase of its energetic efficiency and a considerable reduction of slippage when traversing on rough or uneven surfaces[21].

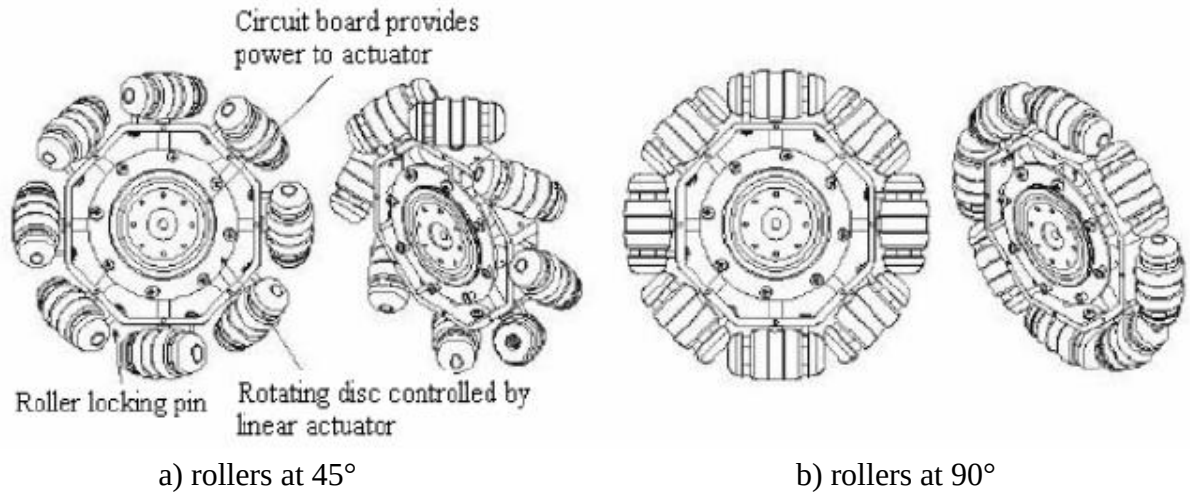


Figure 2.5 Mecanum wheel with dynamically adjustable rollers [21]

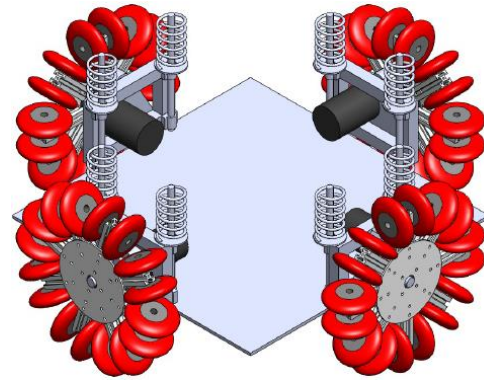
The energy efficiency improvement provided by this solution is given by the ability to reconfigure the rollers arrangement according to the motion direction of the vehicle. In several situations (for example the case of diagonal motion) a conventional Mecanum wheel requires that some of the component's induced motion cancel the motion induced by other components. With the reconfigurable Mecanum wheel it is possible to adjust the rollers such that the components of the induced forces contribute to the overall thrust of the vehicle increasing the efficiency. Due to its complexity, this solution requires the development of control and sensing schema to determine and apply the proper roller configuration. Despite a number of advantages the improvements provided in rough terrain does not extend to diagonal or lateral motion.

The next constructive alternative was developed by the University of Texas at Arlington Research Institute (UTARI) [11]. Their solution is an omnidirectional wheel specifically designed to provide traction and rough surfaces and to have a simple and economical construction. The wheel (Figure 2.6a) is composed by a central hub with 16 peripheral rollers mounted in pairs at 90° with respect to the frame. These rollers are conventional wheels of 100 mm diameter, roughly above 25% of the wheel assembly diameter (380 mm).

The ability to traverse uneven and rough terrain is given by the chosen roller diameter and the grip of the polyurethane used. The wheel was designed to be mounted on a square configuration 4 drive platform with an offset of 90° between each motor as shown in Figure 2.6b.



a) wheel front view



b) omnidirectional platform

Figure 2.6 UTARI omnidirectional wheel design [11]

A prototype of the wheel and the robotic platform was built and tested at flat hard surface and outdoor (uneven surface of grass and dirt) conditions. In both scenarios the vehicle maneuvered as expected, observing full control of all its degrees of freedom. Due to the reduced area of contact provided by the thin rollers the vehicle observed some vibrations which were absorbed by adding a suspension system (Figure 2.6b) [11].

The wheel described above is one of the few alternatives that provides a vehicle with holonomic capabilities on rough terrains. However, as with other designs, the ability to handle rough terrain and some obstacles is determined by the diameter of the rollers. Regardless of this knowledge the precise obstacle overcoming capabilities of the wheel remain unknown. Further testing and analysis are required in order to determine realistically the potential of this wheel.

Following previous developments a multidisciplinary research group from the MIT, Tokyo Institute of Technology and Harvard developed the Omni-ball wheel [9] . This wheel is composed by an assembly of two hemispherical free moving rollers that give a spherical contour to the assembly, and two smaller free rotating barrel rollers at the poles of the assembly (Figure 2.7a). The goal of this design is to enhance the capability of OGVs to overcome obstacles and negotiate ground gaps in holonomic motion [9], a series of tests were performed using the 4 wheel platform shown in Figure 2.7b.

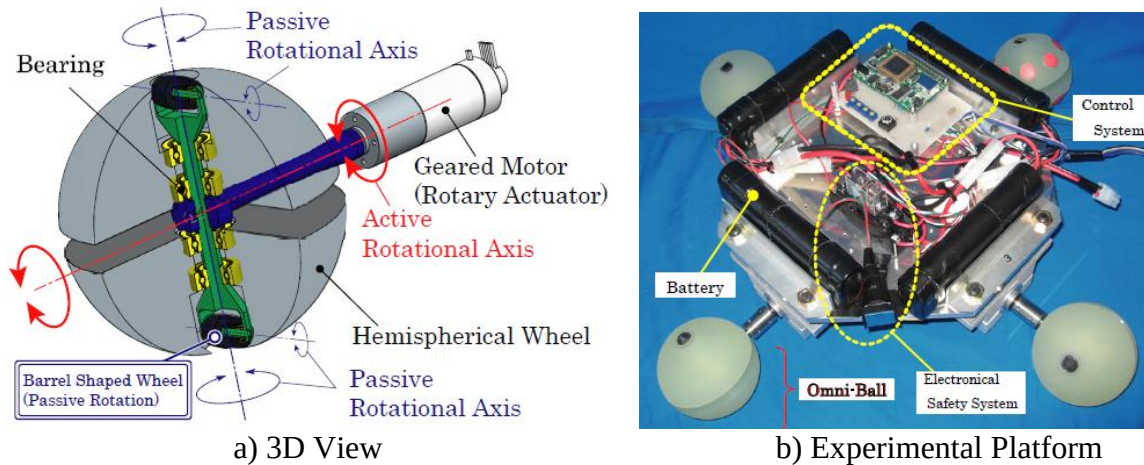


Figure 2.7 Omni-Ball wheel design [9]

The experimental results proved that the omni-ball wheel was able to provide omnidirectional motion (move lateral, diagonal and forward) to the ground robot. The use of this wheel provided a smooth operation with minimal vibrations. Regarding the obstacle negotiation capability, the platform was able to climb obstacles with a height up to 28% of wheel diameter. The gap negotiation ability of the omni-ball wheel was also confirmed by experimental tests where gaps up to 46% of the wheel diameter were smoothly negotiated. Finally the performance of the vehicle was evaluated on steeped surfaces but had difficulties negotiating slopes greater than 8° of inclination. The omni-ball wheel gave to the vehicle the ability to climb the slope in any direction.

The results showed improvements by the omni-ball. When compared to other omnidirectional wheels, its ability to handle obstacles and gaps is remarkable. However, the wheel does not address the rough terrain issues, particularly loose terrain conditions (e.g. sand or mud) and it is incapable of overcoming larger obstacles which traditional wheels can. Moreover, the necessary shaft to transmit the power from the actuator is subjected to high shear stresses if the platform would be scaled up. These two considerations limit the potential use of this solution to very small robots.

In order to increase the capabilities of previous omnidirectional wheels the Elliptical Mecanum Double Wheel (EMDW) and the Semicircular Mecanum Double Wheel (SMDW) were developed at the Autonomous Reconfigurable Robotics Systems Laboratory (AR²S) at the University of Calgary[22][24][26].. Both concepts are devices capable to traverse rough terrain and overcome large obstacles (up to 75% of the wheel diameter).

2.3.1 Elliptical Mecanum Double Wheel

The EMDW (Figure 2.8) is a wheel design constituted by the assembly of two elliptical shaped hubs with peripheral rollers coupled through a special patent pending mechanism. The mechanism has the function to coordinate the rotation and planar motion of the elliptical hubs in order to provide the vehicle with the same movement (direction, velocity, vibrations, etc.) that a conventional Mecanum wheel would provide plus enhanced capabilities to move on rough terrain [22].

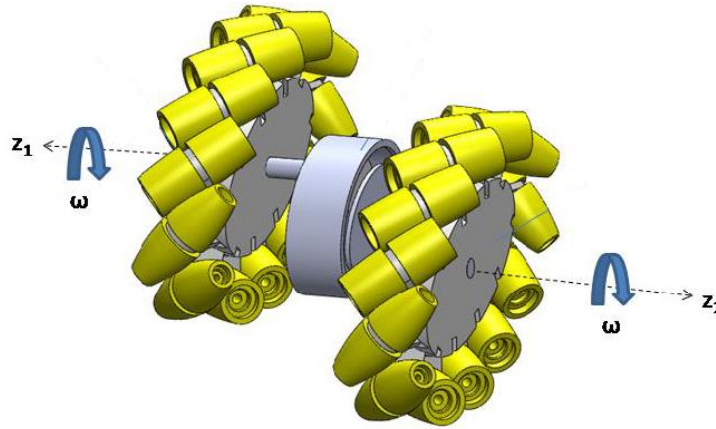


Figure 2.8 Elliptical Mecanum Double Wheel [22]

The EMDW wheels overcome obstacles by stepping on them as the wheel rotates. This process is very similar to the one performed by a person climbing a stairs as illustrated graphically in Figure 2.9. The EMDW can negotiate/overcome large obstacles in more than a motion direction (Figure 2.9). The analysis performed on these wheels determined that they are capable to climb obstacles with a maximum height up to 75% of the elliptical hub's major axis [22], which represents a 300% increase when compared to previous designs. This, however, comes at the expense of a more complex mechanical design and the need to use elaborate control mechanisms.

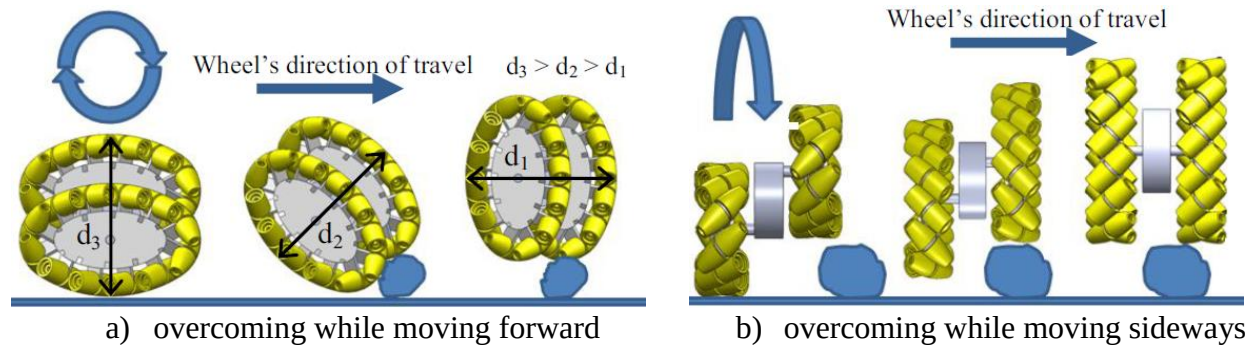


Figure 2.9 Obstacle overcoming principle of an EMDW [22]

While conceptually the EMDW surpasses most of the existing omnidirectional wheels, in practice the design of wheel control mechanisms to achieve the required trajectory is very complex, making its implementation a challenging task.

2.3.2 Semicircular Mecanum Double Wheel

In order to reduce the mechanical complexities associated with the EMDW wheels while still providing the same enhanced capabilities, a second design was developed by the AR²S research group. This new design is the Semicircular Mecanum Double Wheel (SMDW). The SMDW (Figure 2.10) is composed by two mirrored shaped wheel hubs mounted in parallel on the same rotational axis with a shift of 180° between them. Each wheel hub comprises a half circular and half elliptical profiles with custom free moving peripheral rollers attached to each hub (Figure 2.10a).

When fully assembled the wheel possesses the same circular profile as a traditional Mecanum wheel (Figure 2.10c). The elliptical profile of the hubs and the segmented construction of the wheel provide to it the ability to overcome obstacles with a height up to 25% of the assembly diameter [24].

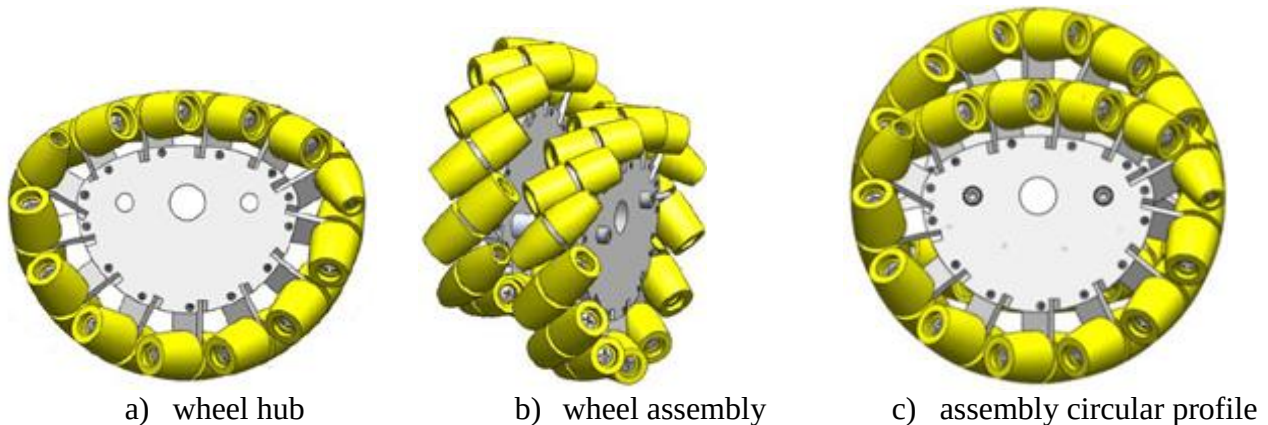


Figure 2.10 Semicircular Mecanum Double Wheel [22]

The SMDW overcomes obstacles using the same principle as the EMDW does, Figure 2.11. The stepping on process and the hub geometry of the SMDW cause a phenomenon of variable torque while an obstacle is being climbed. This phenomenon is the result of a change of the radius of the contact point at a constant turning rate at the transition between the semicircular and the elliptical profile. This phenomenon changes the translational and rotational directions of the OGV motion. Thus controlling OGVs using SMDW requires to take this into account.

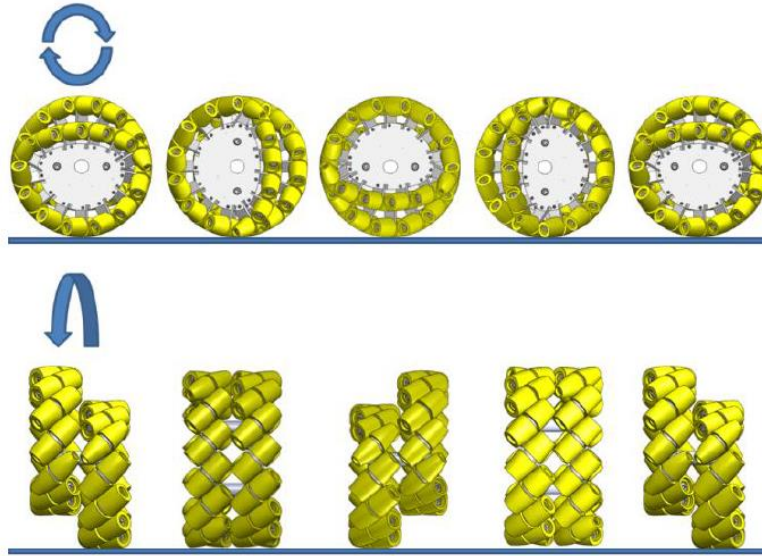


Figure 2.11 SMDW motion configuration along 1 revolution [22]

The SMDW offers several advantages compared to the EMDW. This wheel that can be mounted on most of the existing ground vehicles platforms without modifications. The SMDW design and construction are simpler than the EMDW, since a position mechanism is not required. A fully functional prototype of the SMDW was built and tested by Ramirez, A. and Kuzyk, R. [22]. Several manually controlled tests were performed on different terrain conditions with this prototype. The results of such tests proved the feasibility and operability of this modified Mecanum wheel design to overcome obstacles and navigate on unstructured terrains [24].

One of the challenges when using this wheels is the motion control and navigation of the vehicle. The kinematic model of a conventional Mecanum wheel (Chapter 1 Section 1.4) can still be applied for hard flat surface navigation. However when the obstacle climbing capability is required there are additional considerations that need to be taken into account. The wheels and the vehicle need to be in a specific state or set of states in order to have the appropriate clearance and successfully step on the targeted obstacle.

The task to position the vehicle and the wheels at the appropriate states to climb the obstacle involves additional factors that are not required for the control of vehicles with traditional Mecanum wheels. Some of those additional factors are: the terrain, the required clearance determination, the motion planning of the vehicle, and the trajectory generation. These factors increase the control complexity of a robot using SMDW as first reported in [26].

2.4 Maneuvering Control for Robots with Semicircular Mecanum Double Wheel

In order to make use of SMDW in autonomous ground robots, a maneuvering control strategy to perform obstacle overcoming was developed at the University of Calgary [26]. This solution is based in the approach of treating the maneuvering task as an optimization problem [26] that is required to calculate a desired final state of the OGV frame centre position and wheels angular position. This state is calculated with prior knowledge of the terrain, the obstacle's position and height; and the initial conditions of the vehicle (frame centre position and wheels angular position).

The solution included a trajectory that connects the robot's initial state with its final state using a random search hill climbing algorithm which provides the maneuvering coefficients for a quadratic spline trajectory. The use of spline curves ensures smooth motion and avoids robot discontinuities in velocity and acceleration. The maneuvering algorithm considered an open loop controller which uses the produced spline to determine the velocity of the vehicle's wheels.

The maneuvering control was tested in a physical simulation obtaining results that proved the feasibility of the obtained trajectories in order to overcome the obstacles [26]. Despite the positive results obtained, the maneuvering developed control has a significant shortcoming. The obstacle overcoming can only be performed when the OGV approach obstacles in a sideways direction of travel (Figure 2.12).

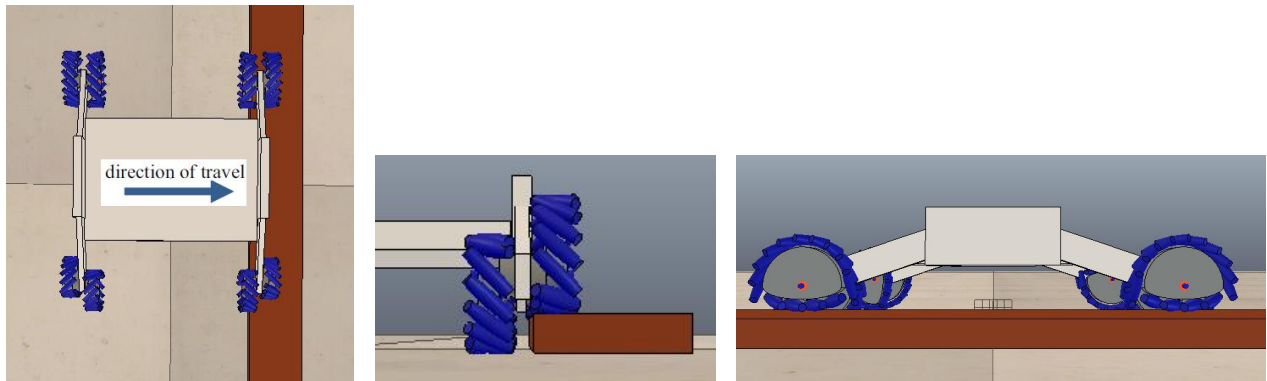


Figure 2.12 Vehicle and SMDWs configuration for obstacle overcoming

The restrictions for obstacle climbing are caused by the maneuvering control developed; not by the design of the SMDWs themselves which can overcome obstacles in multiple directions. From the experimental tests there is evidence that such orientation is not the only suitable possibility to approach obstacles. The position required by the developed maneuvering scheme was chosen in order to simplify the solution of the optimization problem[26].Conversely,

the obstacle overcoming capabilities of the SMDW in different overcoming orientations need to be investigated and analyzed.

2.5 Trajectory Search for Ground Robots

The generation of motion trajectories is essential for the existing UGVs maneuvering algorithm in particular for the vehicles with SMDWs. In this section a review of the most common trajectory generation techniques used in autonomous wheeled robots is presented. The generation of trajectories for UGVs can be performed using a planning or reactive approach [3]. These approaches require information regarding the operational environment of the vehicle and the definition of a goal position. The developed approaches can be used for both holonomic and non-holonomic platforms assuming a proper mathematical model of the vehicle is available.

The reactive approach aims to modulate the vehicle trajectory in order to avoid collisions. The advantage of this approach is that it can be performed in real time with lower computation time requirements and can handle less accurate terrain maps. Under certain circumstances the reactive approach can produce nearly optimal trajectories suitable for indoor applications [27].

The planning approach implies the generation of a trajectory to solve the problem of providing a geometric path suitable to be tracked by a motion controller. The planning approach requires more computation time and generally cannot be performed in real time since the path needs to be defined before the motion. For navigation purposes the combination of both reactive and planning approaches has been developed as a complimentary solution [3]. Such solutions consider the generation of a global path to reach the desired position and a reactive behavior during the execution of the path which accounts for disturbances such as map or sensor inaccuracies.

One of the main challenges of the path planning problem is the generation of smooth paths. The use of polynomial functions for the trajectory has been developed to address this issue [28][29]. For the generation of this polynomials it is often required to produce a set of points along the trajectory which are then fitted into the function. Among the different type of polynomials and fitting approaches, the piece-wise spline curves are a common [30]. More recent applications developed to include the robot's dynamic constraints, the generation of sophisticated curves such as Bezier curves, B-Splines or high order splines used to improve the smoothness and suitability of the paths are available [31][32][33].

Other technique to construct trajectories is the use of primitive motions. The developed techniques use a set of basic motions with continuous velocity and acceleration in order to produce composed and complex trajectories. Some applications[34][35] have been developed which have proved the feasibility of this technique for cluttered environments as well as for real time path planning applications. While the use of primitive motions has been targeted for path planning, in several applications it is also suitable to be used in combination with a reactive approach, to improve the performance of the global trajectory. Compared with the use of polynomials, the primitive motions trajectories can experiment sudden changes in direction for motion and velocity. However, this sudden changes might result in greater control effort and energy consumption [36].

The existing techniques for trajectory search are classified in the following main categories: cell decomposition, roadmap, potential field, mathematical programming, and probabilistic algorithms [3][36]. Some probabilistic algorithms such as Probabilistic Roadmap Method (PRM) and Rapidly-exploring Random Trees (RRT) are commonly used for the generation of entire primitive motion trajectories [36]. These solutions have been combined with genetic algorithms, neural networks, fuzzy logic, random search, particle swarm optimization (PSO), artificial potential fields, linear programming, among others to generate optimal solutions [31][36][37].

Among these developments a stream research work in planning an optimization has focused in the used of genetic algorithms, variants of particle swarm optimization (PSO) and ant colony optimization, or the combination of them [36][37][38][39][40][41]. Some of these solutions have focused and used for omnidirectional wheel robots path planning [38][39]. The PSO and genetic algorithms offer the advantage of incorporating multiple objective parameters for optimization [36] and in several cases can provide solutions faster. The main shortcoming of these algorithms is that they cannot guarantee convergence.

The last step in the trajectory generation and optimization task for OGVs task is the execution of the paths by the platform. The most common trajectory execution systems consider the use of feedback controllers or open loop controllers [3]. Open loop control option has been mainly used for short paths or local navigation because it is prone to accumulate error and deviation as the distance increases. The feedback control navigation implies the use of a trajectory tracker that uses robot position information as feedback in order to compensate for robot's errors in motion.

PID, state space, adaptive and fuzzy logic control schema have been used for feedback position tracking of OGVs[18] [19][42][43][44] .

2.6 Summary

There are diverse and different approaches used to solve the path planning problem of ground robots. However, limited work has been done related to rough terrain navigation and obstacle climbing capabilities of robots equipped with omnidirectional wheels. Current solutions are restricted to a single solution (orientation, position and motion direction) at which the robot can climb obstacles. The limitations in motion direction and vehicle configuration imposed for the single solution restrict significantly the holonomic capabilities during the rough terrain and obstacle overcoming operations. Some devices such as the SMDW provide the potential to handle obstacles in more than a single pose or configuration. However, the maneuvering strategy developed for this device thus far is limited to a single solution for obstacle climbing.

Chapter 3: Problem Statement

The information presented and discussed in the introductory chapter of this thesis, and the literature review section show that one of the persistent problems found in wheeled omnidirectional ground vehicles is the navigation on rough terrain and obstacle climbing capabilities. Diverse approaches have been developed in order to extend the applications and improve the capabilities of vehicles in such conditions. Nevertheless, despite important contributions and achievements the problem to overcome obstacles while retaining holonomic capabilities has not been solved. Some of the constructive solutions address this issue partially; however, the abilities of OGVs to overcome obstacles in omnidirectional motion have not been explored in depth.

The Semi-circular Mecanum Double Wheel developed at the University of Calgary is one of the few devices that can potentially improve the obstacle handling capability in holonomic motion. This solution has already succeeded by climbing relatively large obstacles in lateral motion, unlike any of the other omnidirectional wheel platforms developed thus far. In addition a maneuvering approach for unmanned applications has been developed and tested in simulations for a 4 wheel vehicle equipped with SMDWs. However, such maneuvering control scheme has been restricted to the lateral motion as the only suitable configuration for obstacle overcoming.

3.1 Problem Statement

In order to analyze the true capabilities of the SMDW to overcome obstacles in diverse motion (e.g. lateral or diagonal) further research needs to be performed. Considering the key factors discussed within previous chapters, the problem statement that will be addressed in this thesis is:

Develop a methodology to maneuver a ground vehicle equipped with semi-circular Mecanum double wheels to overcome obstacles such that the vehicle motion (position and orientation) is not restricted to a single solution.

During previously manually controlled tests research work of a wheeled platform with SMDW evidence was collected that such vehicle is able to step on obstacles that are not parallel to the face of the wheels. Nevertheless the capabilities of the wheel under those conditions were

not investigated in detail. The existing maneuvering control technique focuses on the lateral motion step climbing as such particular motion is one of the main differentiators of the SMDW with respect to any other omnidirectional wheel designs.

Determining the clearance that exists between different vehicle/wheel configurations (motion direction and wheel planar and angular position) and the obstacle are key factors to consider of the already developed maneuvering algorithms. However, due to the complex design of the semi-circular Mecanum double, the maximum clearance of the wheel was determined in previous works using a geometric analysis [22][26]. In order to analyze its behavior for different climbing conditions, it is required to perform a more rigorous analysis. The consideration of additional factors such as the contact area required for climbing are needed to complement the existing geometric analysis. As a result the development of a geometric model and the incorporation of a simplified kinematic analysis methodology of the SMDW stepping-on process are presented in this thesis.

The work presented in this thesis is based on the maneuvering control algorithm previously developed in [26]. The main goal of this research is to provide a more extensive use of the SMDW omnidirectional capabilities particularly at the obstacle overcoming stage. To achieve this objective the proposed solution will extend previous solutions to enable the wheel to approach obstacles at more than a single orientation and thus enhance the maneuvering algorithm. By increasing the number of allowable climbing orientations a reduction in the current gap of the holonomic obstacle climbing abilities of omnidirectional wheels can be achieved.

3.2 Thesis Scope and Assumptions

The goal of extending the obstacle overcoming capabilities of robots with SMDW implies multiple aspects with a high degree of complexity such as: better terrain sensing, vehicle motion planning, modeling of the wheel assembly, development and selection of realistic and accurate simulation environments, among others. In order to focus the research effort to the goal of analyzing the wheel capabilities in rough terrain, the proposed solution will use the following assumptions to simplify the endeavor:

- The robot will navigate on a static, homogenous, non-deformable hard surface with constant friction. The study of the wheel's slippage estimation, and its handling and/or modeling is outside the scope of this thesis.
- The primary unevenness of the terrain will be the obstacles to be climbed which will have the same mechanical characteristics as the ground. Obstacles will be approximated as a step of constant height and relatively large length and depth such that once the robot has climbed it, it will remain moving on its surface. Rocky surfaces will not be considered as the frequent terrain geometrical changes will increase the problem.
- The height of the obstacle will always be equal or less than the maximum clearance capabilities of the vehicle's wheel. Higher obstacles can be handled by already existing avoidance algorithms. Obstacle avoidance is not considered here.
- The profile of the terrain including the location of the obstacles and their height is assumed known, constant and accurate. Thus deformations in the terrain are not considered. Terrain information is assumed to be available to the maneuvering algorithm. Vision sensing systems, mapping techniques or other terrain perception algorithms are not within the scope of this work.
- The maneuvering algorithm has accurate information regarding the odometry of the vehicle (linear and angular position of the OGV frame of reference as well as its velocity and rotation), the rotation of the SMDWs (angular velocity and angular position), and the initial conditions. Position estimation/sensing systems, data acquisition and sensor filtering/characterization techniques are outside the scope of this thesis.
- For the analysis of the SMDW, the model of the wheel will be simplified to avoid considering the complex construction and the kinematics of the peripheral rollers.
- The maneuvering methodology requires that the initial position of the OGV is not located more than 3 meters away from the obstacle; and the robot's orientation should not require more than quarter turn rotation to approach the obstacle on a suitable climbing position.

In addition to the above assumptions, the research work developed in this thesis considers only the maneuvering methodology for a ground robot; it excludes: any design modification to the SMDW, as well as any modification of the robotic platform and any improvements/ changes to the robot's auxiliary systems (e.g. sensors, data acquisition, map generation, etc.). Since the main goal is to broaden the omnidirectional capabilities of the SMDW in rough terrain the

proposed solution provide a simple and practical alternative. The scope of this thesis does not include experimental tests, all the evaluation and testing required will be performed in a simulation environment.

3.3 Proposed Solution Overview

In order to increase the capabilities of robots with SMDW to overcome obstacles in more than a single solution (in terms of orientation and wheel states) a new maneuvering methodology is explored and developed in this thesis. The intended final outcome of this new methodology is the generation of a trajectory that drives a given OGV to a desired configuration (orientation and wheels state) that allows it to climb a given obstacle. In order to develop this methodology a series of five research steps will be performed as shown in Figure 3.1.

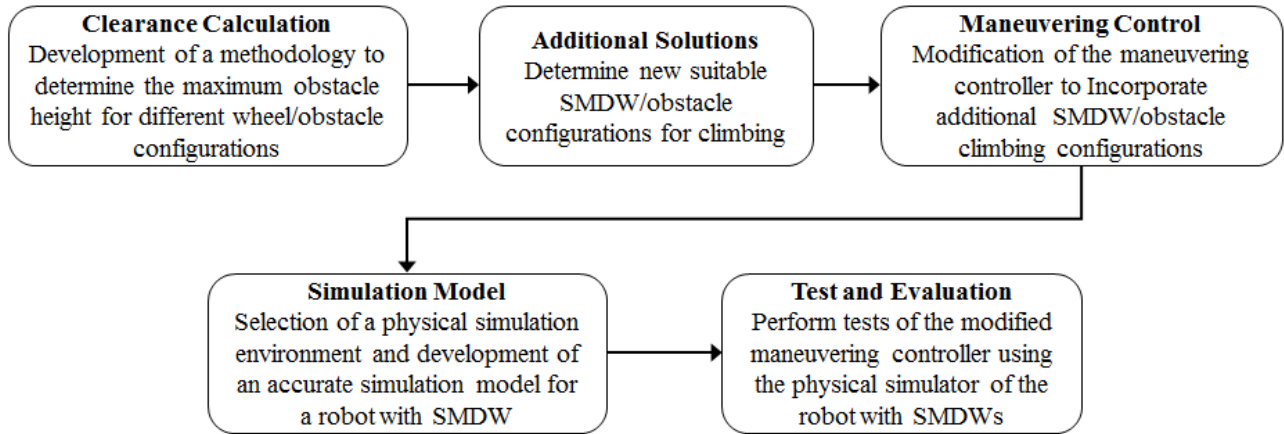


Figure 3.1 Tasks for the development of the maneuvering methodology

The clearance determination will involve an extensive analysis of the SMDW. A new model of the SMDW will be proposed incorporating a more extensive geometric analysis and additional kinematic parameters. The geometric aspects of the model will be used to determine the space clearance of the wheel at different states. In this thesis the wheel clearance is defined as the free space between the wheel hub and the ground in which an obstacle can fit and later be climbed (Figure 3.2). As illustrated in Figure 3.2 the wheel's ground clearance will vary depending on the wheel state with respect to the ground.

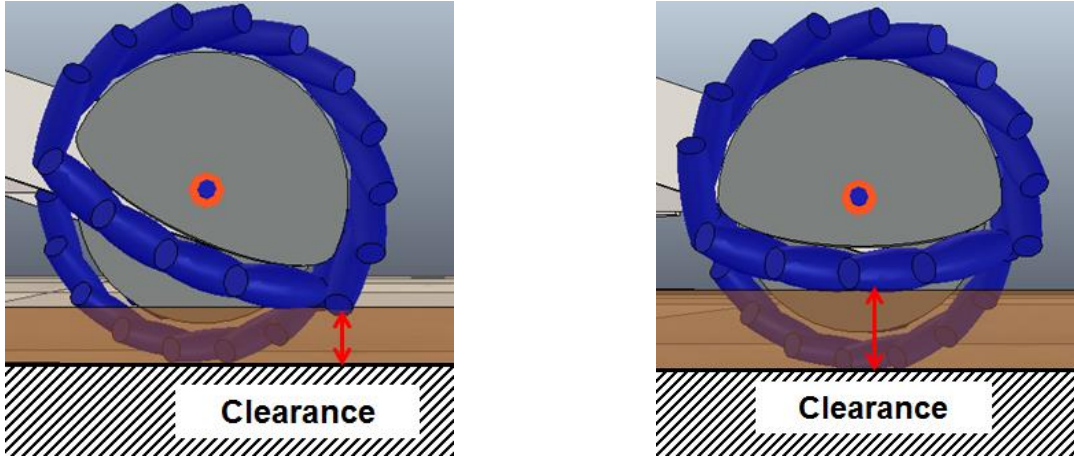


Figure 3.2 SMDW ground clearance

A geometric model is not sufficient to assess accurately the climbing capabilities; such model will only be able to determine whether or not an obstacle can fit physically beneath the SMDW. The kinematic variables to be incorporated into the model have the goal of assessing the climbing suitability of a given SMDW/obstacle configuration. The kinematic aspect of the model will provide knowledge if the given wheel position has the minimum required area of contact to climb on the obstacle.

In order to make possible the incorporation of the kinematic parameters, this work proposes to approximate the wheel assembly and the obstacle as a planar cam and follower mechanism. By the use of this simple yet effective approximation, existing cam theory can be used to estimate the required torque and the minimum area of contact for different wheel/obstacle conditions. Making use of the new improved SMDW geometric model, the clearance determination methodology will provide a tool to estimate different configurations (wheel and vehicle orientation with respect to the obstacle) at which the robot can overcome an obstacle of a defined height and encounter it at a given location along the obstacle.

For this research work a set of suitable climbing robot configurations will be chosen. The existing robot maneuvering planner will be modified to incorporate the additional configurations. The set of the new configurations will be used as valid final positions for the robot trajectory. The modified search module will run to search and find trajectories for those final positions. The trajectories will be evaluated using a pseudo-optimization approach which will assign a cost for each trajectory in function of the final desired states deviation (robot position, orientation and wheels angular position). The generated trajectories for each test will be compared with each other and the one with the lowest cost value will be chosen and tested.

A software based model of a vehicle with SMDWs will be developed to test and evaluate the performance of the trajectories produced by the maneuvering planner. A good simulation with increased accuracy is required in order to evaluate the maneuvering planner and the suitability of the solutions obtained with the new clearance calculation methodology. The use of a software based model (with a physical simulator embedded) will improve the analysis accuracy by considering contact forces and friction to the simulation not used in previous research. The incorporation of such factors will be achieved without the need of further mathematical and physical modeling as a physical engine and library embedded in the simulator will be in charge of those aspects. In addition the simulated model will incorporate the effects of the wheel peripheral rollers, which are a key factor to provide a realistic simulation of the SMDW.

A set of tests will be performed using the physical simulator under different initial and final conditions while varying the location and height of the obstacles. The results of such tests will be analyzed and discussed in order to evaluate the capabilities and feasibility of the trajectories obtained with the maneuvering methodology proposed in this research work.

3.4 Thesis Contributions

The research performed in this thesis intends to provide contributions in the field of obstacle overcoming by robots with omnidirectional wheels. In particular the goal will be to improve the use of semi-circular Mecanum double wheels. A list with the expected contributions of this work is presented below.

- The broadening of the obstacle climbing capabilities of a robot with SMDWs: Prior to the completion of this thesis the state of the art for vehicles with omnidirectional wheels (only for certain wheel types and platform designs) allow them to overcome obstacles at a single and specific configuration. The proposed maneuvering methodology will increase the holonomic capabilities of robots with SMDWs by allowing more than a single configuration for the obstacle climbing operation. Expanding the number of feasible solutions increase the robot's maneuverability and in cases with strong constraints in the navigation space (such as the case of search and rescue applications) increase the chances of the vehicle to find a feasible path (if it exists). While this methodology does not remove entirely the non-holonomic constraints imposed of the

robot, the addition of feasible solutions reduces the gap in order to achieve full holonomic motion at obstacle overcoming.

- The increase of tools and methodologies for the development of SMDW applications: This research work will expand the scope of the first maneuvering controller developed at the University of Calgary[26]. It also proposes new models and methodologies for the analysis of the wheel capabilities. The tools developed for the completion of this thesis will provide more support and background for the development and implementation of autonomous omnidirectional robots.
- The development of a realistic and comprehensive simulation of the semi-circular Mecanum double wheel: In order to evaluate the proposed maneuvering methodology an accurate simulation of a robot with SMDWs is required. This thesis will produce a detailed software based simulator of a SMDW (using the V-REP software) and use it to simulate a 4 wheel omnidirectional platform. The simulation to be developed for the wheel will include the peripheral rollers which are usually neglected. The consideration of the peripheral rollers for simulations of omnidirectional wheels is a key factor since the rollers are the source of its holonomic capabilities and at the same time the reason of several shortcomings such as slippage and energy losses. Moreover a reduced number of research works have incorporated the rollers for the simulation of Mecanum wheels [46] but have only considered the conventional Mecanum wheel where the rollers have a simpler geometry.

Chapter 4: Proposed Solution

Chapter 3 provided a brief introduction and overview to the proposed approach to broaden the omnidirectional capabilities on rough outdoor terrains of robots with SMDWs. The proposed solution implies the modifications on the existing maneuvering algorithm as well as further analysis of the SMDW capabilities. In this chapter such modifications and analysis are described in detail. Also along with the important details of the solution chosen, a discussion of their advantages and shortcomings are covered.

4.1 Obstacle Overcoming Capabilities Estimation for SMDW

Having more than a single feasible configuration to overcome obstacles is fundamental to reduce the restrictions currently imposed to the holonomic capabilities of robots with SMDWs. In order to provide the additional suitable configurations required for the maneuvering methodology to perform the obstacle overcoming, it is necessary to better analyze the capabilities of the SMDW. In the upcoming sections a detailed analysis of different orientations, positions and motions of the SMDW with respect to obstacles are analyzed in order to estimate the wheel abilities and its feasibility to climb over obstacles. This analysis comprises the development of geometric models and the consideration of kinematic parameters used to provide a simple but still realistic assessment.

4.1.1 Geometric Model of a SMDW

The first step to analyze the overcoming capabilities of the SMDW is the development of a simplified geometric model for the wheel. The main goal of this model is to reduce the complexity of the SMDW mathematical model in terms of the kinematic and motion analysis. The simplification begins with the approximation of the exterior profile of the wheel hubs. According to the wheel design the outer edge of each hub can be modeled in a simple way as a jointed semi-circle and semi-ellipse to form a closed curved (Figure 4.1). Inside this profile the peripheral rollers are contained whose contact points with the ground match the trajectory defined by the closed curved profile. Therefore the silhouette inscribing each hub including the peripheral rollers is defined by the composed shape of Figure 4.1.

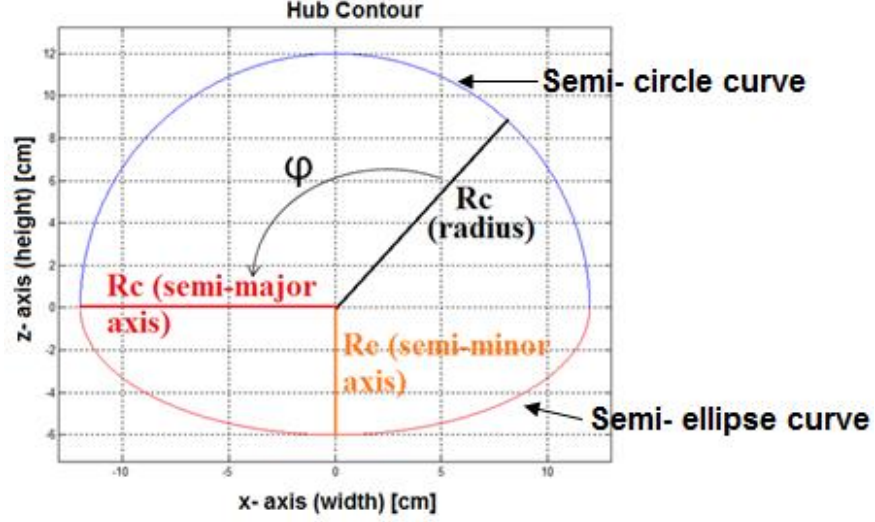


Figure 4.1 SMDW hub contour approximation

The shape of the contour is defined by the following equation in polar coordinates (Eqn. 4.1):

where φ is the angle of construction for the hub such that:

$$f(\varphi) = \begin{cases} R_c \cos(\varphi) & 0 \leq \varphi < \pi \\ R_c \sin(\varphi) & \pi \leq \varphi \leq 2\pi \end{cases} \quad \varphi = \left\{ 0, \frac{2\pi}{n}, \frac{4\pi}{n}, \dots, \frac{(n-1) \cdot 2\pi}{n}, 2\pi \right\} \quad (\text{Eqn. 4.1})$$

n = angle intervals

The φ angle of construction is measured from the R_c semi-major axis as shown in Figure 4.1.

The hub contour can be represented in Cartesian coordinates using (Eqn. 4.2)

Such that:

$$H \in \mathbb{R}^{3 \times n} \quad \varphi = \left\{ 0, \frac{2\pi}{n}, \frac{4\pi}{n}, \dots, \frac{(n-1) \cdot 2\pi}{n}, 2\pi \right\} \quad (\text{Eqn. 4.2})$$

$$H = \begin{bmatrix} f(\varphi) = R_c \cos(\varphi) & 0 \leq \varphi \leq 2\pi \\ 0 \\ f(\varphi) = \begin{cases} R_c \sin(\varphi) & 0 \leq \varphi < \pi \\ R_e \sin(\varphi) & \pi \leq \varphi \leq 2\pi \end{cases} \end{bmatrix}$$

Having reduced the modeling complexity of the wheel hubs, a 3D geometric assembly of the rigid wheel profile was proposed. Such assembly is composed by a wireframe of the two hub contours, where both hubs have the same axis of rotation but shifted 180° with respect to each other (Figure 4.2). The inner profiles faces are separated by an offset distance which corresponds to the space required by the wheels attachment arm mounted on a skid vehicle.

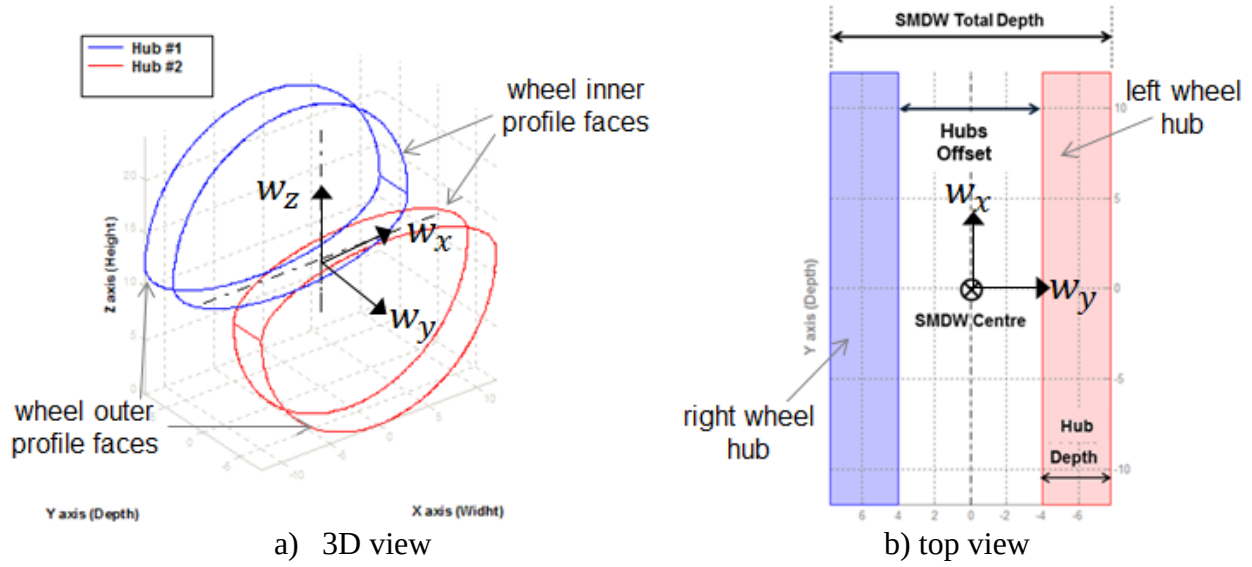


Figure 4.2 SMDW wireframe geometrical model

For the geometrical analysis of the SMDW a wheel's frame of reference has been established in the middle of the offset between the hub profiles which corresponds to the rotational centre of the wheel assembly. The Cartesian coordinates of this frame of reference with respect to the ground are $(0,0,R_c)$. The z-axis component is translated by the corresponding radius of the circular component of the hub, assuming the flat ground will be considered as zero height.

The neutral position or default orientation of the SMDW model for its geometrical study and analysis is shown in Figure 4.3. This neutral position corresponds to the arrangement at which the wheel achieves its maximum obstacle overcoming capability in lateral motion.

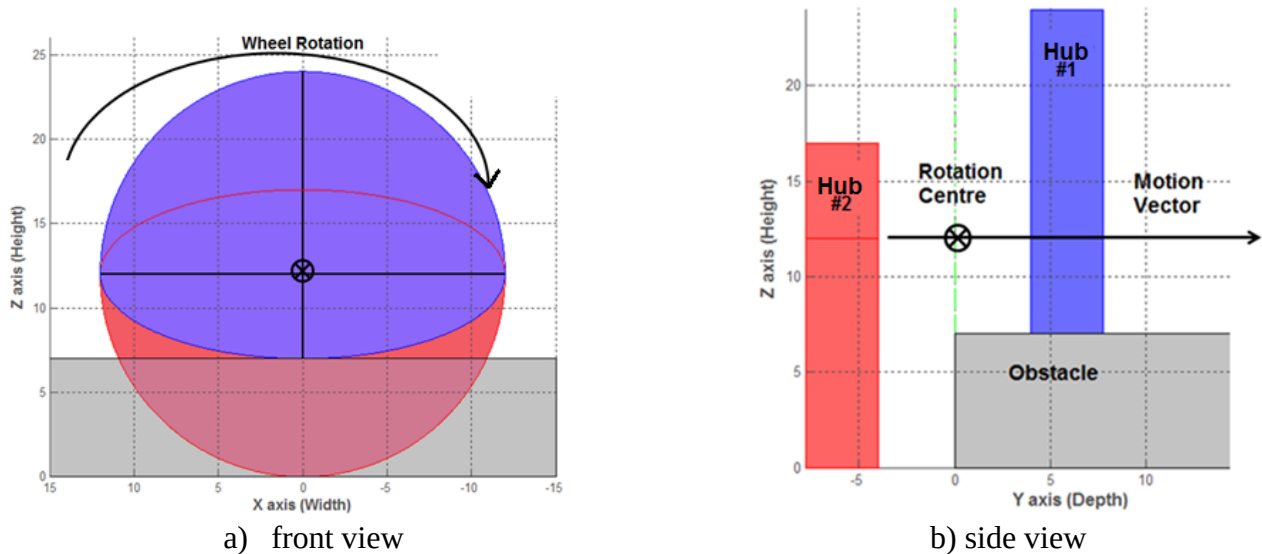


Figure 4.3 SMDW simplified geometrical model neutral position

The simplified geometric model of the SMDW at the defined neutral position can be expressed as a matrix Γ_0 containing the point's cloud of the wire frame structure. The matrix of the geometric model is shown in (Eqn. 4.3) where an additional fourth point with value 1 was incorporated in order to allow the use of geometric transformations matrices.

Where :

$$\Gamma_0 = \begin{bmatrix} P_x \\ P_y \\ P_z \\ 1 \end{bmatrix} \quad \Gamma_0 = T_c \cdot [H_1 \quad || \quad H_2]$$

(Eqn. 4.3)

[$P_x \ P_y \ P_z$] = Point cloud
data of wheel's wireframe
 T_c = Centre of rotation
translation matrix
 H_1 =Hub #1 wireframe matrix
 H_2 =Hub #2 wireframe matrix

The wheel model is composed by the concatenation (||) of the hubs' points matrices. Similarly the hubs matrices are given by the concatenation of the hub contours. For Hub #1 a translation on the y -axis is applied to the contours in order to incorporate the separation between the hubs and the depth between the inner faces as represented below in (Eqn. 4.4).

$$T_c = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & R_c \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad H_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0.5d \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot H \quad || \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0.5d + p \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot H \quad (\text{Eqn. 4.4})$$

Such that $H_1 \in \mathbb{R}^{4 \times 2n}$

Where: d = Offset between hubs p = depth of the hub

The Hub 2 matrices include an additional rotation about the w_y axis accounting for the 180° rotational shift between hubs (Eqn. 4.5).

$$H_2 = \begin{bmatrix} \cos(\pi) & 0 & \sin(\pi) & 0 \\ 0 & 1 & 0 & -0.5d \\ -\sin(\pi) & 0 & \cos(\pi) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot H \quad || \quad \begin{bmatrix} \cos(\pi) & 0 & \sin(\pi) & 0 \\ 0 & 1 & 0 & -(0.5d + p) \\ -\sin(\pi) & 0 & \cos(\pi) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot H \quad (\text{Eqn. 4.5})$$

Such that $H_2 \in \mathbb{R}^{4 \times 2n}$

Finally the original hub contour is redefined as a 4 x n matrix (Eqn. 4.6) needed to make its structure suitable for the use of geometrical transformation matrices.

$$H = \begin{bmatrix} f(\varphi) = R_c \cos(\varphi) & 0 \leq \varphi \leq 2\pi \\ 0 \\ f(\varphi) = \begin{cases} R_c \sin(\varphi) & 0 \leq \varphi < \pi \\ R_e \sin(\varphi) & \pi \leq \varphi \leq 2\pi \end{cases} \\ 1 \end{bmatrix} \quad \begin{array}{l} \text{Such that:} \\ H \in \mathbb{R}^{4 \times n} \\ \varphi = \{0, \frac{2\pi}{n}, \frac{4\pi}{n}, \dots, \frac{(n-1) \cdot 2\pi}{n}, 2\pi\} \end{array} \quad (\text{Eqn. 4.6})$$

The simplified geometric model (Eqn. 4.7) is to be used to determine the maximum clearance of the wheel when it approaches a known obstacle located in a different orientation than parallel to robot's lateral view. In order to account for the effects of the wheel's pose and orientation, geometric transformations need to be applied to the wheel's wireframe matrix Γ_0 (Eqn. 4.3). The required two geometric transformations are:

- Rotation of θ degrees about the w_y axis (Figure 4.4a) to provide different states of the wheel angular position, θ
- Rotation of α degrees about the w_z axis (Figure 4.4b) to yaw the OGV and change approaching angle α between the obstacle and the wheel's face

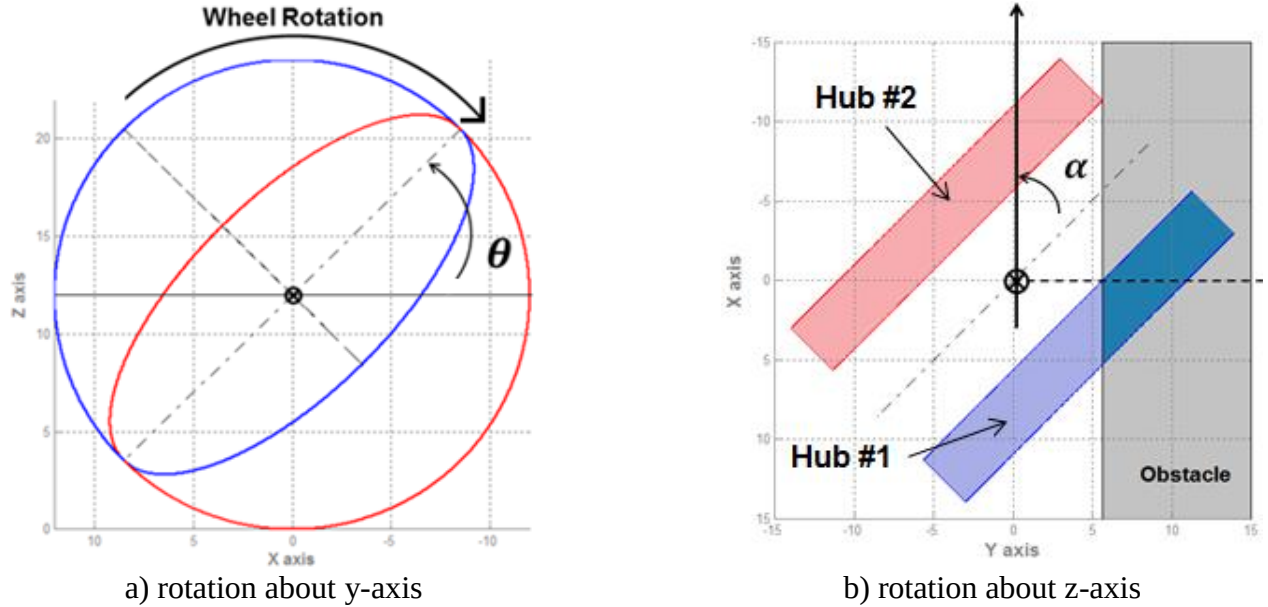


Figure 4.4 SMDW geometric model with 3D rotation transformations

The final geometrical model of the SMDW considering the transformations indicated above is given by (Eqn. 4.7):

$$\Gamma = R_z(\alpha) \cdot R_y(\theta) \cdot \Gamma_0$$

Such that: $\Gamma \in \mathbb{R}^{4 \times 4n}$

$$\Gamma = \begin{bmatrix} \cos(\alpha) & -\sin(\alpha) & 0 & 0 \\ \sin(\alpha) & \cos(\alpha) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} \cos(\theta) & 0 & \sin(\theta) & 0 \\ 0 & 1 & 0 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \Gamma_0 \quad (\text{Eqn. 4.7})$$

4.1.2 Wheel's Parameters (Frames of Reference, Neutral Configuration and Dimensions)

In order to analyze the overcoming capabilities of the SMDW the wheel parameters need to be defined. At the neutral configuration (also known as the default configuration) of the SMDW the elliptical profile of the wheel is positioned parallel to the ground as shown in Figure 4.3b. In addition the outer face hubs are oriented parallel to the obstacle ($\alpha = 0^\circ$).

The wheel's angular position θ and orientation α have both a value of 0° (zero degrees) positioned. The positive rotation direction for θ and α is clockwise about the y-axis counterclockwise about the z-axis, respectively. Thus a positive velocity $\dot{\theta}$ induces forward vehicle motion. A schematic representation of the wheel's frames of reference, neutral configuration and geometric dimensions is presented in Figure 4.5. The geometric dimensions have been taken from a wheel prototype.

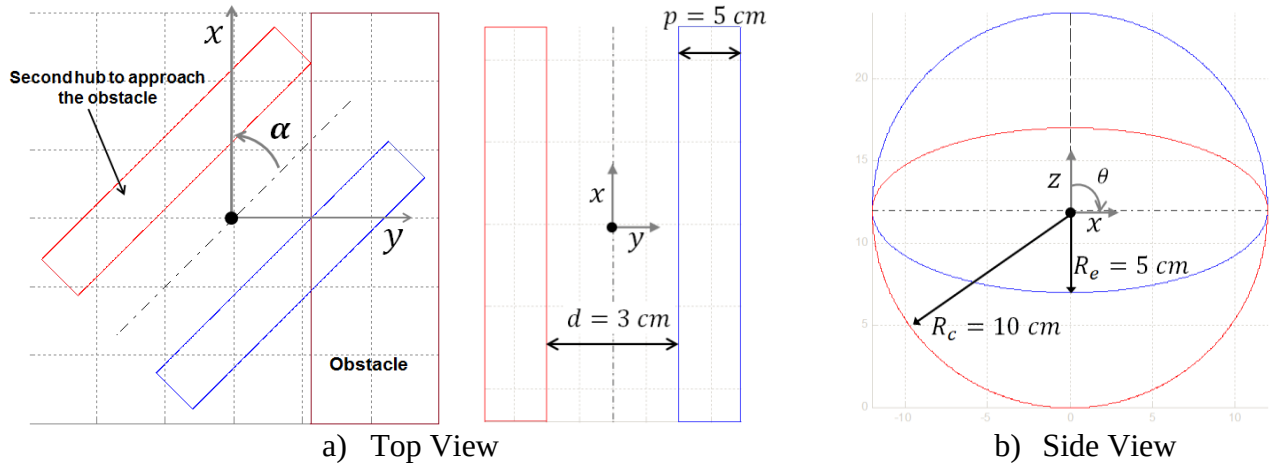


Figure 4.5 SMDW dimensions, frames of reference and default configuration

4.1.3 Geometrical Analysis for Ground Clearance Calculation

In order to determine the available wheel ground clearance for a SMDW using the geometric model described previously it is required to estimate the section of the wheel's hub that is placed above the obstacle's top surface. Recalling that the wheel's ground Clearance changes as the wheel rotates, the determination of the ground clearance is achieved by calculating the point at which the second hub (Figure 4.5a) of the wheel assembly will collide with the obstacle during the wheel rotation. The minimum z-axis value of the SMDW wire frame point cloud corresponding to the hub section above the obstacle defines the wheel's ground clearance.

Such process can be clearly visualized for the case at which the angle of approach between the wheel and the obstacle, α , is equal to zero. For this particular case shown in Figure 4.6, the entire Hub #1 is located above the obstacle and the point at which Hub #2 will collide with the obstacle corresponds to the entire inner face of the hub (Figure 4.6a). The clearance is the lower point of the Hub#1 contour at the given angular position θ (Figure 4.6b).

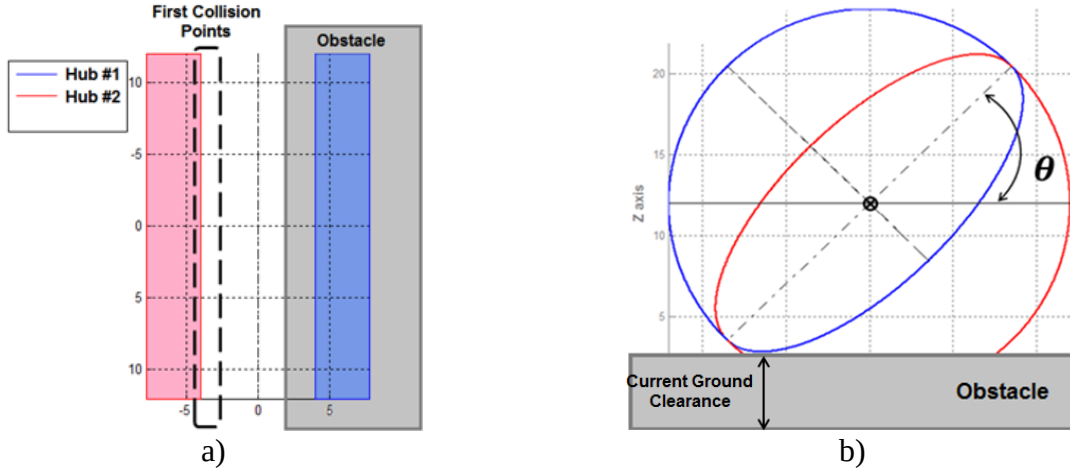


Figure 4.6 SMDW geometric model at $\theta = -45^\circ$ and $\alpha = 0^\circ$

This ground clearance process can be extrapolated for the cases when the approach angle α is greater than zero and less than 90° (Figure 4.7). For the illustration depicted in Figure 4.7 where $\alpha = -45^\circ$ the first point of collision between Hub #2 and the obstacle corresponds to the right hand side corner of the inner face of the hub (Figure 4.7a). The section of hub #1 that is located above the obstacle corresponds to the portion located right to the horizontal projection of the hub #2 collision line. The clearance can be found by determining by the lower point of the Hub #1 and its section located above the obstacle (Figure 4.7b).

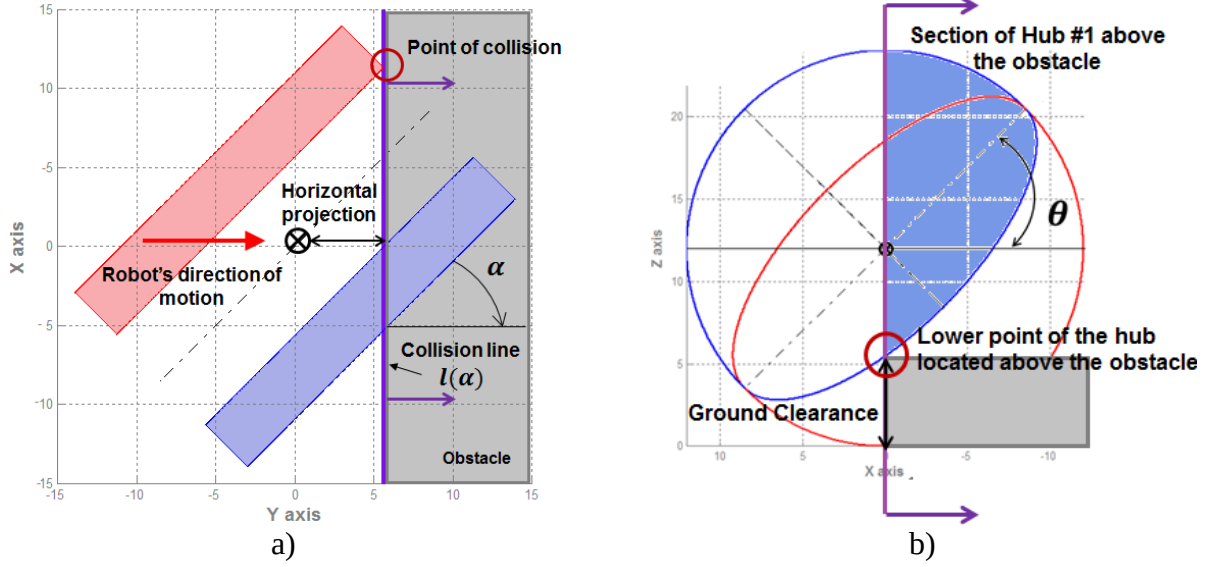


Figure 4.7 SMDW geometric model at $\theta = -45^\circ$ and $\alpha = -45^\circ$

From the description presented above, the horizontal projection of the collision line can be determined by (Eqn. 4.8).

Where:

$$l(\alpha) = \begin{cases} -\frac{d}{2} + R_c \cdot \cos\left(\frac{\pi}{2} + \alpha\right) & -\frac{\pi}{2} < \alpha < 0 \\ -\frac{d}{2} + R_c \cdot \cos\left(\frac{\pi}{2} - \alpha\right) & 0 < \alpha < \frac{\pi}{2} \end{cases} \quad \begin{array}{l} d = \text{Offset between hubs} \\ R_c = \text{SMDW assembly radius} \\ l = \text{collision point projection} \end{array} \quad (\text{Eqn. 4.8})$$

The determination of the first collision point of the hub in contact to the ground is the last calculation required to provide all the necessary input data to calculate the ground clearance of the SMDW using the wheel's geometric model. The next step in the process is to obtain a subset Γ_u of the wireframe matrix Γ_0 that contains all the elements P_x, P_y and P_z for which the P_y elements are greater than the first collision point. Finally given an obstacle, the effective ground clearance value $h(\alpha, \theta)$ of a SMDW is given by the minimum of the elements P_z of Γ_u . An overview of the SMDW clearance determination process is provided in Figure 4.8 .

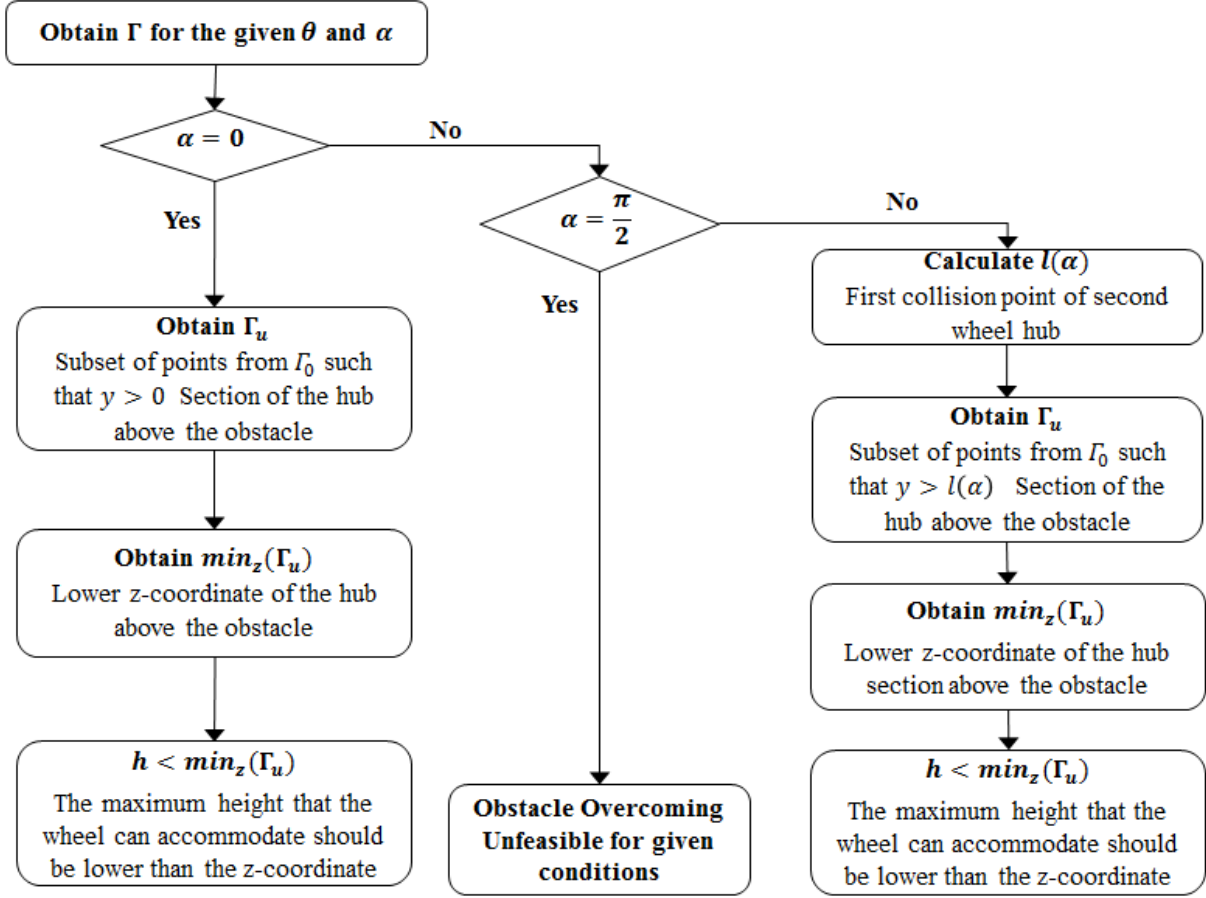
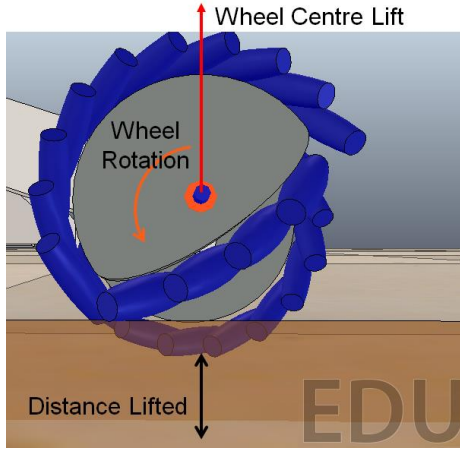


Figure 4.8 SMDW ground clearance determination process

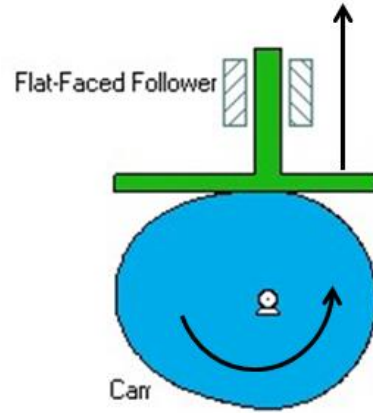
The ground clearance determines the free space between the obstacle and the portion of the wheel hub located above it. Such analysis does not evaluate if that portion of the wheel hub located above the obstacle provides enough contact surface to step on the obstacle as the SMDW rotates. As a result there is a need to include the assessment for the region of wheel contact, such analysis is presented in the next section.

4.1.4 Wheel-Obstacle Region of Contact during Obstacle Climbing

The SMDW exhibits a particular behaviour while climbing obstacles in lateral motion under such conditions the hub of the wheel in contact with the obstacle observes almost exclusively a rotational motion while the centre of the wheel assembly experiences an up linear motion producing the lifting of the vehicle (Figure 4.9a). These motions resemble very closely to the behaviour of a cam mechanism with a flat-faced follower (Figure 4.9b).



a) SMDW

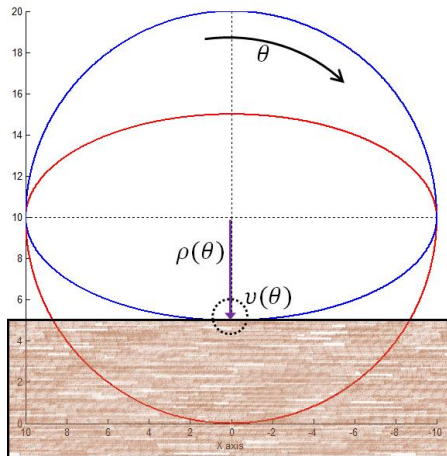


b) flat-faced follower cam

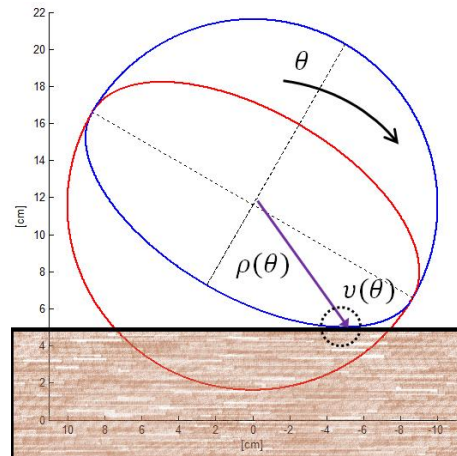
Figure 4.9 SMDW and cam mechanism comparison

“A cam mechanism is a mechanical system consisting of three basic components: a driving element, called the cam; a driven element, termed the follower; and a fixed frame” [45]. The motion induced in the follower is responsible for the transmission of a force from the rotation (produced by an input torque) of the driven mechanism (cam/wheel). The trajectory of the follower is known as motion program and it is determined directly by the profile (contour) of the cam. These similar elements can be seen in the SMDW.

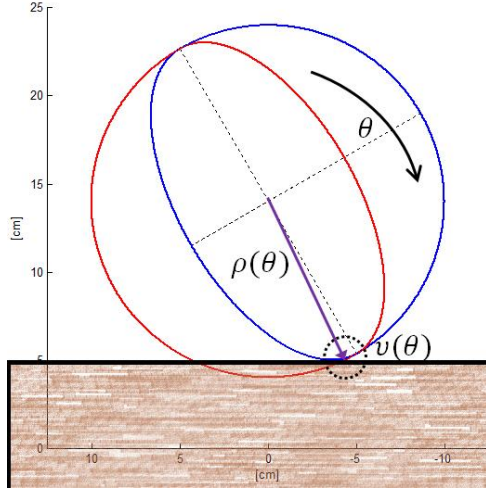
Approximating the SMDW as a cam-follower mechanism, it can be calculate the point at which the wheel makes contact with the obstacle $v(\theta)$ and the distance from the wheel rotational center to this point (θ) . During the obstacle climbing the location of the contact point $v(\theta)$ of the semi-elliptical hub profile varied in function of the wheel angular position θ as can be seen in Figure 4.10.



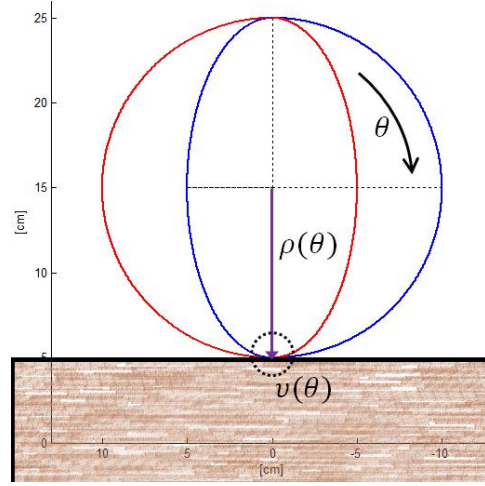
a) $\theta = 0^\circ$ $v(\theta) = 0$ [cm]



a) $\theta = 30^\circ$ $v(\theta) = 4.63$ [cm]



c) $\theta = 60^\circ$ $v(\theta) = 3.49[cm]$



d) $\theta = 90^\circ$ $v(\theta) = 0 [cm]$

Figure 4.10 SMDW obstacle overcoming process

The point of contact can be calculated using the SMDW geometrical model. The computation process implies the generation of a SMDW point cloud $\Gamma(\theta, \alpha)$ using (Eqn. 4.7) which is then translated along the z-axis by the distance lifted by the wheel at such angular position θ . Next using the translated point of cloud is used to determine the region of the point cloud that is over the obstacle Γ_u . Finally from Γ_u the contact point coordinates can be determined as the coordinates of the point which has the minimum value of the z-axis components P_z . A summary of the contact point estimation process is provided in Figure 4.11.

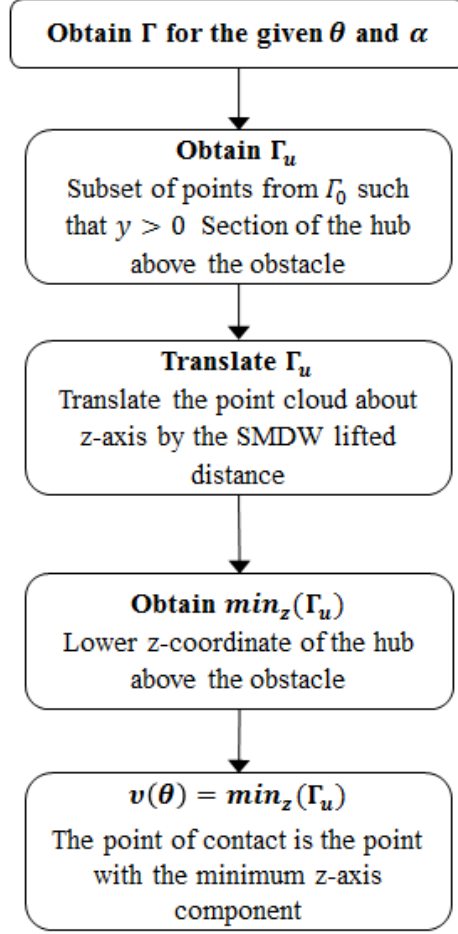


Figure 4.11 SMDW contact point estimation process

The wheel/obstacle region of contact λ can be determined by calculating the locations of the contact points $v(\theta)$ for the climbing interval $\theta_0 \leq \theta \leq \frac{\pi}{2}$. Where θ_0 is the wheel's initial position for the climbing operation and also angular position at which the ground clearance required for a given obstacle height is achieved. The region of contact is dependent on the initial angular position of the SMDW θ_0 and is defined by (Eqn. 4.9).

$$\lambda(\theta_0) = v(\theta) \quad \text{Such that: } \lambda(\theta_0) \in \mathbb{R}^{1 \times n} \quad (\text{Eqn. 4.9})$$

$$\theta = \left\{ \theta_0 \cdots \frac{\pi}{2} \right\}$$

The region of contact (Figure 4.12) is the portion of the SMDW located above the obstacle and delimited by the minimum and maximum points of contact along the rotation interval, v_{min} and v_{max} , respectively.

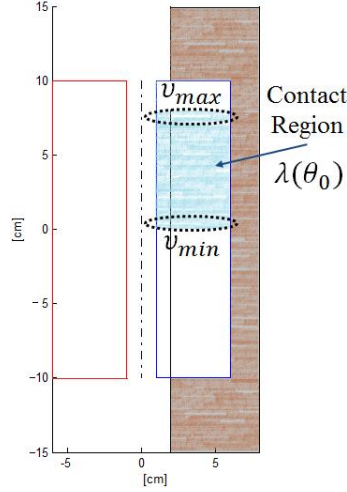


Figure 4.12 SMDW Region of Contact

Similarly to the horizontal projection of the collision line in the geometrical analysis for the clearance estimation, the region of contact λ can be projected over the obstacle proportionally to the wheel orientation α . This projection allow to analyze the feasibility of climbing an obstacle at different wheel configurations (α, θ) . In order to identify if a configuration is suitable for obstacle climbing such projection should show that the whole range of $\{v_{min} \cdots v_{max}\}$ is contained within the projection and the majority of λ located above the obstacle.

The key aspect for this is to ensure that the whole range of contact points $v(\theta)$ is contained within the λ projection. Different orientations are portrayed in Figure 4.13 with the objective to illustrate this issue. At an orientation of $\alpha = -10^\circ$ the projection of λ allows to contain the entire region of contact before the second hub makes contact with the obstacle. At a larger angle of orientation $\alpha = -30^\circ$ the situation changes dramatically as the projection of λ no longer contains the full range of contact points $v(\theta)$. At one of the most critical points of contact, $v\left(\frac{\pi}{2}\right)$, when the transition between the semi-elliptical and semi-circular profile occurs, the v_{min} is completely outside the λ projection. In other words at such orientation the wheel will lose contact when the climbing stage is about to be completed. Thus, the wheel will not be able to overcome the obstacle.

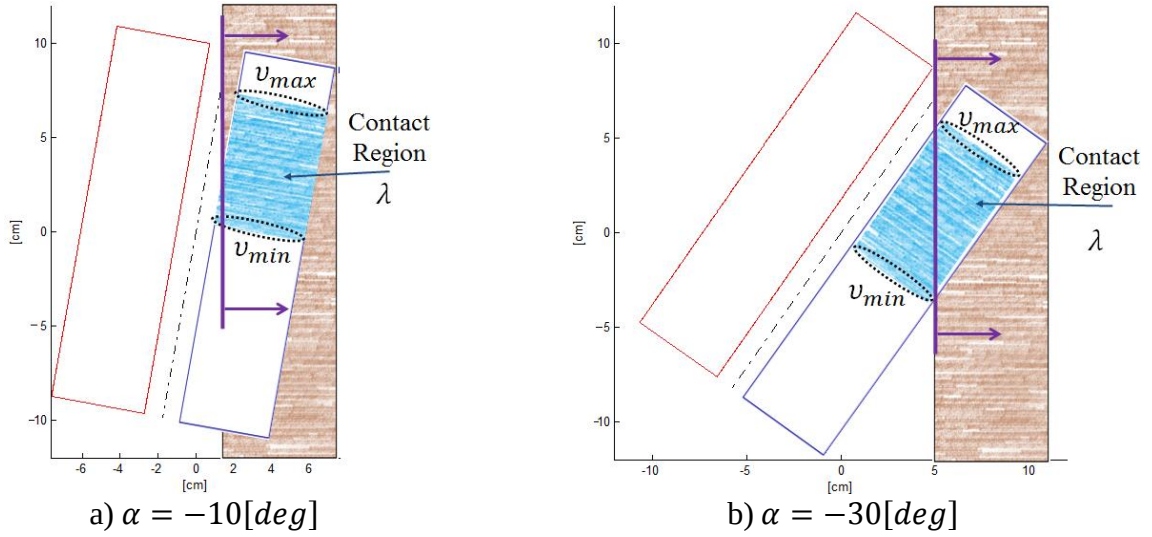


Figure 4.13 Wheel Orientation Effect on Region of Contact

In summary the of definition the contact region projection helps to determine if a SMDW moving laterally can overcome an obstacle at a given orientation α and wheel position θ . However, the effective range of action is considerably limited. In Figure 4.13 already at $\alpha = -30^\circ$, the climbing is no longer possible since the hub does not cover the required region of contact. It was observed that the climbing abilities start to reduce at a relatively small angles such as $\alpha = -10^\circ$

Given the limited range of orientation angles at which the obstacle climbing operation is feasible; it was concluded that the wheel positioned parallel to the obstacle with a small degree of deviation, constitutes the most suitable configuration for obstacle overcoming in lateral motion. In order to achieve the goal of this thesis and provide more than a single configuration for obstacle overcoming the following section provides a solution for analysis.

4.1.5 Angled Obstacle Approach in Forward Motion

A ground robot equipped with conventional wheels (non-omnidirectional) travelling in forward motion can overcome an obstacle located perpendicularly to its wheel frame. The overcoming capabilities on a conventional wheel depend on the height of the obstacle (must be lower than the wheel radius) and on the torque/force applied to the wheel. As shown in Figure 4.14 in order to successfully overcome an obstacle the force exerted on the conventional wheel should be large enough to counteract the reacting torques produced by the weight and normal forces.

The obstacle climbing in forward motion for omnidirectional wheels (including Mecanum wheels) presents several challenges and limitations due to the passive motion of the peripheral rollers. The passive rollers motion causes significant losses while intending obstacle overcoming. Figure 4.14 is also illustrates the forward approaching of an obstacle by a traditional Mecanum wheel. It can be appreciated that for this case, the normal force produced by the obstacle and the friction of the wheel observe a different vectorial components to those on a conventional wheel. Such difference are a consequence of the passive rollers oriented at 45° .

The new vectorial components contribute to an increase amount of torque required to overcome the obstacle. The height of the obstacle increases the normal and the obstacle force tends to be focalized on the z-axis direction due to the titling of the roller. In a more rigorous analysis the Mecanum wheel complexity increases since not only the height of the obstacle but also the position of the wheel determine the normal components. For such reasons with special emphasis on the energy loses the forward obstacle overcoming tends to be avoided and limited in current robots having traditional omnidirectional wheels.

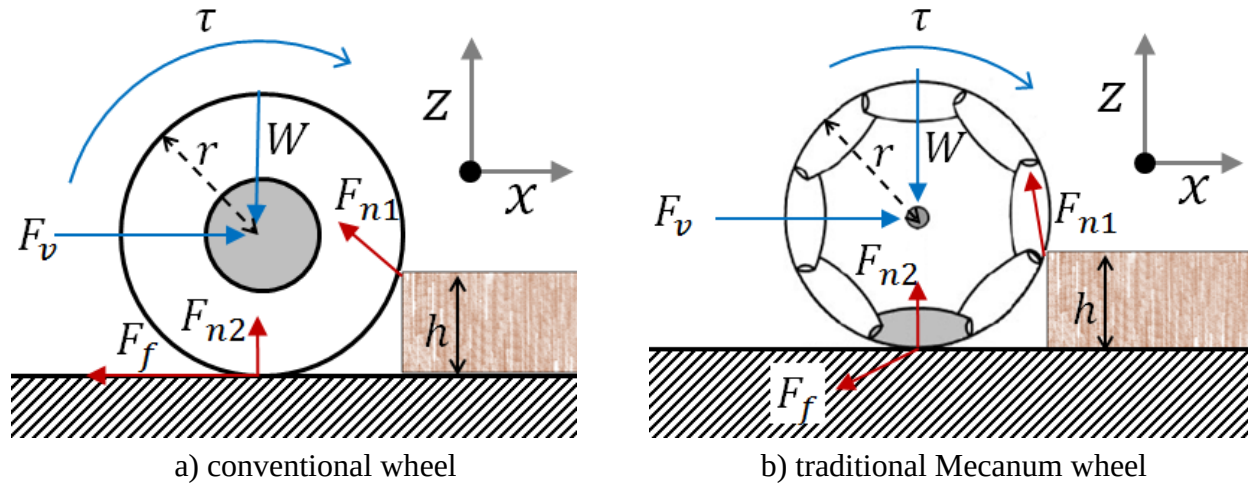


Figure 4.14 Obstacle overcoming in forward motion

A SMDW has the same issues (energy losses, high torque demand and limited height) as a conventional Mecanum wheel for overcoming obstacles in forward motion. Nevertheless the situation changes when the vehicle (SMDW) approaches the obstacle at an orientation different to perpendicular to the obstacle. As previously discussed at different configurations (orientations angles α and wheels positions θ) a portion of the SMDW hub can be located with enough clearance above an obstacle.

The analysis presented in Section 4.1.4 proved that at many angled configurations (specially anyone where $\alpha \geq 30^\circ$) the obstacle overcoming in lateral motion is not possible since the portion of the wheel located above the obstacle cannot guarantee the required contact surface. While the portion of the wheel above the obstacle (Figure 4.15) cannot ensure the required contact in many cases, it might provide enough surface area when considering a forward motion for the obstacle overcoming.

In order to analyze the SMDW's forward obstacle overcoming capabilities in a non-perpendicular approach the case when $\alpha = -45^\circ$ will be used. As can be seen in Figure 4.15b (free body diagram) for this case, the forces acting on the wheel hub are the same as in the traditional Mecanum wheel case with an important change in the obstacle normal force. The normal force F_{n1} acts underneath the face of the wheel hub which can generate (depending on the tangent at the contact point) a torque in opposite direction to the one exerted by the wheel's actuator. This should enable the SMDW to climb the obstacle. This effect only occurs when the edge of the hub in contact to the obstacle is the semi-elliptical edge of the wheel. On the other hand, if the semi-circular edge is the one in contact the reactions produced will be the as the ones showed for a conventional Mecanum wheel (Figure 4.14).

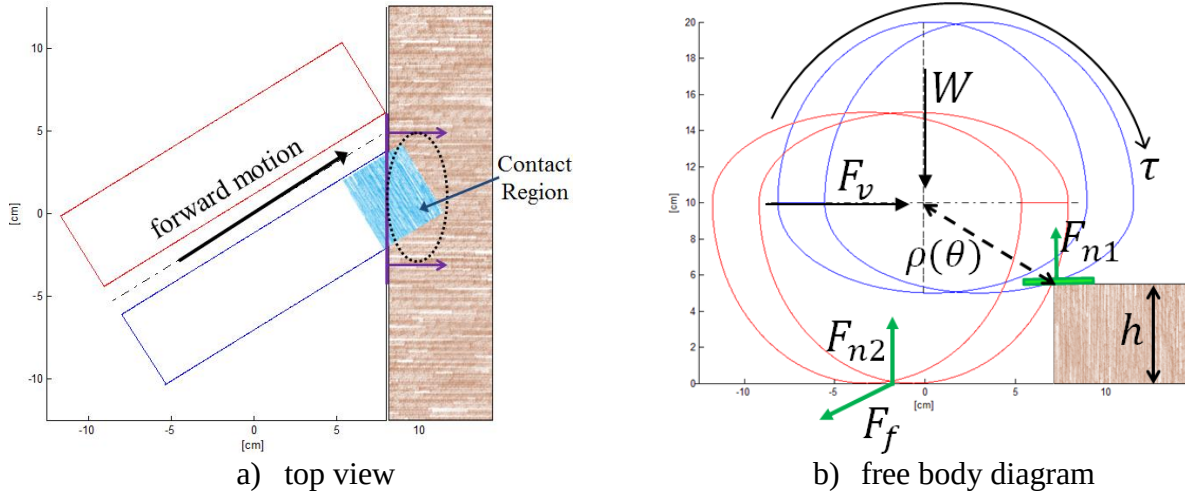


Figure 4.15 SMDW angled approach and forward motion for obstacle overcoming

At certain angles of approach, α , and wheel positions, θ , the SMDW can observe a quasi-static behavior at the instant in which the obstacle climbing starts, at such operation mode the wheel behaves in a fashion similar to a leg on a step. In quasi-static operation the normal force F_{n2} and the friction force F_f disappear. In addition the velocity produced by the driving force F_v and the SMDW's angular velocity become zero. The only remaining forces are the ones exerted

by the vehicle-wheel weight W , the normal force at the obstacle point of contact F_{n1} ; and the driving force F_v which becomes a driving torque τ (provided by the actuator). At this particular instance the SMDW can be represented by a revolute joint and single link with a sliding end in contact with the obstacle as illustrated in Figure 4.16 .

The key parameters to determine whether or not the SMDW will achieve the desired quasi-static operation and be able to climb the obstacle are the point of contact $v(\theta)$ and the angle ζ_m of the tangent line at the point of contact (Figure 4.16b). The tangent angle is important since it determines if the contact between the wheel and the obstacle can neglect the roller passive motion as the roller's angle approach the horizontal surface (i.e. $\zeta_m \leq 0^\circ$). In addition by using this simplified model quasi-static model it is possible to obtain a rough estimation of the torque needed to lift the wheel and its load using the distance from the axis of rotation to the point of contact $\rho(\theta)$.

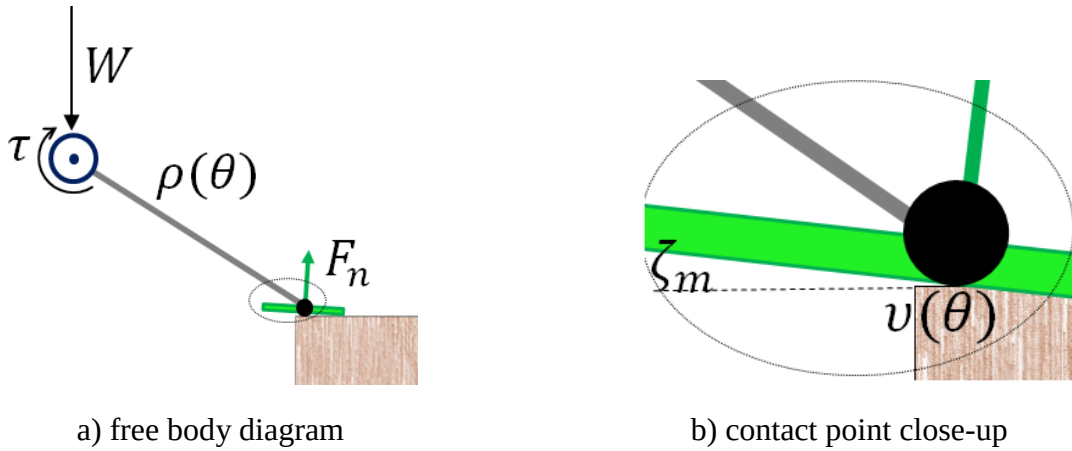


Figure 4.16 SMDW Quasi-static Obstacle Climbing Simplification

The quasit-static model is based on the geometrical simplification of the SMDW and considers that the climbing occurs in a short time interval such that the passive motion of the rollers and any potential slippage of the vehicle can be neglected for that time interval. The transient behavior of the wheel and the dynamic analysis of the wheel-obstacle interaction are not addressed by this research work. To obtain the point of contact $v(\theta)$ at a given configuration $\Gamma_0(\theta, \alpha)$, the previously described geometric model is employed. The developed process is similar to the ground clearance estimation with the additional steps required to compute the tangent angle ζ_m . The developed new process as a flow chart is presented in Figure 4.17.

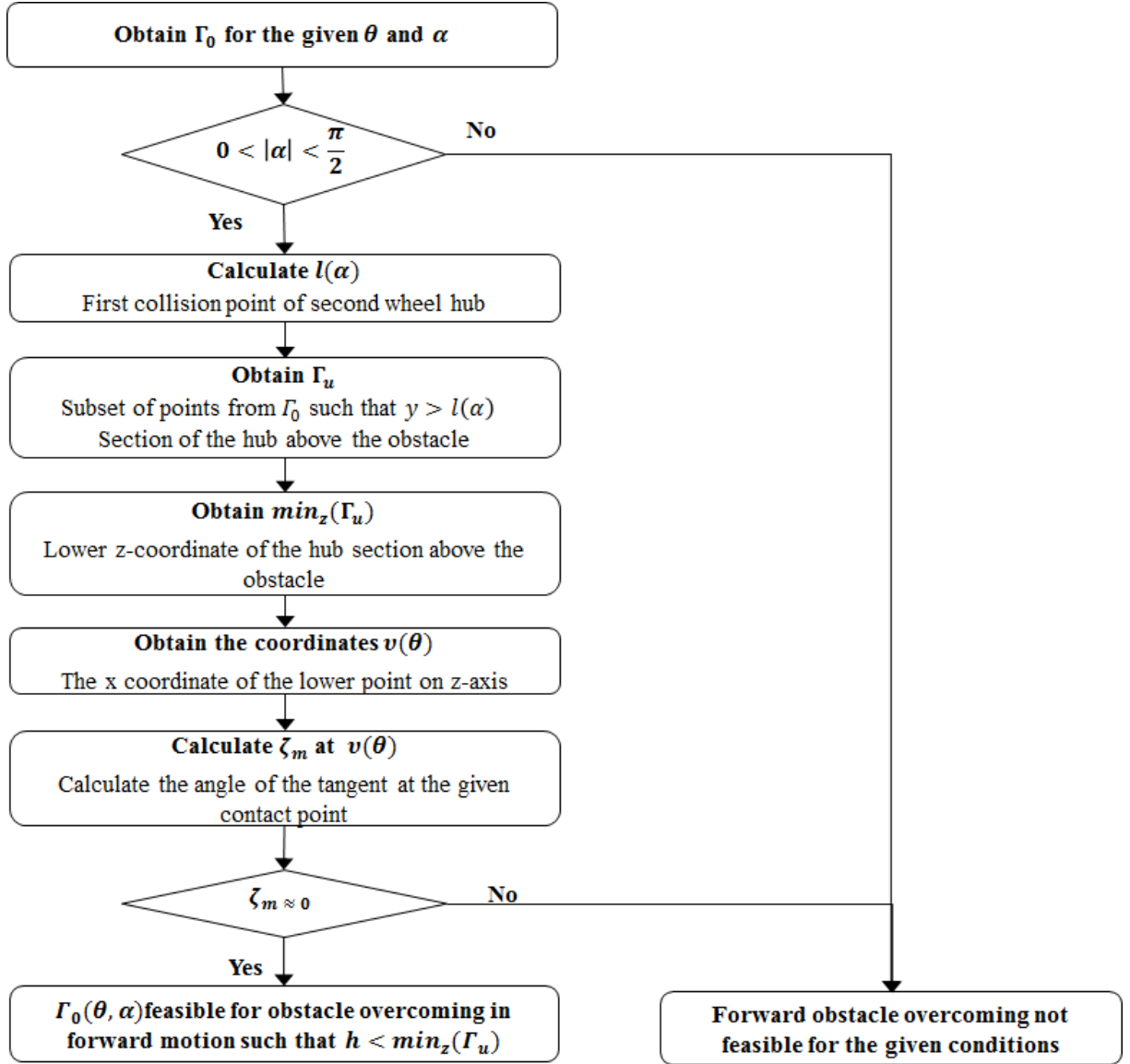


Figure 4.17 Wheel Configuration Evaluation for Obstacle Climbing in Forward Motion

For the calculation of ζ_m it is necessary to estimate first the coordinates of the first point of contact $v(\theta)$ between the SMDW and the slope of the tangent at that point (first derivative). The coordinates of $v(\theta)$ are obtained following the process described in Figure 4.11. The slope can be obtained from the derivative of the curve that describes the elliptical profile at the given $v(\theta)$.

The equation accounts for a whole ellipse despite the fact that the SMDW has just a semi-elliptical profile. The geometric model handles such issue by providing a point of contact just for the range of angular positions θ at which the semi-elliptical profile is in contact with the obstacle. The elliptical profile of the SMDW hub is described by (Eqn. 4.10) and (Eqn. 4.11).

$$\frac{(x \cdot \cos(\theta) + z \cdot \sin(\theta))^2}{R_c^2} + \frac{(x \cdot \sin(\theta) - z \cdot \cos(\theta))^2}{R_e^2} = 1 \quad (\text{Eqn. 4.10})$$

$$A \cdot x^2 + 2B \cdot xz + C \cdot z^2 = 1 \quad (\text{Eqn. 4.11})$$

where:

$$A = \frac{\cos^2(\theta)}{R_c^2} + \frac{\sin^2(\theta)}{R_e^2} \quad B = \sin(\theta) \cdot \cos(\theta) \cdot \left(\frac{1}{R_c^2} - \frac{1}{R_e^2} \right) \quad C = \frac{\sin^2(\theta)}{R_c^2} + \frac{\cos^2(\theta)}{R_e^2}$$

(Eqn. 4.11) considers that centre of the hub is located at the origin (0,0,0) and has a zero angle of approach ($\alpha = 0^\circ$). In order to account for these conditions, two operations need to be applied to the coordinates of $v(\theta)$. The first one is a rotation of $-\alpha$ angle which reverts the approach angle. The second and last operation is a translation about the z-axis of $-R_c$ which locates wheel profile frame of reference at the origin. Using the implicit derivative technique on the manipulated elliptical profile equation (Eqn. 4.12) is obtained for the tangent slope.

$$z' = \frac{-(A \cdot x + B \cdot z)}{B \cdot x + C \cdot z} \quad (\text{Eqn. 4.12})$$

The slope is obtained by evaluating the modified coordinates of $v(\theta)$ in (Eqn. 4.12). Once the slope has been calculated ζ_m can be obtained using the inverse tangent of such slope (Eqn. 4.13).

$$\zeta_m = \tan^{-1}(z') \quad (\text{Eqn. 4.13})$$

The value obtained from the angle of the tangent is subsequently used to evaluate whether or not a given configuration $\Gamma_0(\theta, \alpha)$ can overcome an obstacle in forward motion. If the angle is close to zero or slightly negative (i.e. $-10^\circ < \zeta_m \leq 0^\circ$) the configuration can be considered feasible since the wheel hub can engage contact with the obstacle almost at a horizontal plane allowing the quasi-static climbing operation.

Finally in order to analyze the effect of the approaching angle α on the forward motion a comparison between the ground clearance $h(\theta, \alpha)$ and the slope ζ_m at different values of α was investigated (Figure 4.18). The ground clearance and slope were calculated using the neutral wheel position, $\theta = 0^\circ$, at which the maximum overcoming capabilities are expected.

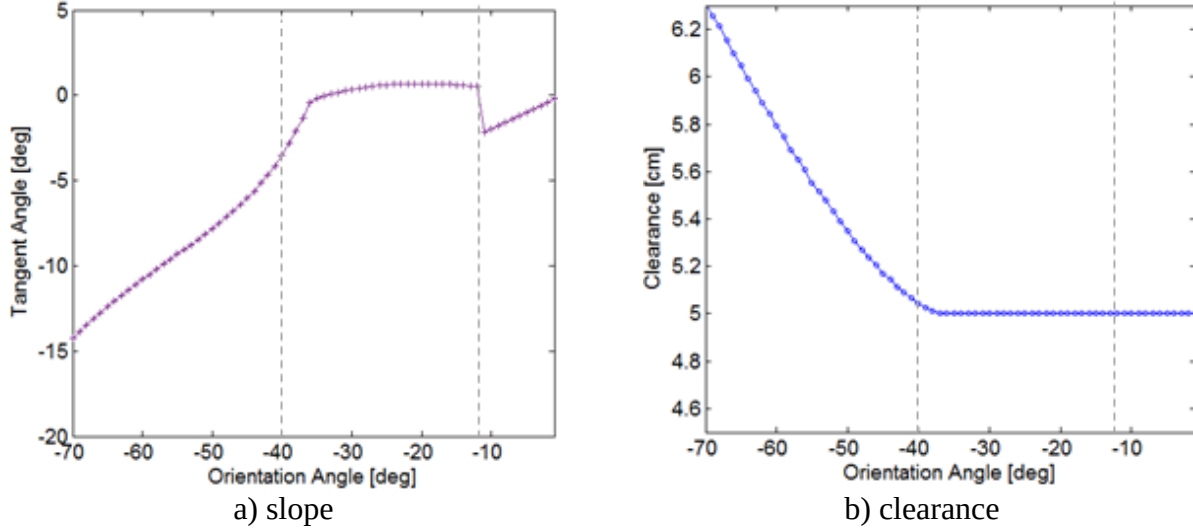


Figure 4.18 Clearance and Slope Angle Comparison for $\theta = 0^\circ$ and $-70^\circ \leq \alpha \leq -1^\circ$

From Figure 4.18 it is observed that while at almost any orientation the wheel, α , provides a quite comprehensive clearance, the effective range of operation is limited due to the slope angle. The analysis developed for the forward motions case implies that the angle of the tangent at the point of contact should approach zero in order to allow the quasi-static operation. However, the range of the approach angle at which obstacle overcoming can be performed in forward motion is $-40^\circ \leq \alpha \leq -13^\circ$. At this given range obstacles of $h \leq 5 \text{ cm}$ in height can be handled. This results provide a set of additional wheel configurations that the wheel can overcome obstacles, contributing significantly to the main goal of this research work.

4.1.6 Selection of Obstacle Overcoming Configurations

In order to test the obstacle overcoming capabilities based on the discussed analysis three motion configurations were chosen to be used by the maneuvering methodology. The first of the chosen configurations is the combination given by the wheel's face oriented parallel to the obstacle ($\alpha = 0^\circ$) while the robot moves laterally (perpendicular to the obstacle). Although this configuration, Figure 4.19, has been investigated in previous research [22][24][26] was selected to compare the more generic work developed in this thesis which should provide comparable results of previous work.

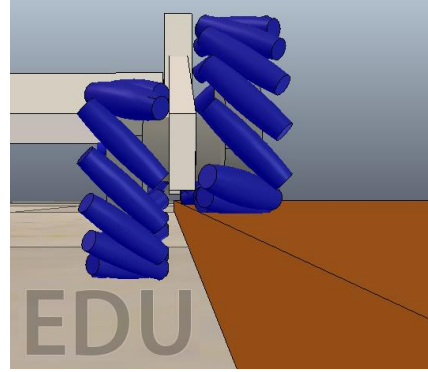
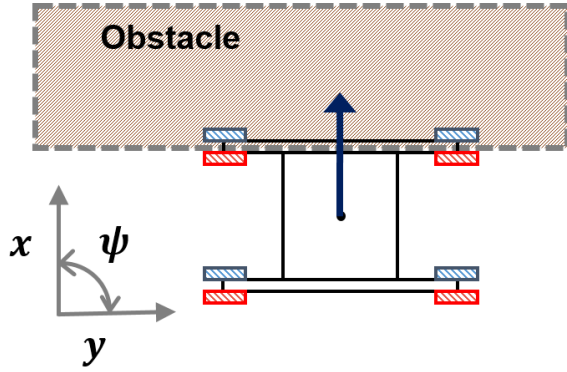


Figure 4.19 SMDW configuration for lateral motion and $\alpha = 0^\circ$

The second of the configurations chosen is a more generic motion when the SMDW approaches the obstacle at an angle $\alpha = \pm 35^\circ$ while the robot moves forward as shown in Figure 4.20. This configuration has not been tested prior to this research, however, the analysis presented in this thesis provided evidence of its feasibility. The sign of the robot's orientation angle will depend on the initial orientation of the robot, ψ , such that the whole rotation required by the robot does not exceed a quarter turn (90°).

The last selected configurations for the simulations and test uses an angle of approach $\alpha = \pm 45^\circ$ while the robot moves forward. Despite the fact that the geometric analysis performed previously (Section 4.1.5) indicates that such orientation might not be suitable for the obstacle climbing process, previous tests [12][24] provide evidence that indicates that such orientation provides favorable conditions for obstacle overcoming. This will be formally investigated herein (Chapter 6).

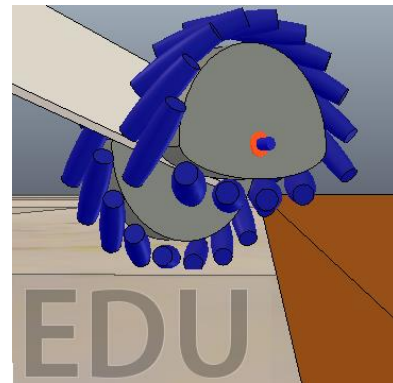
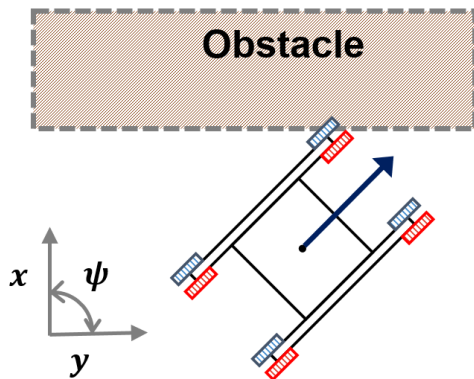


Figure 4.20 SMDW configuration for forward motion and $\alpha = \pm 35^\circ$ or $\alpha = \pm 45^\circ$

4.2 Maneuvering Planning

In order to increase the omnidirectional abilities of robots with SMDWs during the obstacle overcoming operation a new maneuvering approach was developed. The methodology considers a planning algorithm with constraint navigation space similar to the reference maneuvering controller reported in [26]. The goal of the proposed maneuvering algorithm is to position the robot in a state where it will be capable of overcoming the given obstacle at diverse approaching angles. The methodology determines a sequence of actions that drive a robot equipped with SMDW from an initial state to a position and configuration where the robot can climb a known height obstacle.

The methodology performs three main tasks to achieve its goal: i) determines a set of orientations and wheel states required to overcome an obstacle of a given height and a desired point of contact along the obstacle; ii) once the required positions and wheel states are determined the methodology uses them as target states in order to generate a trajectory that connects the initial position of the robot with them; iii) finally the methodology generates and refines the required trajectory. Figure 4.21 illustrates this methodology as a flow chart.

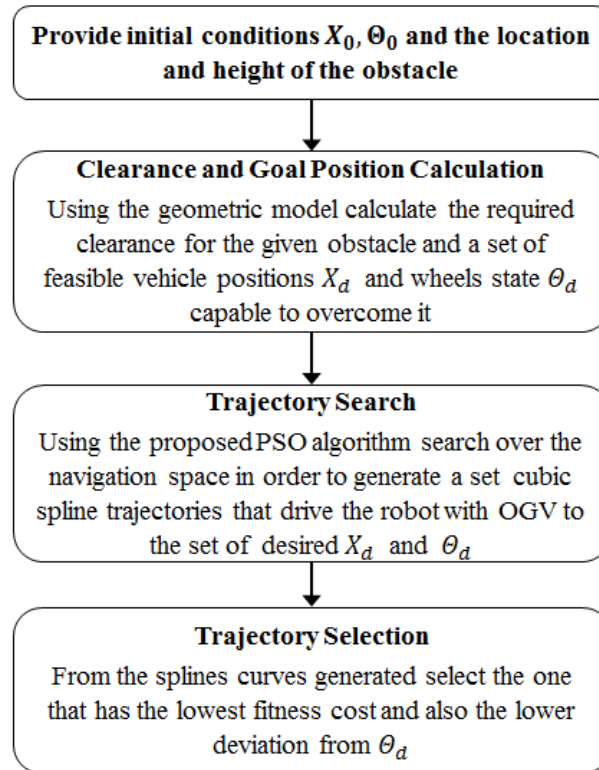


Figure 4.21 Maneuvering methodology flowchart

4.2.1 Cubic Spline Trajectory for Ground Robot Maneuvering

In order to provide a smooth motion for the path that connects the initial position of the robot to the desired climbing configuration a cubic piece-wise spline polynomial as a function of time was chosen (Figure 4.22). As mentioned previously the use of spline curves is one of the most common and basic approaches for the definition of the path to be followed by a mobile robot [30][46]. Spline polynomials offer the advantage of providing velocity and acceleration continuity (depending on the order of the curve) as well as a simple algorithm to generate paths based on a set of control points and boundary conditions. The Chapra and Canale fitting approach [58] was selected for the spline interpolation in this thesis.

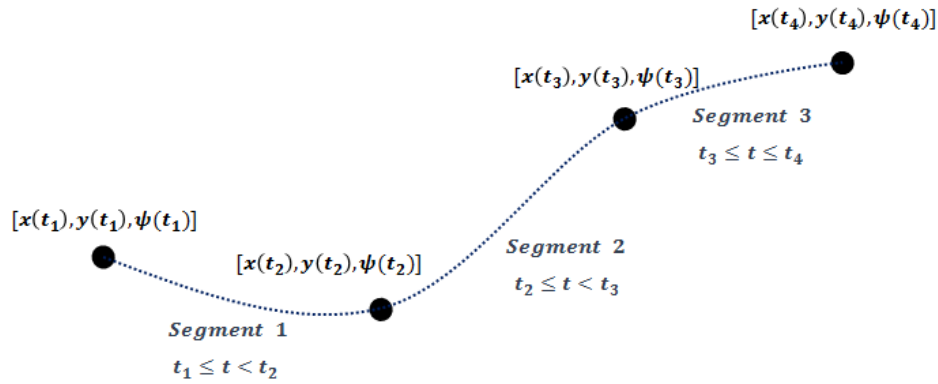


Figure 4.22 Piecewise cubic spline robot trajectory

The overall spline trajectory developed in this thesis is composed by a segmented cubic function for each one of the dimensions of the robot position (x , y and ψ). To generate this function, the fitting technique requires 4 control points over 3 intervals. The equations for each of the position dimensions as a function of time are given by (Eqn. 4.14) as per the fitting methodology definition.

$$\begin{aligned}
 x(t) &= \begin{cases} a_{x1} \cdot (t_2 - t)^3 + b_{x1} \cdot (t - t_1)^2 + c_{x1} \cdot (t_2 - t) + d_{x1} \cdot (t - t_1) & t_1 \leq t < t_2 \\ a_{x2} \cdot (t_3 - t)^3 + b_{x2} \cdot (t - t_2)^2 + c_{x2} \cdot (t_3 - t) + d_{x2} \cdot (t - t_2) & t_2 \leq t < t_3 \\ a_{x3} \cdot (t_4 - t)^3 + b_{x3} \cdot (t - t_3)^2 + c_{x3} \cdot (t_4 - t) + d_{x3} \cdot (t - t_3) & t_3 \leq t \leq t_4 \end{cases} \\
 y(t) &= \begin{cases} a_{y1} \cdot (t_2 - t)^3 + b_{y1} \cdot (t - t_1)^2 + c_{y1} \cdot (t_2 - t) + d_{y1} \cdot (t - t_1) & t_1 \leq t < t_2 \\ a_{y2} \cdot (t_3 - t)^3 + b_{y2} \cdot (t - t_2)^2 + c_{y2} \cdot (t_3 - t) + d_{y2} \cdot (t - t_2) & t_2 \leq t < t_3 \\ a_{y3} \cdot (t_4 - t)^3 + b_{y3} \cdot (t - t_3)^2 + c_{y3} \cdot (t_4 - t) + d_{y3} \cdot (t - t_3) & t_3 \leq t \leq t_4 \end{cases} \quad (\text{Eqn. 4.14}) \\
 \psi(t) &= \begin{cases} a_{\psi1} \cdot (t_2 - t)^3 + b_{\psi1} \cdot (t - t_1)^2 + c_{\psi1} \cdot (t_2 - t) + d_{\psi1} \cdot (t - t_1) & t_1 \leq t < t_2 \\ a_{\psi2} \cdot (t_3 - t)^3 + b_{\psi2} \cdot (t - t_2)^2 + c_{\psi2} \cdot (t_3 - t) + d_{\psi2} \cdot (t - t_2) & t_2 \leq t < t_3 \\ a_{\psi3} \cdot (t_4 - t)^3 + b_{\psi3} \cdot (t - t_3)^2 + c_{\psi3} \cdot (t_4 - t) + d_{\psi3} \cdot (t - t_3) & t_3 \leq t \leq t_4 \end{cases}
 \end{aligned}$$

The spline fitting methodology [58] considers the four boundary conditions shown below for the calculation of the polynomial coefficients.

1. The first control point is given by the initial position at time zero

$$[x(t_1), y(t_1), \psi(t_1)] = [x_0, y_0, \psi_0] \quad t_1 = 0 \quad (\text{Eqn. 4.15})$$

2. The fourth and last control point is given by the desired final position

$$[x(t_4), y(t_4), \psi(t_4)] = [x_d, y_d, \psi_d] \quad t_4 = t_f \quad (\text{Eqn. 4.16})$$

3. The interior control points have the same acceleration

$$[\ddot{x}(t_2), \ddot{y}(t_2), \ddot{\psi}(t_2)] = [\ddot{x}(t_3), \ddot{y}(t_3), \ddot{\psi}(t_3)] \quad (\text{Eqn. 4.17})$$

4. The acceleration of the initial and final control points is zero

$$[\ddot{x}(t_1), \ddot{y}(t_1), \ddot{\psi}(t_1)] = [\ddot{x}(t_4), \ddot{y}(t_4), \ddot{\psi}(t_4)] = [0 \ 0 \ 0] \quad (\text{Eqn. 4.18})$$

The Chapra and Canale cubic-spline fitting algorithm provides a solution for the polynomial coefficients on a fixed interval. Thus, each time that the time span changes it is required to recalculate the polynomial coefficients. Since time is one of the parameters that can be changed depending on the overall distance that the robot has to maneuver; a parameterization of the spline functions was proposed. The parameterization consists basically in a variable change that makes the polynomial coefficients solutions not dependent of the time intervals. The proposed variable change and the adjustments in the intervals and boundary conditions are shown in Table 4.1.

Table 4.1 Time Parameterization and Adjusted Boundary Conditions

Parameter “s” time mapping and spline intervals $s(t) = \frac{t}{3} \quad 1 \leq s \leq 4 \quad [s_1 \ s_2 \ s_3 \ s_4] = [1 \ 2 \ 3 \ 4]$	
Initial Position $[x(1), y(1), \psi(1)] = [x_0, y_0, \psi_0]$	Final Position $[x(4), y(4), \psi(4)] = [x_d, y_d, \psi_d]$
Acceleration at interior points $[\ddot{x}(2), \ddot{y}(2), \ddot{\psi}(2)] = [\ddot{x}(3), \ddot{y}(3), \ddot{\psi}(3)]$	Acceleration at initial and final position $[\ddot{x}(1), \ddot{y}(1), \ddot{\psi}(1)] = [\ddot{x}(4), \ddot{y}(4), \ddot{\psi}(4)] = 0$

After applying the variable change and the new intervals, the parameterized robot trajectory becomes:

$$\begin{aligned}
x(s) &= \begin{cases} a_{x1} \cdot (2-s)^3 + b_{x1} \cdot (s-1)^2 + c_{x1} \cdot (2-s) + d_{x1} \cdot (s-1) & 1 \leq s < 2 \\ a_{x2} \cdot (3-s)^3 + b_{x2} \cdot (s-2)^2 + c_{x2} \cdot (3-s) + d_{x2} \cdot (s-2) & 2 \leq s < 3 \\ a_{x3} \cdot (4-s)^3 + b_{x3} \cdot (s-3)^2 + c_{x3} \cdot (4-s) + d_{x3} \cdot (s-3) & 3 \leq s \leq 4 \end{cases} \\
y(s) &= \begin{cases} a_{y1} \cdot (2-s)^3 + b_{y1} \cdot (s-1)^2 + c_{y1} \cdot (2-s) + d_{y1} \cdot (s-1) & 1 \leq s < 2 \\ a_{y2} \cdot (3-s)^3 + b_{y2} \cdot (s-2)^2 + c_{y2} \cdot (3-s) + d_{y2} \cdot (s-2) & 2 \leq s < 3 \\ a_{y3} \cdot (4-s)^3 + b_{y3} \cdot (s-3)^2 + c_{y3} \cdot (4-s) + d_{y3} \cdot (s-3) & 3 \leq s \leq 4 \end{cases} \quad (\text{Eqn. 4.19}) \\
\psi(s) &= \begin{cases} a_{\psi1} \cdot (2-s)^3 + b_{\psi1} \cdot (s-1)^2 + c_{\psi1} \cdot (2-s) + d_{\psi1} \cdot (s-1) & 1 \leq s < 2 \\ a_{\psi2} \cdot (3-s)^3 + b_{\psi2} \cdot (s-2)^2 + c_{\psi2} \cdot (3-s) + d_{\psi2} \cdot (s-2) & 2 \leq s < 3 \\ a_{\psi3} \cdot (4-s)^3 + b_{\psi3} \cdot (s-3)^2 + c_{\psi3} \cdot (4-s) + d_{\psi3} \cdot (s-3) & 3 \leq s \leq 4 \end{cases}
\end{aligned}$$

This cubic spline trajectory allows the control of a robot with SMDW by using the first derivative of the motion polynomials to calculate the independent wheels' rotational speed using the basic kinematic model of a traditional Mecanum wheel vehicle. For the calculation of the spline coefficients, the initial and the desired final positions of the robot are the input data for the maneuvering planner. However the interior control points need to be defined. The selection of these points is the main task of the trajectory searcher, explained in detail in the following section.

4.2.2 Particle Swarm Optimization Search for Spline Control Points

The methodology chosen for the spline coefficients calculation requires the definition of 4 control points along the trajectory as shown in Figure 4.23. Two of those points are given by the initial and the final positions of the OGV. In order to complete the set of required control points, two intermediate points (P_1, P_2) need to be defined (Figure 4.23).

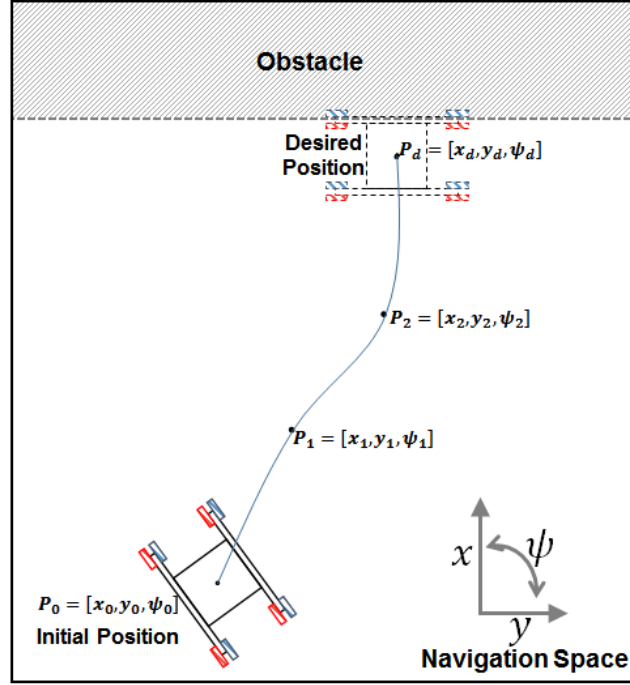


Figure 4.23 Omnidirectional ground robot trajectory space

For such task a Particle Swarm Optimization (PSO) algorithm was selected for the exploration and selection of suitable combinations of control point locations $P = [x, y, \psi]$ over the navigation space. The PSO is a technique based on a social-psychological model of social influence and social learning [47]. It is influenced by evolutionary computation and genetic algorithm techniques[48]. The algorithm makes use the basic search entities called particles and a fitness function which provides the metrics for evaluating the suitability of the function/problem being explored.

In the PSO algorithm a given number of particles are distributed over the search space of a given problem; each particle evaluates the objective function (fitness function) at its current location. The particles update their position on a combination of their current location and the history of their own best position and the best locations of the other members of the swarm with some random perturbations. The algorithm updates the position of the particles at each iteration and eventually the swarm as a collective entity moves close to an optimum of the objective function [49].

Each particle of the swarm has a given number of dimensions (j dimensions) on the search space, also it contains three vectors corresponding to the particle location, the velocity of update and the previous best location (personal and global) for each dimension. The most basic

PSO variations are divided in global best (gbest) and local best (lbest) approaches. The global best approach implies the information exchange among all the particles of the swarm; this is the algorithm that was selected to be used in this thesis. The basics structure for a gbest particle is given the by the expressions contained in Table 4.2 .

Table 4.2 PSO Gbest particle equations

$p_{ij}(k+1) = p_{ij}(k) + v_{ij}(k+1)$ $v_{ij}(k+1) = v_{ij}(k) + c_1 \cdot r_{1ij}(k) \cdot (p_{bij}(k) - p_{ij}(k)) + c_2 \cdot r_{2ij}(k) \cdot (\hat{p}_{bj}(k) - p_{ij}(k))$ $y_{ij}(k+1) = \begin{cases} p_{bij}(k) & \text{If } S(p_{ij}(k+1)) \geq S(p_{bij}(k)) \\ p_{ij}(k) & \text{If } S(p_{ij}(k+1)) < S(p_{bij}(k)) \end{cases}$ $\hat{p}_{bj}(k+1) = \begin{cases} \min\{S(p_{1j}(k)), \dots, S(p_{ij}(k))\} & \text{If } S(\min\{S(p_{1j}(k)), \dots, S(p_{ij}(k))\}) < S(\hat{p}_{bj}(k)) \\ \hat{p}_{bj}(k) & \text{Else} \end{cases}$	
Where:	c_2 = social knowledge acceleration constant
k = iteration index	r_{1ij}, r_{2ij} = stochastic components (random values from a uniform distribution in the range [0,1])
j = number of dimensions in the search space	p_{bij} = personal best position for particle i on j dimension
i = number of particles of the swarm	$\hat{p}_{bj}$ = global best position for particles on j dimension
p_{ij} = location for particle i on j dimension	$S(p_{ij})$ = fitness function
v_{ij} = velocity for particle i on j dimension	
c_1 = cognitive knowledge acceleration constant	

As can be seen from the particle equations shown above in a global best implementation the location of the particle is determined simultaneously by the own particle historical best location and the historical best position of the swarm .The pseudo-code of the whole general algorithm for a basic global best PSO implementation is provided in Table 4.3.

Table 4.3 Gbest PSO Pseudo-code [47]

```

Create and initialize a swarm of  $i$  particles with  $j$  dimensions
repeat
for each particle  $n = 1 \dots i$  do
    //Set the personal best position
    if  $S(p_n) < S(p_{bn})$  then
         $p_{bn} = p_n$ ;
    end
    //Set the global best position if  $S(p_{bn}) < S(\hat{p}_{bn})$  then
         $\hat{p}_{bn} = p_{bn}$ ;
    end
end
for each particle  $n = 1 \dots i$  in each dimension  $m = 1 \dots j$  do
    update the particle velocity  $v_{nm}$  for each dimension;
    update the particle location  $p_{nm}$  for each dimension;
end
until stopping condition is true;

```

A global best basic PSO algorithm was selected to explore the navigation space and find a suitable location ($P = [x, y, \psi]$) for the control points P_1, P_2 (Figure 4.23) of the vehicle's frame with the cubic spline trajectory. The location of P_1, P_2 , should contribute to have an overall trajectory such that the deviation in terms of the SMDWs states Θ and the robot final position X is contained within a defined range that allows the overcoming of the given obstacle with known height.

In order to simplify the problem and reduce the search effort; it was decided to place the first control point P_1 in an arbitrary and constant location in function of the search space ranges. The PSO will only be used to explore the navigation space and select a suitable location for the second control point P_2 between P_1 and the desired final position. For this the navigation space was divided such that the space search for P_2 comprises only the second upper half of the entire navigation space. The trajectory control points layout and the proposed space search described above are depicted in Figure 4.24.

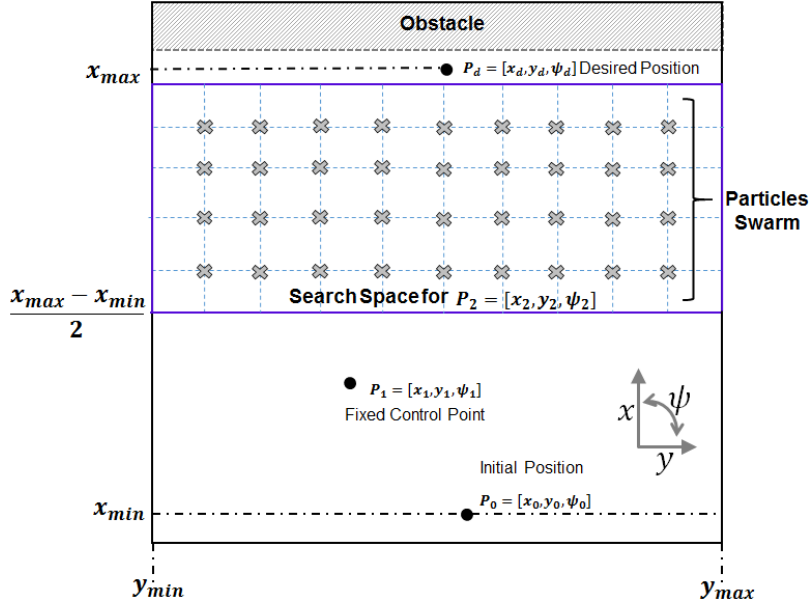


Figure 4.24 PSO search space for P_2 location

A swarm of 50 particles exploring over 3 dimensions (x, y planar position and ψ orientation) was selected for the implementation of the algorithm. For the initialization of the search, the particles were distributed uniformly along the search space in an array 5 by 10 in the x -axis and the y -axis respectively (Figure 4.24). The number of particles was chosen based on the guidelines provide in [47] which indicate that for many applications a swarm of 50 particles provide a good convergence and a near optimal solution. The performance of this swarm size is evaluated in Chapter 5. The resulting equations for the particles are given by (Eqn. 4.20).

$$p(k+1) = \begin{bmatrix} x(k+1) \\ y(k+1) \\ \psi(k+1) \end{bmatrix} \quad v(k+1) = \begin{bmatrix} v_x(k+1) \\ v_y(k+1) \\ v_\psi(k+1) \end{bmatrix}$$

$$\begin{bmatrix} v_x(k+1) \\ v_y(k+1) \\ v_\psi(k+1) \end{bmatrix} = \begin{bmatrix} v_x(k) + c_1 \cdot r_{x1}(k) \cdot (x_b(k) - x(k)) + c_2 \cdot r_{x2}(k) \cdot (\hat{x}_b(k) - x(k)) \\ v_y(k) + c_1 \cdot r_{y1}(k) \cdot (y_b(k) - y(k)) + c_2 \cdot r_{y2}(k) \cdot (\hat{y}_b(k) - y(k)) \\ v_\psi(k) + c_1 \cdot r_{\psi1}(k) \cdot (\psi_b(k) - \psi(k)) + c_2 \cdot r_{\psi2}(k) \cdot (\hat{\psi}_b(k) - \psi(k)) \end{bmatrix} \quad (\text{Eqn. 4.20})$$

$$\begin{bmatrix} x(k+1) \\ y(k+1) \\ \psi(k+1) \end{bmatrix} = \begin{bmatrix} x(k) + v_x(k+1) \\ y(k) + v_y(k+1) \\ \psi(k) + v_\psi(k+1) \end{bmatrix}$$

In order to consider a location $P_2 = [x_2, y_2, \psi_2]$ as a suitable solution with the overall robot's trajectory should drive the vehicle to a final position $X_f = [x_f, y_f, \psi_f]$ with wheels position $\Theta_f = [\theta_{1f}, \theta_{2f}, \theta_{3f}, \theta_{4f}]$ within the range of the predefined configuration suitable to

perform the obstacle climbing. The fitness function of Table 4.4 was developed in terms of the deviation (error) in the robot's position, X , and wheels state, Θ .

To obtain the final position X_f and final wheels state Θ_f for a given trajectory a numerical simulation of the OGV along the trajectory was performed using the robot's kinematic model (described in Chapter 1). In order to improve the estimation of the fitness cost, saturation and dead-zone limits were imposed to the kinematic model in order to mimic the behavior of an electric actuator driving each wheel. The deviation is calculated by using the absolute value norm of the difference between the desired final position and the actual final position. The expression of the fitness function described is given in Table 4.4.

Table 4.4 Defined fitness function for P_2 search algorithm

$S(X, \theta) = \gamma \cdot X_d - X_f + \beta \cdot \Theta_d - \Theta_f $ $S(X, \theta) = [\gamma_x \ \gamma_y \ \gamma_\psi] \cdot \begin{bmatrix} x_d - x_f \\ y_d - y_f \\ \psi_d - \psi_f \end{bmatrix} + [\beta_1 \ \beta_2 \ \beta_3 \ \beta_4] \cdot \begin{bmatrix} \theta_{1d} - \theta_{1f} \\ \theta_{2d} - \theta_{2f} \\ \theta_{3d} - \theta_{3f} \\ \theta_{4d} - \theta_{4f} \end{bmatrix}$	
Where:	Θ_d =actual SMDWs final state
X_d =desired robot final position	γ =final position deviation penalty coefficient
X_f =actual robot final position	β =SMDWs final state deviation penalty coefficient
Θ_d =desired SMDWs final state	$\beta \gg \gamma \quad \gamma_x \geq \gamma_\psi \gg \gamma_y$

For the calculation of the wheel states Θ deviation the modulus of the final function with respect to π is used. For the clearance estimation and desired position the range of a SMDW over one revolution $0 \leq \theta \leq 2\pi$ is used. Therefore, for the calculation of the wheels deviation a range of $0 \leq \theta \leq \pi$ was used.

A convergence criterion based on the number of iterations was selected for the PSO search algorithm. The selection of this approach was taken in order to address the possible existence of local minima; which in a piecewise function like the proposed trajectory it is very likely to occur. The main shortcomings of the fixed iterations number as the only convergence criterion is that in order to perform an extensive search the number of iterations performed can be larger than the iterations required in many cases. The additional unnecessary iterations increase the computation time and can degrade the performance of the search algorithm.

However, despite these tradeoffs, the simplicity of the fixed iteration criterion makes it an appropriate choice for the intended purpose. Figure 4.25 provides a flow chart diagram that summarizes the overall operation of the proposed PSO search algorithm for the selection the second trajectory control point, P_2 .

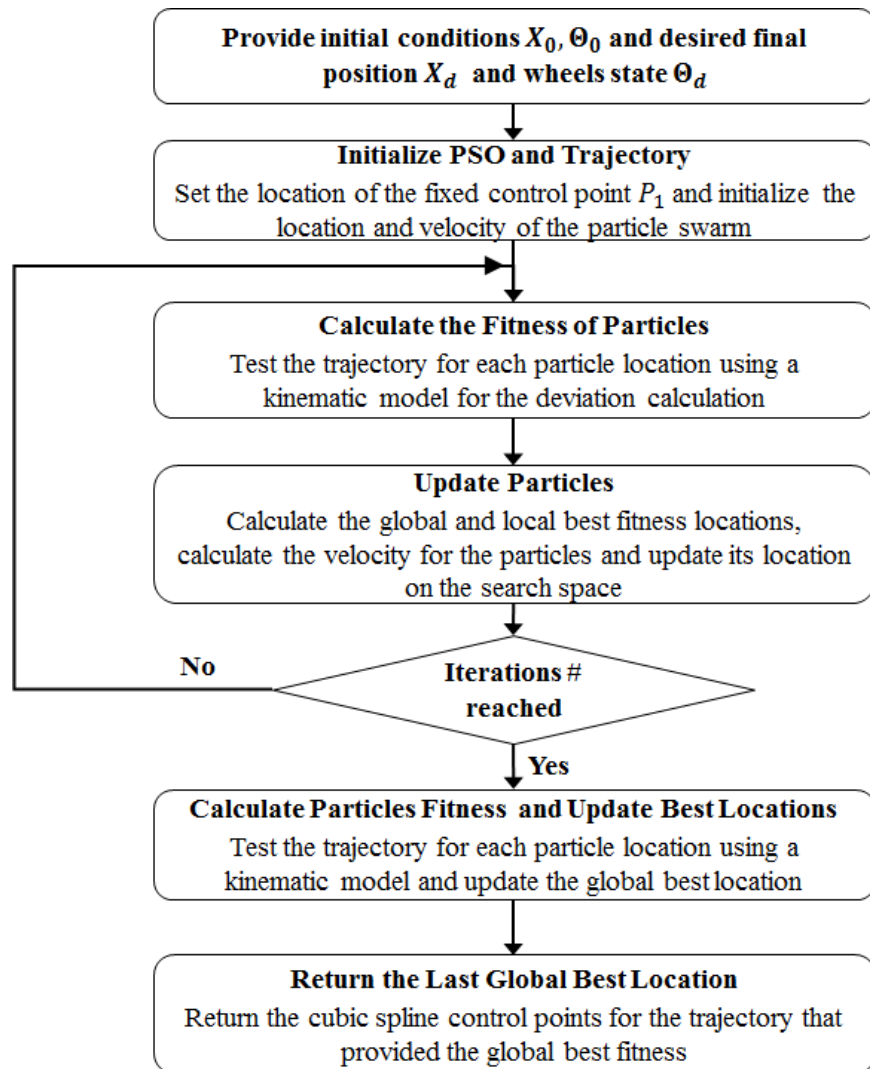


Figure 4.25 Summary of the PSO search algorithm for spline trajectory control point selection

Chapter 5: Simulation Setup

5.1 Simulation Environment

In order to evaluate the performance of the proposed maneuvering methodology, a series of simulations and tests were conducted for each one of the key concepts involved in this research work. The overall evaluation used a physical simulator where different trajectories were generated by the maneuvering methodology. A series of shorter scope sub-tests were also performed to assess the validity of the simulator model, tune of the PSO search algorithm, and validate the proposed geometric model for the SMDW.

The clearance estimation, the PSO search algorithm, the motion controller and the data collection algorithms were implemented in the MATLAB environment. The robot and the SMDWs software base models (also known as physical models since they are developed using a physical engine) were developed in Virtual Robot Experimentation Platform (V-REP) using the Open Dynamics Engine (ODE) and the Bullet Physics library. A link between the MATLAB environment and the robot's physical model was established using the remote API functionality of the V-REP simulation environment.

V-REP is a robotic simulator that provides a distributed control architecture with a hybrid approach for the dynamics simulation [50]. The simulator provides a set of different relative independent modules for specific functions with a distributed control approach [51]. The functionality of the modules is active as needed and can interact with functionalities or external applications.

Unlike the conventional dynamic rigid body applications V-REP is a hybrid platform which has the ability to select between a simple kinematic model and a more realistic physics model depending on the intended application[51]. For cases where collisions and contact forces are not relevant such as trajectory planning the simulator can be set to use a pure kinematic model reducing the computation effort and error produced by the models and numeric solvers of the physical engines. For the simulations performed in this thesis, the simulator was set to use a physics engine that considered collisions, contact forces, etc.

The professional version of the software with full functionality and all the modules is available under the GNU GPL license free of charge for educational purposes [52]. The physics or dynamic module of V-REP provides the tools for testing control and navigation algorithms; as well for the analysis of the interaction between the robot and its operating environment.

5.2 Simulation Settings and Robot Specifications

For the simulations a 4 wheel omnidirectional vehicle physical simulator was developed. The vehicle reference for this thesis and its simulations, is the unmanned ground robot LOC8 (Figure 5.1). This platform was selected because it has been designed and built specifically for the use of SMDWs and it is available at the University of Calgary's AR²S laboratory. In addition to manual controlled tests, some previous research studies were performed using such platform as reference; therefore, its utilization provides consistency and a common frame of reference for results to be compared.

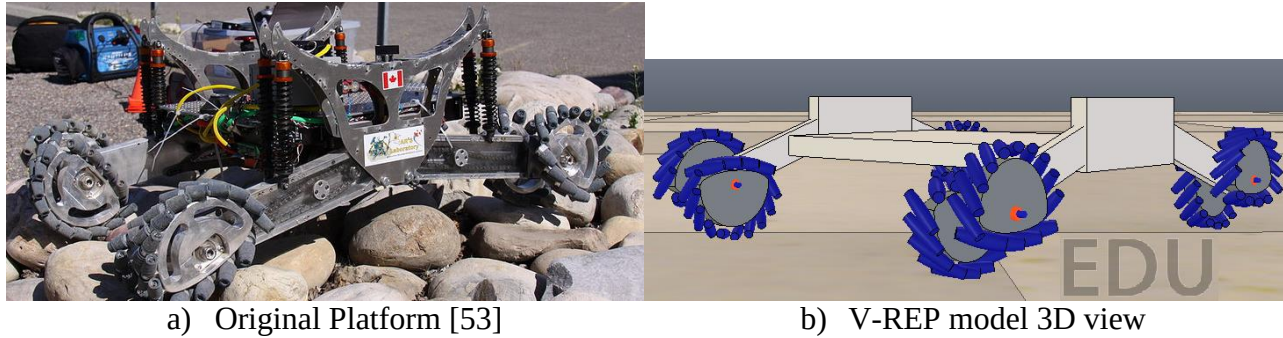


Figure 5.1 LOC8 omnidirectional ground vehicle

The basic omnidirectional ground platform (Figure 5.1b) was implemented in the V-REP simulator considering the use of SMDW and a chassis with the dimensions and physical parameters of the LOC 8 platform [53]. The model developed in V-REP used a simpler geometry than the ones found in the original chassis (Figure 5.1a). The model however, retains the basic functionality. The suspension and the transmission systems of the robot were not included. Instead a welded link between the wheels, the frame of the robot, and the revolute joints / actuators was used. The mass of the frame was considered to be 14.5 kilograms and the SMDW assembly including rollers comprised a total mas of 1.44 kilograms per wheel. The physical dimensions and default orientation of the robot are depicted in Figure 5.2. The SMDW dimensions used are provided in Figure 4.5. The frame of reference of the vehicle was located at the geometric centre of the chassis frame. The rotation of the robot, ψ , is considered to be positive in the counterclockwise direction.

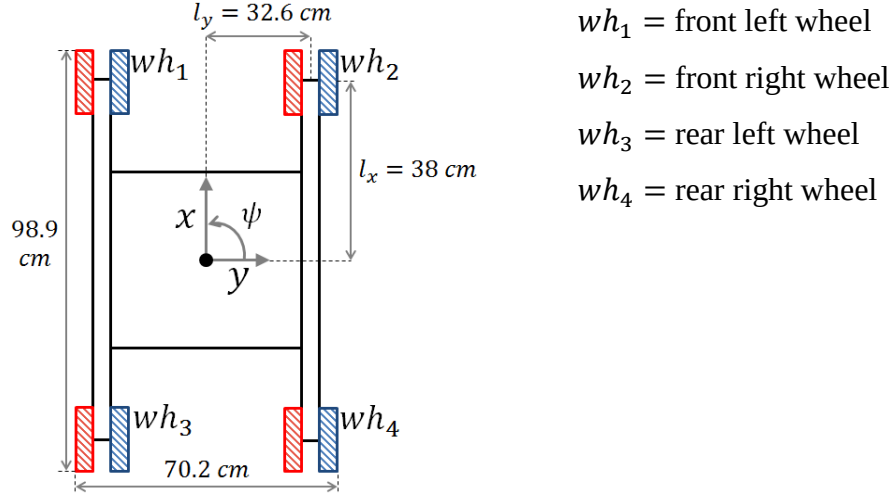


Figure 5.2 Robot dimensions and default orientation

The V-REP version used was 3.1.3R2 for 32 bits while the release of the physical engines used were Bullet 2.78 and ODE 0.12. For the climbing over obstacles the Bullet physical engine was used due to its capability to simulate soft contacts. For the navigation and trajectory tracking simulations the ODE engine was used. For both engines the time step used was 10 milliseconds and for the accuracy settings the default option was selected. The MATLAB release used was 2014a (8.3.0.532) for 64 bits. None special toolboxes or MATLAB plugins were used.

5.3 Physical Simulator Development

As described previously a physical simulator of a robot with SMDWs was developed for evaluating and testing the maneuvering methodology produced by this thesis. A key component of this simulator is the approach used for modeling the SMDWs. The most common approach for the simulation and design of Mecanum wheel applications, is modeling the motion of the entire vehicle neglecting the individual behavior of the rollers. Several kinematic and dynamic models have been developed for several configurations of OGV with Mecanum wheels[16][54][55]. Those models have proven to work well when contact with ground by all the wheels is assumed. Such models can only be applied when navigating on flat hard surfaces. As a result they cannot be used to study the behavior of the wheel when interacting with obstacles or navigating on rough terrain.

One of the key advantages of the physical simulation used is that it model the contact between the wheel rollers and the ground/obstacles; it also introduces slippage due to the

frictions of the rollers. The development of the simulator implied the modeling of conventional Mecanum wheel in order to validate the simplifications proposed and proof the validity of the model against the existing models. After the original assumptions and the performance of the chosen software were validated with the conventional Mecanum wheels, a physical simulation for the SMDW was built using the V-REP software. The design was refined and tested incorporating the wheel simulation to a ground robot and evaluating its trajectory for a set of given motion inputs. The results and details for both development stages are shown in the next sections.

5.3.1 Physical Simulation of a Traditional Mecanum Wheel

The first step in development of a SMDW physical simulator was to produce an accurate simulator of traditional Mecanum wheel. The objective of this first task was to analyze the performance and capabilities in collision detection, motion constraints and contact forces of the V-REP environment. According to the documentation of V-REP [56] a simple design using as much as possible primitive shapes, is the best approach to develop an accurate and efficient physical simulation.

A physical model of the traditional Mecanum was developed in V-REP. The V-REP model considered the rollers as independent shapes connected to the wheel frame through revolute joints (Figure 5.3) as proposed in [57]. In order to provide the appropriate passive motion constraints each roller joint of the model was oriented at 45° or -45° .

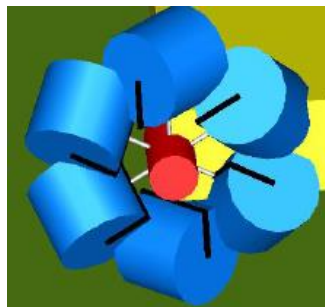
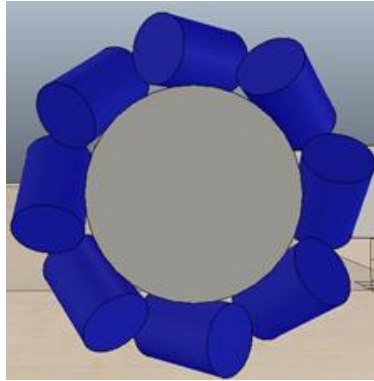


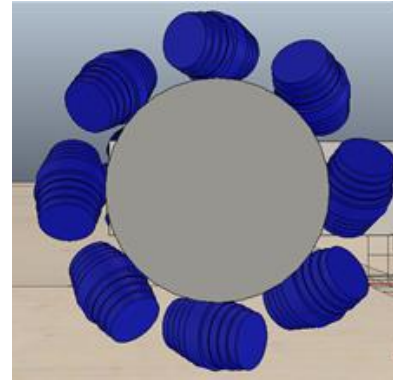
Figure 5.3 Mecanum wheel Kalman model in Dymola [57]

Two different geometrical representations of the rollers were considered for the physical simulator of the conventional Mecanum wheel. The first representation was the cylindrical shape (Figure 5.4a). This shape was chosen based on the research presented in [57] which indicates that a model of the Mecanum wheel considering the rollers as cylinders produced results consistent with the behavior of the real device. The second representation was the rollers as truncated

cones built by a series of successive thin cylindrical sections of decreasing radii (Figure 5.4b). The use of both roller representations has the goal to improve the model performance by using primitive shapes instead of the complex barrel shape of the Mecanum wheel's rollers.



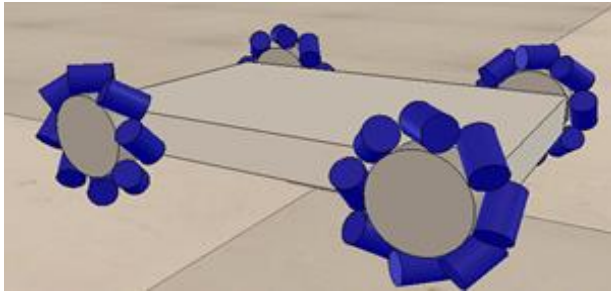
a) Cylindrical rollers



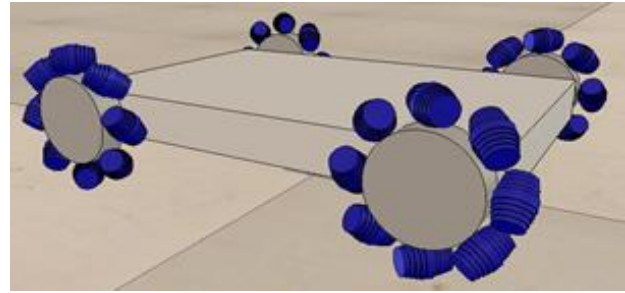
b) Conical rollers

Figure 5.4 Mecanum wheel V-REP model

The behavior of the physical simulator considering the two different roller shape representations was tested using a 4 wheel ground platform (Figure 5.5). A series of open loop test considering different angular velocities were performed and the trajectories described for the robots under such velocity inputs were used to evaluate the validity and performance of the proposed Mecanum wheel simulators.



a) cylindrical rollers wheel model



b) conical rollers wheel model

Figure 5.5 Omnidirectional Ground Vehicle V-REP model

The three motion inputs shown in Figure 5.6 were chosen for the open loop tests. For each one of the test scenarios different wheel rotational velocity values were considered with the objective to analyze if the outcome of the models was consistent under different conditions. A total of 10 simulations for 5 different values were run. The results of those simulations were as expected. Therefore the results for just one velocity value in each motion fashion are presented (Figure 5.7).

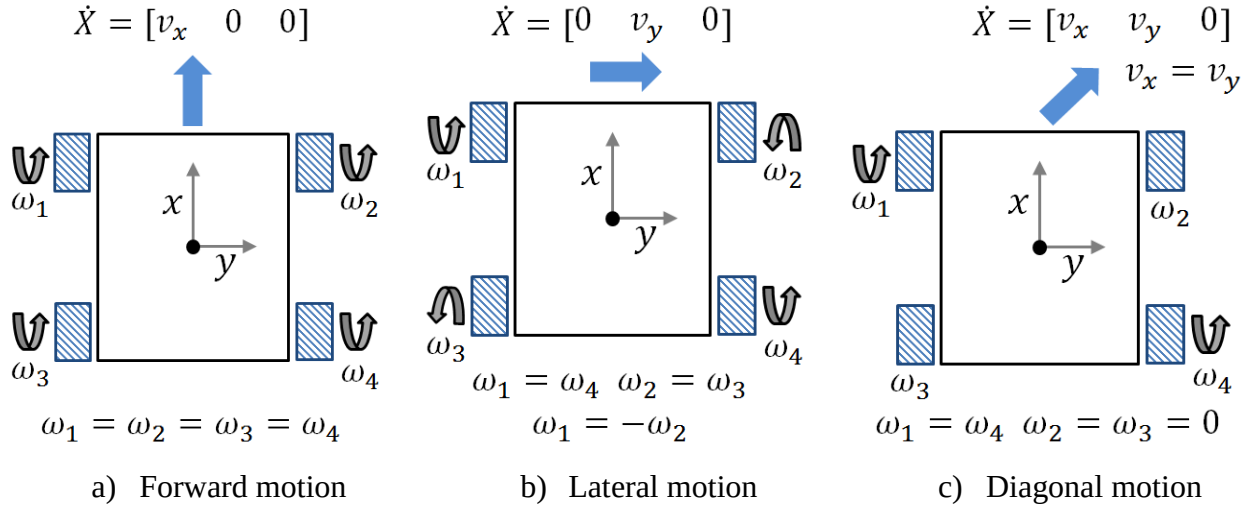


Figure 5.6 Open loop tests velocity inputs and motion directions

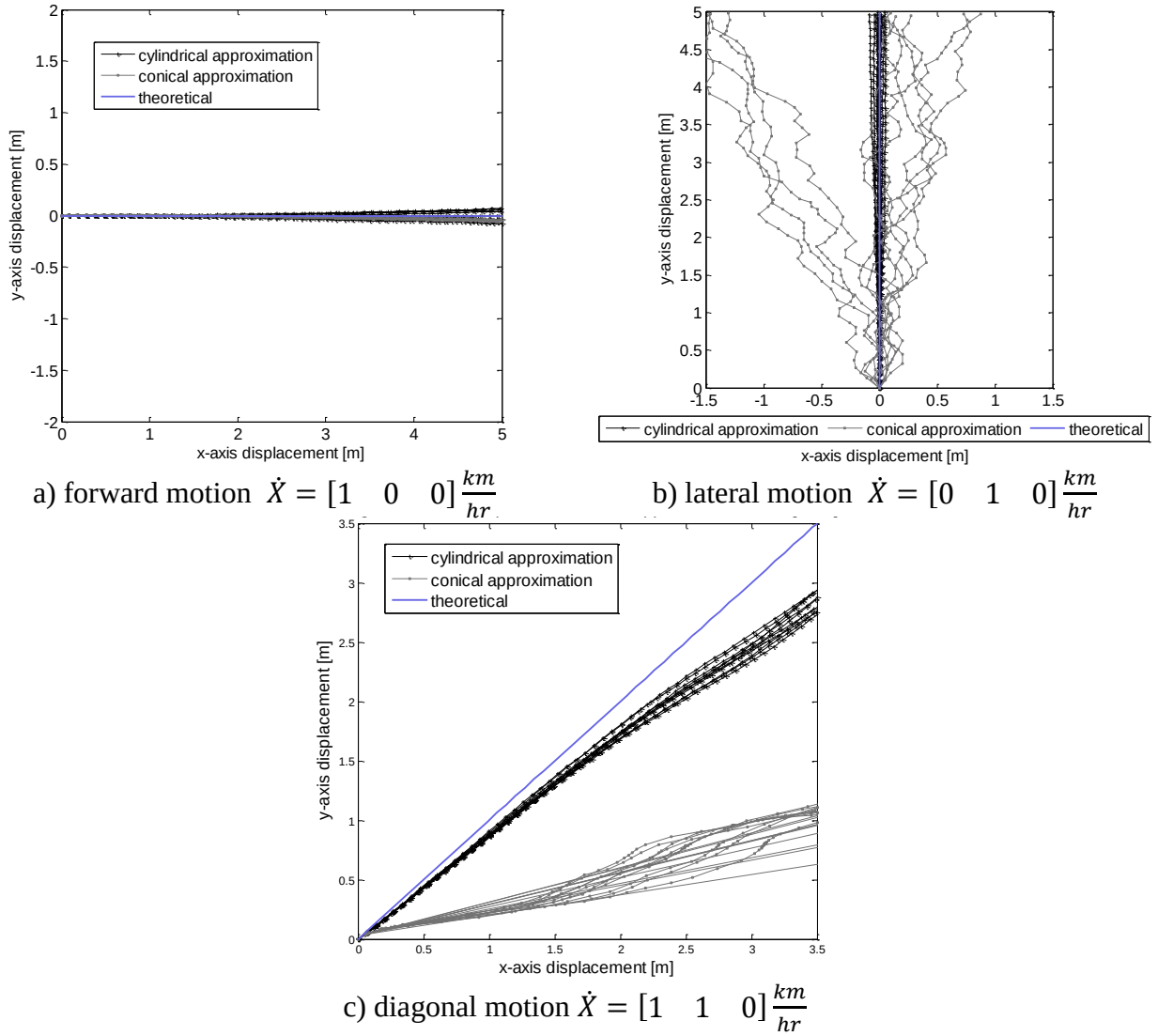


Figure 5.7 Mecanum wheel V-REP models simulation results

A) *Forward Motion*. In this case the simulation was run for a travel distance of 5 metres in the x-axis direction. The results for both wheel models (Figure 5.7a) were consistent with the expected theoretical motion. Minor variations (up to of 7 cm) were observed in the final position and trajectory of the vehicle as expected due to the slippage present on a real Mecanum wheel as well as on a physical simulated model.

B) *Lateral Motion*. The simulation was run until the OGV simulator reached a total travel distance of 5 metres in the y-axis direction. In the lateral motion simulation results (Figure 5.7b) the conical roller representation observed an unexpected behavior. For this wheel model, the motion of the vehicle was not close to the theoretical expected motion. A large deviation on the x axis final position was observed in all tests. Moreover the vehicle with the conical rollers exhibited a diagonal displacement instead of a lateral displacement trajectory. Conversely the model using the cylindrical roller approximation produced a trajectory that follows closely the expected theoretical trajectory.

C) *Diagonal Motion*. The robot was commanded to move for 5 meters at a 45° angle, $\frac{\Delta y}{\Delta x} = 1$, with respect to the vehicle's frame of reference. Similar to the lateral motion results, the diagonal motion (Figure 5.7c) exhibited a larger deviation with respect to the theoretical trajectory. The cylindrical rollers representation followed the theoretical trajectory very close up to a distance of 1.5 m of horizontal component. Around half way of the simulation the deviation between the trajectory of the vehicle and the expected motion started to increase. For the conical roller wheel model, the simulation results diverged significantly from the theoretical trajectory from the start of the simulation.

Based on the observed results the conical roller approximation does not provide an accurate representation of a traditional Mecanum wheel. In 2 out of 3 analyzed cases (the cases that make use of the holonomic capabilities of the Mecanum wheel) the model with this roller approximation performed badly, diverging significantly from the expected trajectory. Therefore, the representation of the rollers as truncated cones was discarded for subsequent work.

The cylindrical roller approximation on the other hand provided satisfactory results on all of the open loop tests. However, for the diagonal case, after a travel of approximately 2.1 m the deviation with respect to the theoretical trajectory increased significantly. Despite the increase in the divergence on the diagonal motion trajectory and a large final position deviation, the

cylindrical roller model followed the theoretical behavior trend. As a result this roller approximation was selected to be used for in the SMDW model development.

5.3.2 Simulation of a Semi-circular Mecanum Double Wheel

Once the cylindrical roller approximation proposed in [57] was tested and validated in a traditional Mecanum wheel. The next step was the development of a SMDW model. This section describes the development and testing of a SMDW model in V-REP approximating the rollers as cylinders.

The first SMDW physical model was built following the dimensions and construction layout of the prototype built and tested at the University of Calgary [24]. The peripheral rollers were approximated as cylinders of two different diameters following the theoretical design reported in [24]. For the rollers attached to the semi-circular profile cylinders of 19 mm of diameter were used. For the semi-elliptical profile the rollers were modeled by cylinders with a 25 mm diameter. The wheel hubs were imported as a mesh from a CAD model generated in SolidWorks according to the refined design (Figure 5.8). The rollers were attached to the wheel's hub through a revolute joints that allow them to rotate freely. The wheel model is composed by an assembly of 2 wheel hubs rotated 180° with respect to each other (Figure 5.8b).

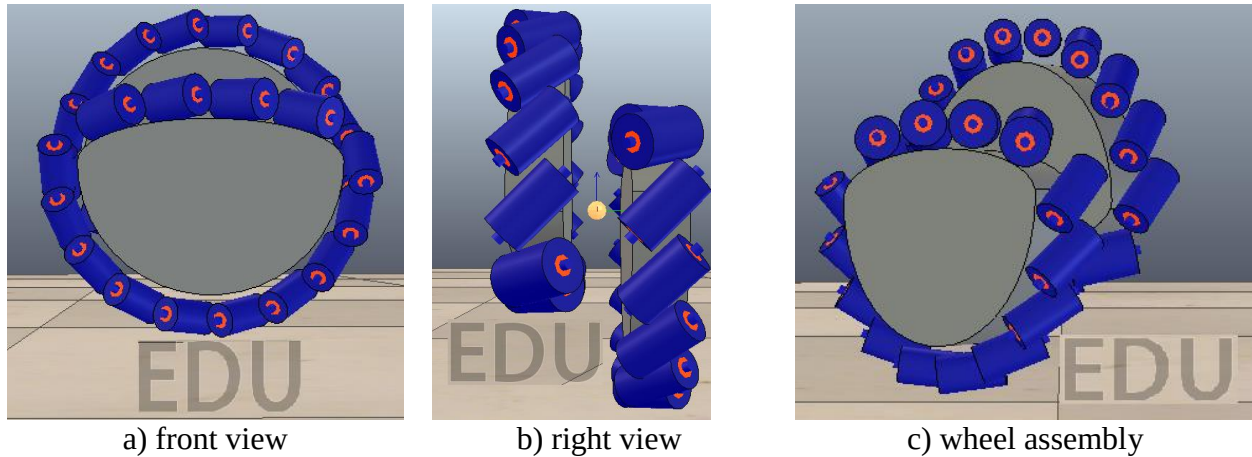


Figure 5.8 SMDW V-REP model with cylindrical rollers

A validation of the SMDW physical model with cylindrical rollers was performed. Similar to the traditional Mecanum wheel tests, a set of open loop tests was performed. The same three motion cases used in Section 5.3.1 were analyzed considering 50 different simulations (sets of 10 simulations for 5 different velocity values). The simulations were run until the robot traveled a distance of 5 meters. The robot velocity values chosen for the SMDW physical model are shown in Table 5.1.

Table 5.1 SMDW Tests Velocity Inputs

Forward motion velocity $\left[\frac{km}{hr}\right]$	Lateral motion velocity $\left[\frac{km}{hr}\right]$	Diagonal motion velocity $\left[\frac{km}{hr}\right]$
$\dot{X}_1 = [1 \ 0 \ 0]$	$\dot{X}_1 = [0 \ -1 \ 0]$	$\dot{X}_1 = [1 \ -1 \ 0]$
$\dot{X}_2 = [2 \ 0 \ 0]$	$\dot{X}_2 = [0 \ -2 \ 0]$	$\dot{X}_2 = [2 \ -2 \ 0]$
$\dot{X}_3 = [3 \ 0 \ 0]$	$\dot{X}_3 = [0 \ -3 \ 0]$	$\dot{X}_3 = [3 \ -3 \ 0]$
$\dot{X}_4 = [4 \ 0 \ 0]$	$\dot{X}_4 = [0 \ -4 \ 0]$	$\dot{X}_4 = [4 \ -4 \ 0]$
$\dot{X}_5 = [5 \ 0 \ 0]$	$\dot{X}_5 = [0 \ -5 \ 0]$	$\dot{X}_5 = [5 \ -5 \ 0]$

A) *Forward Motion*. The results (Figure 5.9a) show that the proposed SMDW model exhibits a moderate deviation as the vehicle reaches the final position. The maximum deviation observed was of approximately 0.6 m over the 5 m trip along the x- axis (forward trajectory). This deviation is equivalent to a 12% of the x component of the displacement. It is possible to appreciate that in short distances up to 1 m, the model followed very close the expected trajectory. Nevertheless as the distance increased the vehicle exhibited a consistent increment in the deviation of the vehicle trajectory with respect to the theoretical trajectory.

B) *Lateral Motion*. The observed behavior (Figure 5.9b) shows a noticeable discrepancy between the simulation and the theoretical trajectory. For the slowest velocity $\dot{X}_1 = [0 \ -1 \ 0] \frac{km}{hr}$, the simulations did not correlate at all with the expected theoretical behavior. For this velocity, the simulations did not reach the 5 m y-axis displacement after 300 seconds (6000 simulation cycles) despite that under ideal conditions such travel should take 18 seconds. These results are considered as a divergence of the physical model for those conditions. The results for the remaining velocity values show a large final position deviation. In all the simulations the time required to complete the 5 m travel exceeded the 50 seconds (1000 simulation cycles) despite that at the given velocity inputs should not have exceeded 9 seconds (under ideal conditions). Additionally, the render tool of V-REP showed the wheels rotating in a non-smooth manner suggesting that they were subjected to either a considerable friction or experiencing some obstruction.

C) *Diagonal Motion*. From the simulation results shown Figure 5.9c it is observed that the vehicle's trajectory deviated considerably from the theoretically expected trajectory in each one of the numerous simulations. Once again all the simulations exceeded the theoretical time

for trajectory completion (21 seconds at the slowest velocity), they took at least twice the expected time to complete the travel. The considerable deviation in the desired trajectory and the significantly larger time required to complete the travel, might indicate the presence of high friction or slippage issues with the physical model. In contrast to the lateral motion case, the trend of the diagonal motion trajectories was more consistent. Finally the render tool of V-REP showed that the vehicle was not traveling smoothly; instead the wheels rotation and frame motion displayed perturbations similar to the ones caused by small obstructions on the vehicle's path.

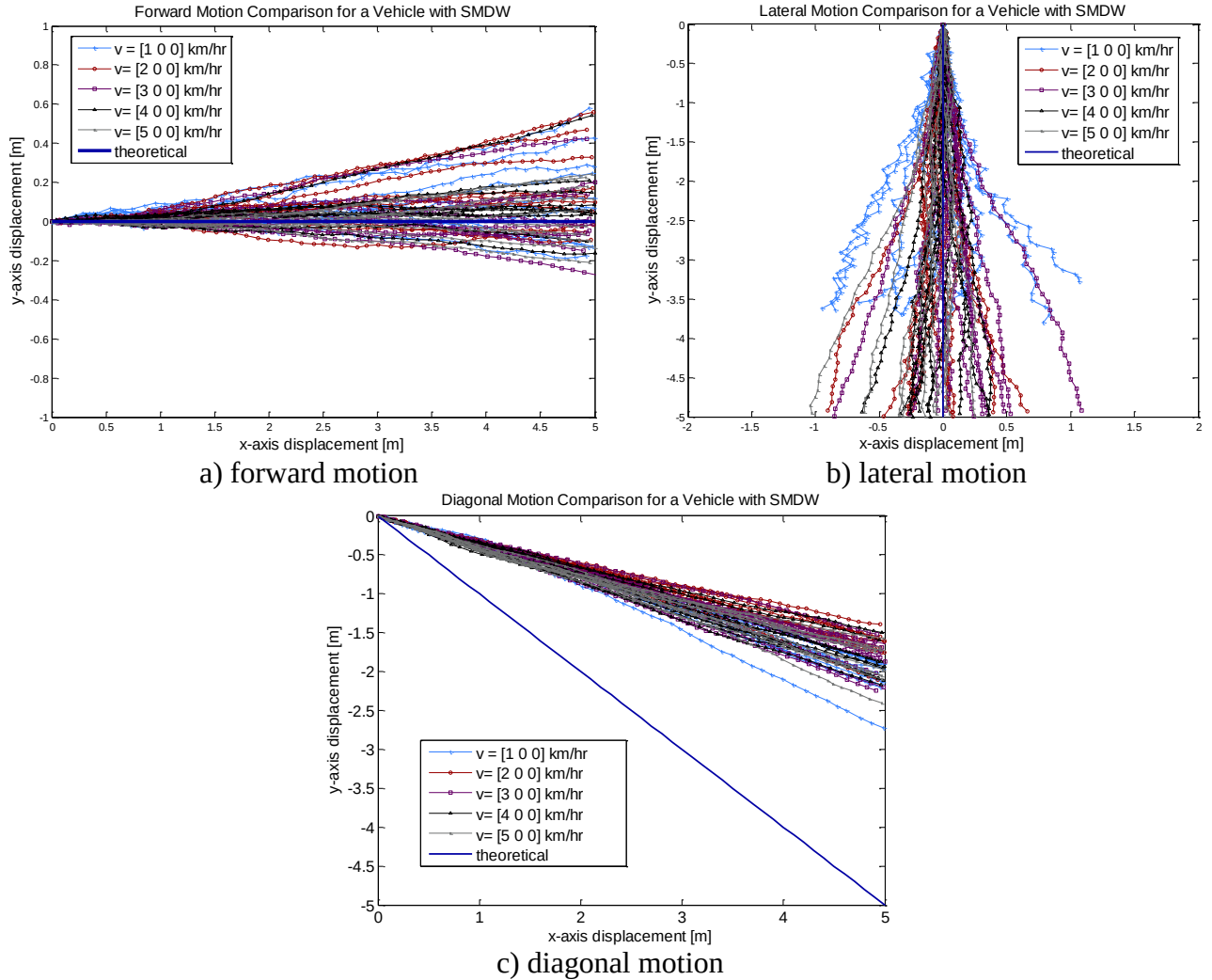


Figure 5.9 SMDW with cylindrical rollers simulation results

The simulation results for the lateral and diagonal motion indicate that the V-REP model developed does not provide an accurate representation of the SMDW device. Such simulations provided evidence that the wheels did not rotate uniformly despite that the tests were done with

constant velocity inputs. Further analysis into the wheel joints velocities and torques and the simulation let to conclude that the cylindrical rollers were causing a situation of multiple rollers in simultaneous contact at the transition (see Figure 5.10). From the joints torques analysis, it was concluded that the multiple contact was generating energy loses and altering the rotation of the wheels. It was assumed that the effect of the multiple points of contact originated the deviation of the vehicle's trajectory and the slower motion of the vehicle.

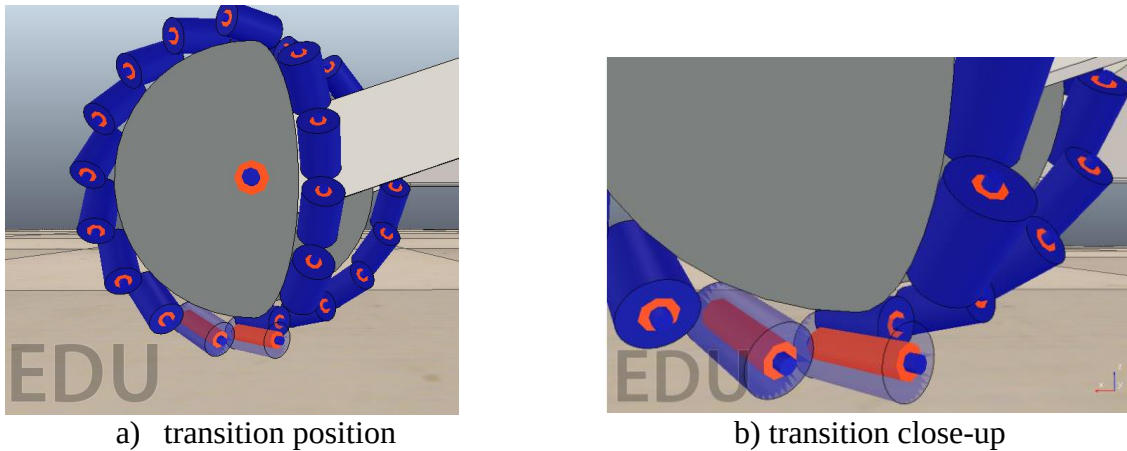


Figure 5.10 SMDW V-REP model hub transition

The study included in the Kalman doctoral dissertation was developed considering a traditional Mecanum wheel. Therefore, the implications in the use of cylindrical rollers for a SMDW model were overlooked. Since the V-REP model with cylindrical rollers cannot be used to represent properly a SMDW it is necessary to refine the proposed physical simulator considering the actual shape for the wheel rollers.

A new physical model was developed considering a complex roller shape calculated to provide a circular silhouette to the assembly hubs of the SMDW. The shape and length of the rollers used was taken from [24]. Subsequently a 3D CAD model of the barrel shape rollers was generated in SolidWorks (Figure 5.11). A mesh was generated from the SolidWorks file and imported into V-REP as shown in Figure 5.12.

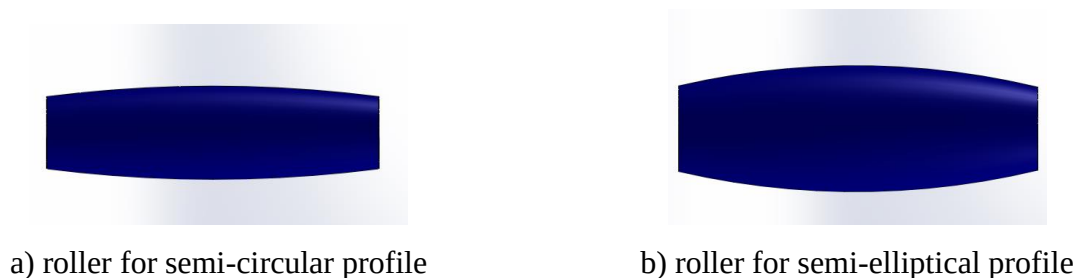


Figure 5.11 Barrel shaped SMDW rollers

In addition it was required to increase the gap between the SMDW hubs (Figure 5.12b) in order to accommodate the new barrel shaped rollers in a similar way as the experimental robot.

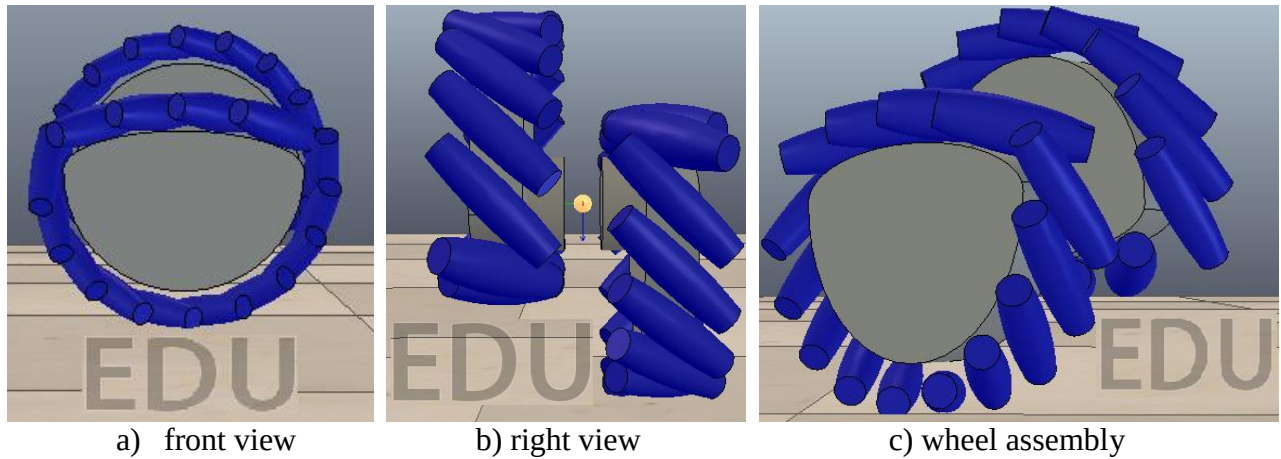


Figure 5.12 SMW V-REP model with barrel shape rollers

The new V-REP model of the enhanced wheels was incorporated to the LOC8 platform model and the motion tests performed with the cylindrical rollers were repeated under exactly the same conditions and velocity input. The simulation results for the three motion cases forward, lateral and diagonal motions are shown in Figure 5.13.

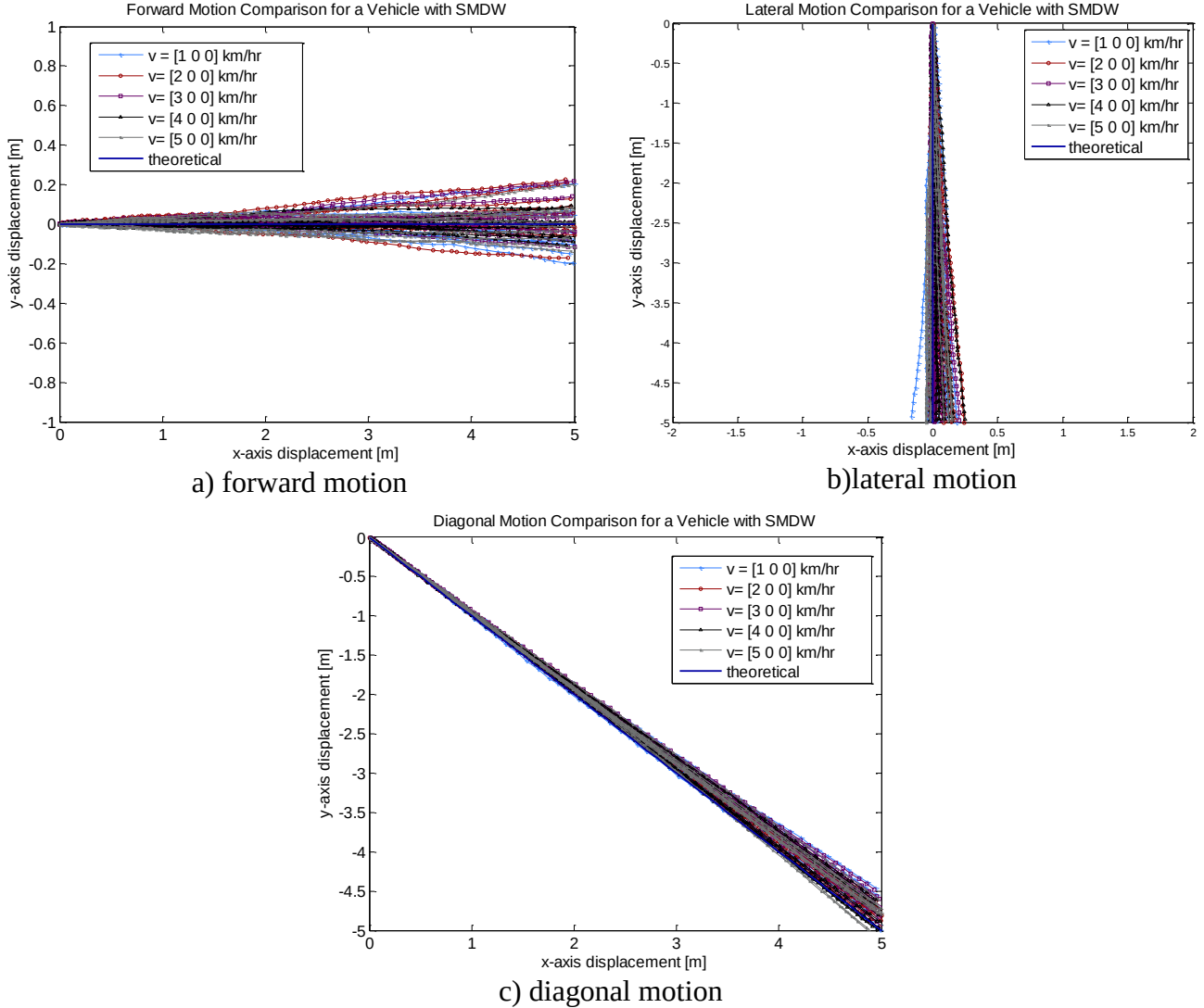


Figure 5.13 V-REP Simulation results for barrel rollers SMDW

As can be seen from the results shown above the use of barrel rollers improved significantly the performance of the SMDW model as the trajectory of the vehicle closely follows the theoretical trajectory.

The improvement of the SMDW is not just observed in the vehicle's trajectory. As can be seen in Figure 5.14 there is also a significant difference at the rotational velocity of the joint that connects the wheel with the vehicle's frame. The multiple contact points originated by the cylindrical rollers exert an effect on the wheel rotation that does not allow it to achieve the motion commanded to the actuators. Instead an erratic rotation with a lot of variations in its velocity is observed by the SMDW. Conversely for the model with the barrel shape rollers, the

wheel assembly exhibited a rotation with a steady velocity according to the commands sent to the actuator (V-REP joint of the vehicle frame).

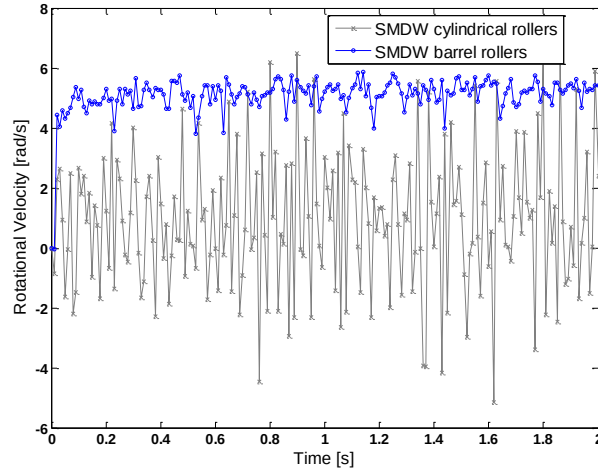


Figure 5.14 Rotational velocity comparison with different roller shapes

As a result the time required to complete the simulations reduced drastically and the vehicle model completed each run in a time very close to the expected theoretical time for the given velocity input.

5.4 Trajectory Search PSO Module

With a proper SMDW model the proposed PSO searching program was implemented in MATLAB for the generation of a cubic spline trajectory to drive the omnidirectional robot to the give desired position and wheel states. The constants and tuning parameters required for the PSO algorithm were chosen according to the guidelines provided in [47] . A series of preliminary tests were run in order to evaluate the values chosen and perform adjustments if required.

For the tuning and testing of the PSO search program the initial conditions and desired states were taken from the data used at the University of Calgary [26]. The following sections show the simulation results for the different parameters selected. Modifications to the originally proposed algorithm were also performed.

5.4.1 PSO Search Algorithm Initial Implementation Tuning

The initial implementation of the PSO search algorithm considered a swarm of 50 particles over a tridimensional space $[x, y, \psi]$ for the location of the control point P_2 of a cubic spline trajectory. The parameters chosen for: the acceleration coefficients (social and cognitive

components), the particle initialization, the location of P_1 , the convergence criterion , the fitness function gain, and the time interval for the spline are shown in Table 5.2.

Table 5.2 PSO search algorithm parameters for 1st implementation

Acceleration Coefficients $c_1 = c_2 = 0.45$	Time Interval $0 \leq t \leq 5 \text{ seconds}$	Convergence Criterion 500 iterations
Particle Initialization $P_{ij} = \left[\left(x_b + \frac{x_{max} - x_b}{6} \right) \quad \left(y_{min} + j \cdot \frac{y_{max} - y_{min}}{11} \right) \quad (\psi_f - \psi_0) \cdot r_{ij} \right]$ Where: $i = \{1, 2 \dots 5\} \quad j = \{1, 2 \dots 10\} \quad x_{max} = x_f \quad x_b = \frac{x_f - x_0}{2}$ r_{ij} = random number from normal distribution ; such that: $0 \leq r_{ij} \leq 1$ y_{max} = maximum value of the y dimension search space ; such that: $y_{max} > y_0, y_{max} > y_f$ y_{min} = minimum value of the y dimension search space; such that: $y_{min} < y_0, y_{min} < y_f$		
Control Point P_1 Location $P_1 = \left[\left(x_{min} + \frac{x_b}{2} \right) \quad \left(y_0 - \text{sign}(y_0) \cdot \frac{y_{max} - y_{min}}{4} \right) \quad \left(\frac{\psi_f - \psi_0}{3} \right) \right]$ Where: $x_{min} = x_0 \quad x_b = \frac{x_f - x_0}{2}$ y_{max} = maximum value of the y dimension search space ; such that: $y_{max} > y_0, y_{max} > y_f$ y_{min} = minimum value of the y dimension search space; such that: $y_{min} < y_0, y_{min} < y_f$		
Fitness Function Gains $\gamma = [1 \ 0.125 \ 1] \quad \beta = [10 \ 10 \ 10 \ 10]$		

The parameters described above were used to evaluate the performance of the PSO search technique. A set of tests were performed considering the initial conditions and target states shown in . The three main parameters to evaluate were the deviation of the trajectory, the navigation space explored (exploration and exploitation) by the particle swarm, and the location of control point P_2 .

Table 5.3 First implementation PSO search algorithm test parameters

Wheels States					Robot Position			
	$\theta_1[\text{rad}]$	$\theta_2[\text{rad}]$	$\theta_3[\text{rad}]$	$\theta_4[\text{rad}]$		$x \text{ [m]}$	$y \text{ [m]}$	$\psi \text{ [rad]}$
$\Theta_0 =$	0	0	0	0	P_0	0	0.25	0.3927
$\Theta^* =$	0	0	0	0	P_d	2	0.75	1.5708
Navigation Space								
$x_{min} = 0 \text{ [m]}$			$x_{max} = 2 \text{ [m]}$		$y_{min} = -1.5 \text{ [m]}$		$y_{max} = 1.5 \text{ [m]}$	

A total of 20 tests were performed Figure 5.15 to Figure 5.18 show illustrative results.

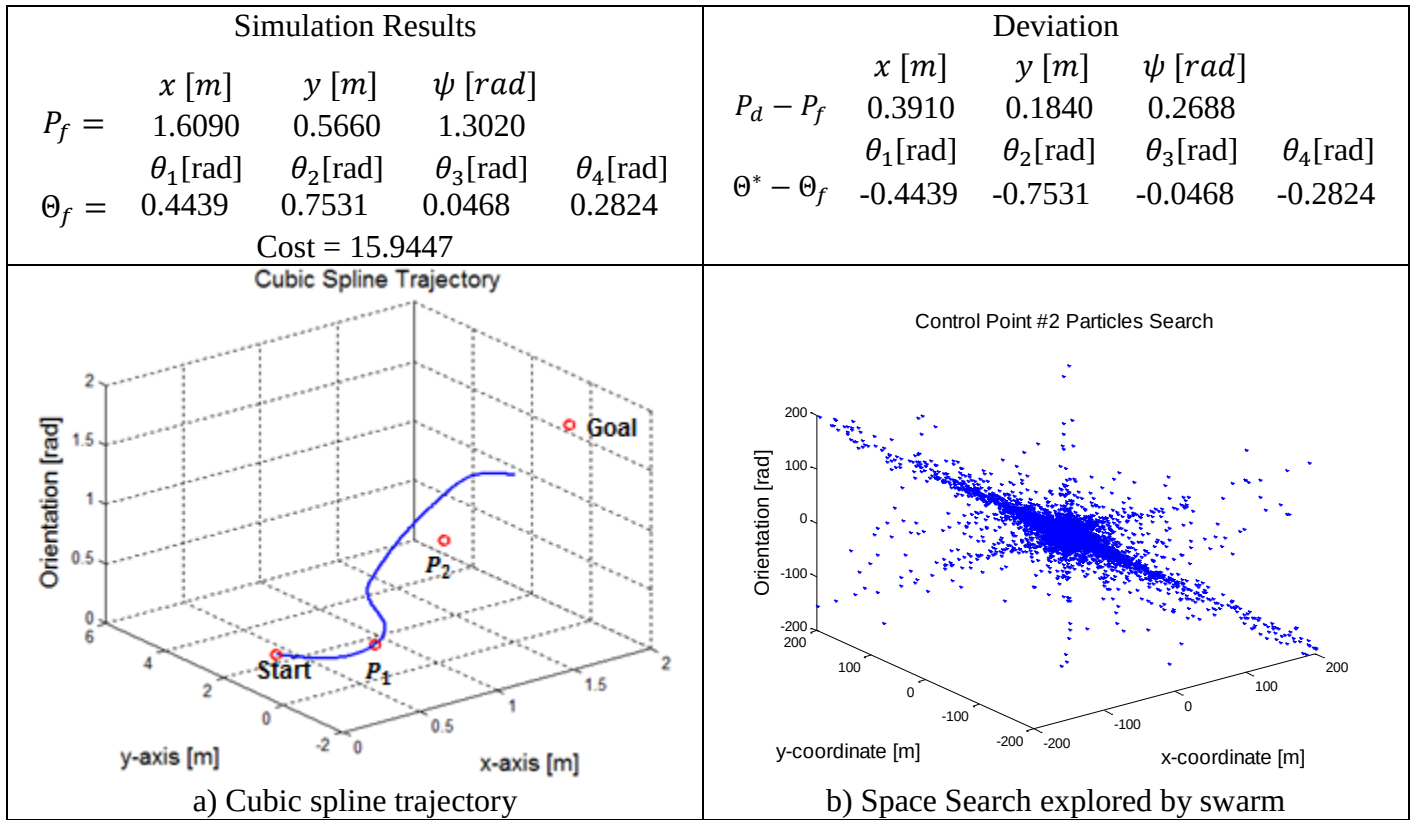


Figure 5.15 PSO search algorithm simulation results #1

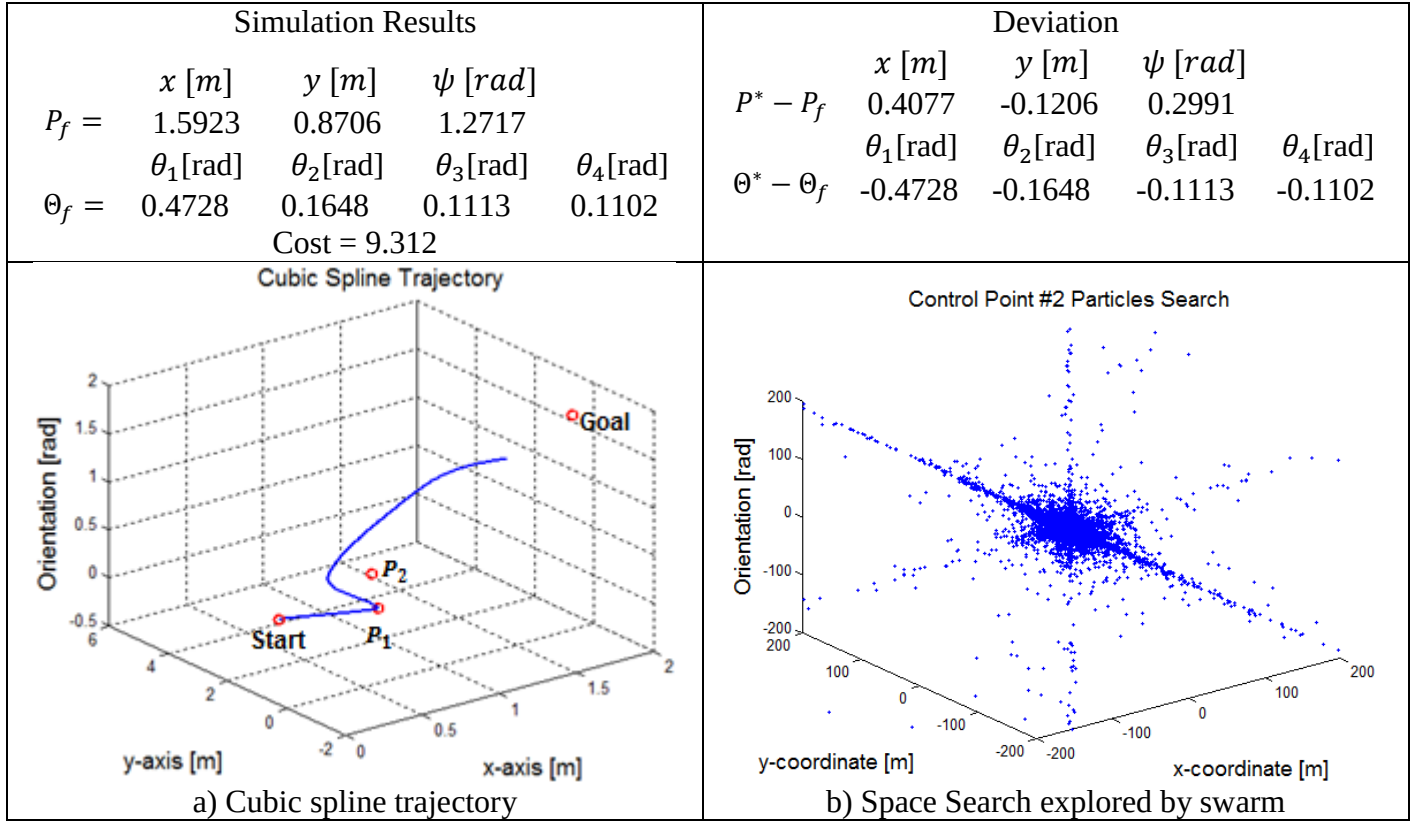


Figure 5.16 PSO search algorithm simulation results #2

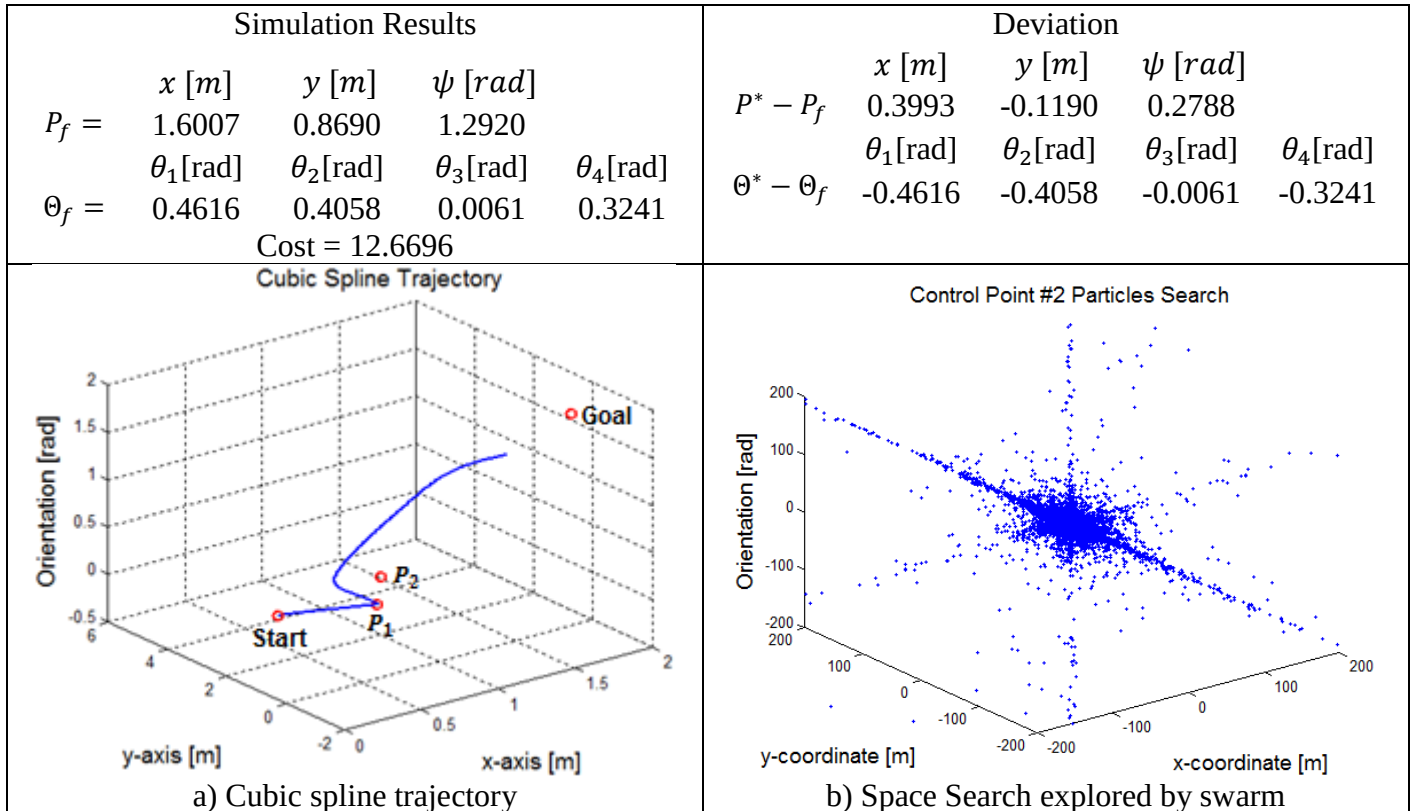


Figure 5.17 PSO search algorithm simulation results #3

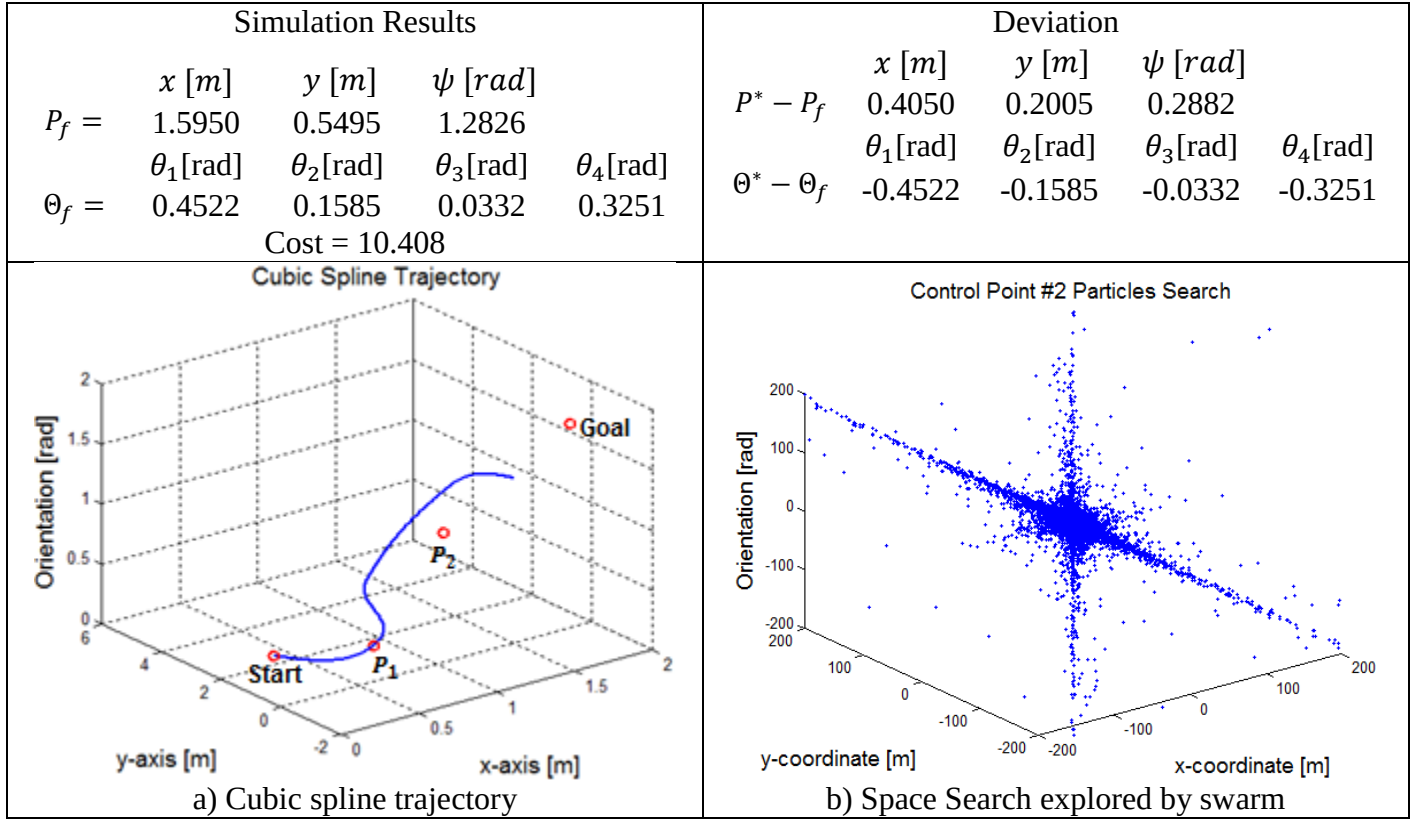


Figure 5.18 PSO search algorithm simulation results #4

The first implementation of the PSO algorithm produced trajectories that converge to a similar location for the control point P_2 as be seen in Figure 5.15 to Figure 5.18. Nevertheless the solutions produced observed a significant deviation on the robot's position and a smaller but still noticeable deviation for the wheels state. In each case the kinematic simulated trajectory did not pass through P_2 neither by P^* , where P^* is the desired robot's position. This issue in the trajectory is likely the result of the sharp turns in the produced trajectories which caused the trajectory to fall within the saturation and dead-zone established for the kinematic model. In addition the space explored by the swarm extended beyond the navigation limits (Table 5.3) in the tests.

In order to improve the quality of the trajectories obtained additional tests considering a larger number of iterations were performed. Tests considering up to 5000 iterations were run with no improvements (Figure 5.19).

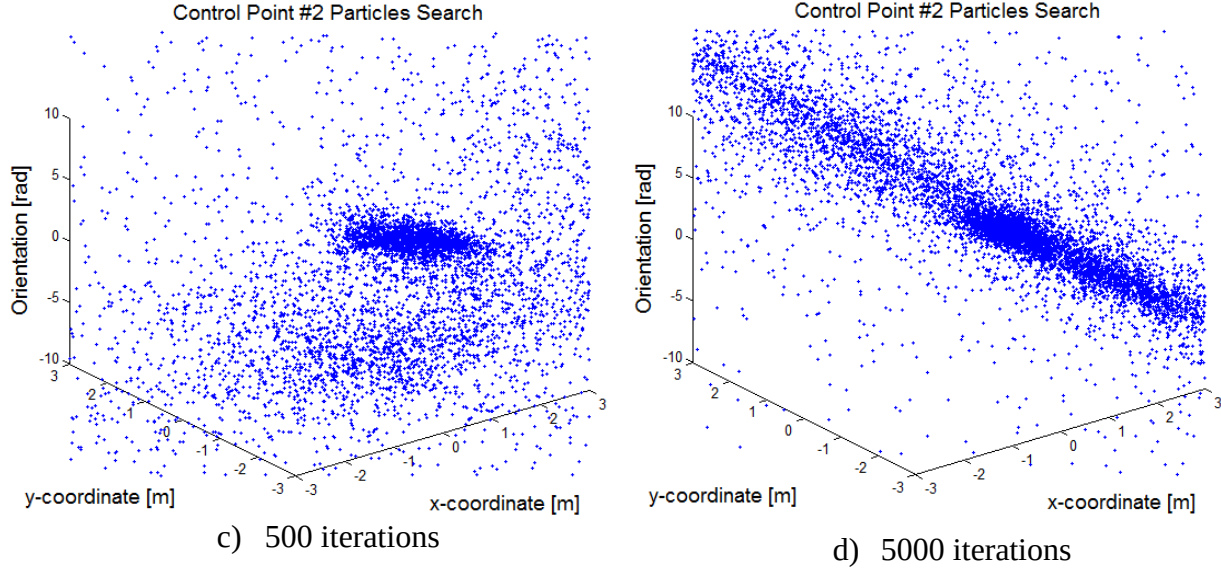


Figure 5.19 Particle Swarm Exploration Close-up

A series of major modifications following [47] were performed to the PSO search algorithm in order to improve the quality of the generated trajectories. The changes aimed to increase the search ability of the swarm by increasing the variety on the explored locations contained within the navigation space. .

5.4.2 Enhanced PSO Search Algorithm

The first adjustment made to improve the exploitation was the reduction of the acceleration coefficients. A lower value allowed the particles to explore more points within the navigation space before starting to explore outside of it. Another modification was the incorporation of particle's reset to the algorithm. This reset takes place when a particle is located outside the search space.

The redefinition of the trajectory structure was also used. Originally a tridimensional cubic spline trajectory $[x(t), y(t), \psi(t)]^T$ was proposed. However, having three dimensions to explore over the search space imposes great challenges for convergence and to produced consistent (similar) trajectories. As a result a new trajectory structure was used. The new structure considers the use of a first order polynomial for the robot orientation ψ with constant rotational velocity. For the motion in the x and y directions the structure keeps the previous piecewise cubic spline polynomials without modifications. With the use of this new structure the PSO algorithm reduces its dimensions to just two since the rotations velocity is fixed and constant.

While the trajectories generated by the PSO search do not have any direct action over the $\psi(t)$ trajectory, the overall motion of the vehicle as well as its final position and wheels state are affected by the changes in ψ . The new changes on the trajectory structures motivated the use of a different location of the control point P_1 with to reduce the PSO search space such that the navigation space is more uniformly divided between the internal control points. The changes to the particles equations and tuning parameters are summarized in Table 5.4.

Table 5.4 Enhanced PSO search algorithm parameters and trajectory redefinition

Parameterized Robot Trajectory $x(s) = \begin{cases} a_{x1} \cdot (2-s)^3 + b_{x1} \cdot (s-1)^2 + c_{x1} \cdot (2-s) + d_{x1} \cdot (s-1) & 1 \leq s < 2 \\ a_{x2} \cdot (3-s)^3 + b_{x2} \cdot (s-2)^2 + c_{x2} \cdot (3-s) + d_{x2} \cdot (s-2) & 2 \leq s < 3 \\ a_{x3} \cdot (4-s)^3 + b_{x3} \cdot (s-3)^2 + c_{x3} \cdot (4-s) + d_{x3} \cdot (s-3) & 3 \leq s \leq 4 \end{cases}$ $y(s) = \begin{cases} a_{y1} \cdot (2-s)^3 + b_{y1} \cdot (s-1)^2 + c_{y1} \cdot (2-s) + d_{y1} \cdot (s-1) & 1 \leq s < 2 \\ a_{y2} \cdot (3-s)^3 + b_{y2} \cdot (s-2)^2 + c_{y2} \cdot (3-s) + d_{y2} \cdot (s-2) & 2 \leq s < 3 \\ a_{y3} \cdot (4-s)^3 + b_{y3} \cdot (s-3)^2 + c_{y3} \cdot (4-s) + d_{y3} \cdot (s-3) & 3 \leq s \leq 4 \end{cases}$ $\psi(s) = \frac{\psi_f - \psi_0}{3} \cdot s \quad \text{where: } s = \frac{t}{3}$	
PSO Particles Equations $\begin{bmatrix} x(k+1) \\ y(k+1) \end{bmatrix} = \begin{bmatrix} x(k) + v_x(k+1) \\ y(k) + v_y(k+1) \end{bmatrix} \quad c_1 = c_2 = 0.15$ $\begin{bmatrix} v_x(k+1) \\ v_y(k+1) \end{bmatrix} = \begin{bmatrix} v_x(k) + c_1 \cdot r_{x1}(k) \cdot (x_b(k) - x(k)) + c_2 \cdot r_{x2}(k) \cdot (\hat{x}_b(k) - x(k)) \\ v_y(k) + c_1 \cdot r_{y1}(k) \cdot (y_b(k) - y(k)) + c_2 \cdot r_{y2}(k) \cdot (\hat{y}_b(k) - y(k)) \end{bmatrix}$	
Control Point P_1 Location $P_1 = \left[\left(x_{min} + \frac{x_{max} - x_{min}}{2.5} \right) \quad \left(y_0 + \frac{y_f - y_0}{3} \right) \right] \quad \text{Where: } x_{min} = x_0 \quad x_{max} = x_f$	
Particle Initialization $P_{ij} = \left[\left(x_b + \frac{x_{max} - x_b}{6} \right) \quad \left(y_{min} + j \cdot \frac{y_{max} - y_{min}}{11} \right) \right]$ Where: $i = \{1, 2 \dots 5\} \quad j = \{1, 2 \dots 10\} \quad x_{max} = x_f \quad x_b = \frac{x_f - x_0}{2}$	
Particle Location Reset $\text{If } \left((x_i > x_{max}) \text{ or } (x_i < x_{min}) \text{ or } (y_j > y_{max}) \text{ or } (y_j < y_{min}) \right)$ $P_{ij} = \left[\left(x_b + 0.9 \cdot r_i \cdot (x_{max} - x_b) \right) \quad \left(y_{min} + \frac{y_{max} - y_{min}}{2} + 0.9 \cdot (r_j - 0.5) \cdot (y_{max} - y_{min}) \right) \right]$	

In order to evaluate the effect of the modifications to the PSO search algorithm a set of 20 simulations were performed. For these simulations the same conditions of the first PSO

implementation (Table 5.3) were considered in order to facilitate comparison between both implementations. Herein 3 representative tests are presented.

As can be seen in Figure 5.20 to Figure 5.22 the modification of the PSO parameters improved the trajectories produced. The deviation between the robot's position and the wheel states decreased significantly. The space explored also improved significantly as the reset action achieved to keep the swarm within the defined search space. In Figure 5.20b to Figure 5.22b one can observe important changes in the exploration and exploitation capabilities of the swarm. While the particles targeted their search in the region surrounding the location of P_2 , it is possible to observe that the system also analyzed locations all over the different regions of the search space. This more extensive search over the navigation space provided a major reduction in the trajectory deviation and a significant increase in the quality of the trajectories.

It can be concluded that as intended, the re-definition of the trajectory scheme (which eliminates one dimension of the PSO search) had a significant impact on the consistency of the produced trajectories.

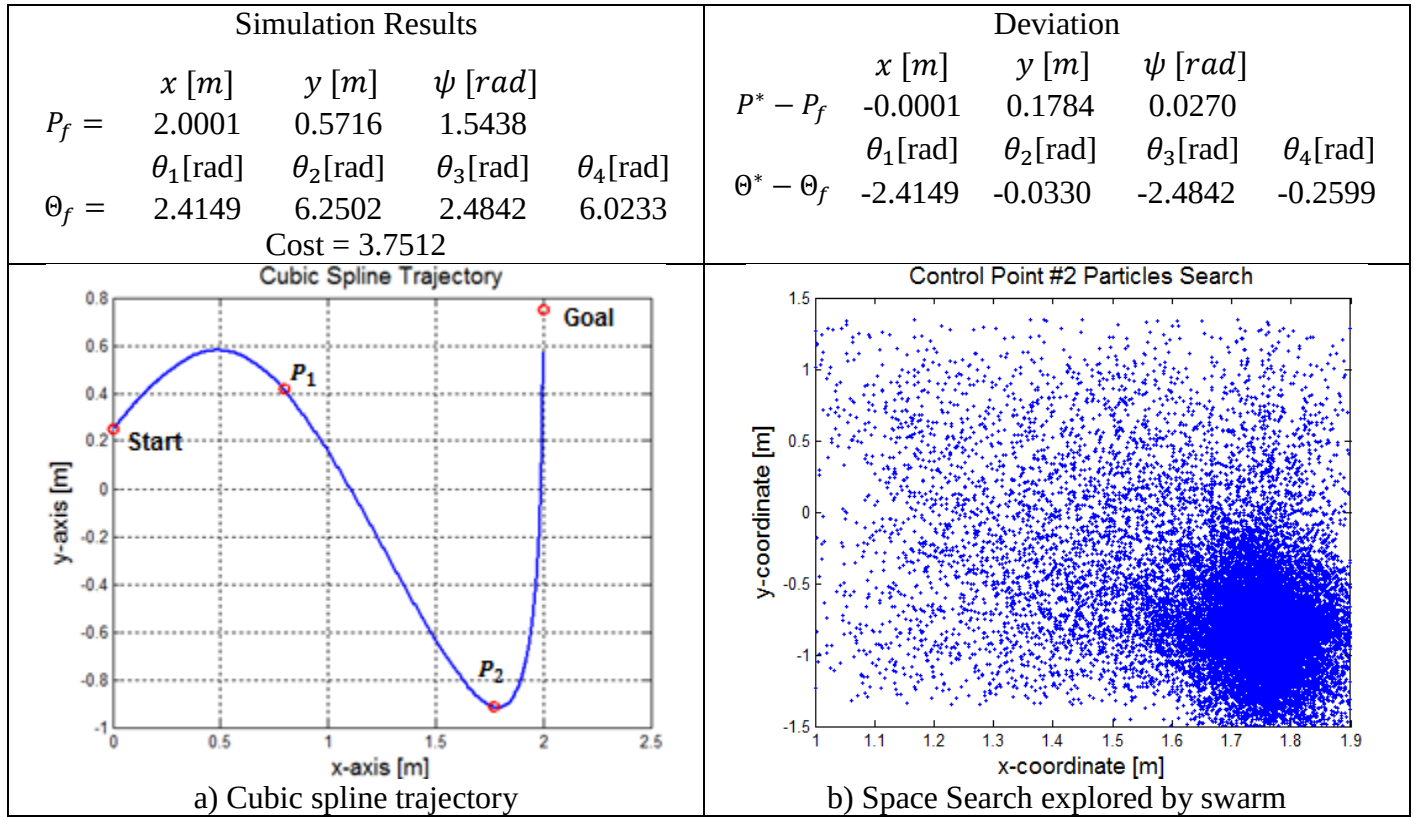


Figure 5.20 Enhanced PSO search algorithm simulation results #1

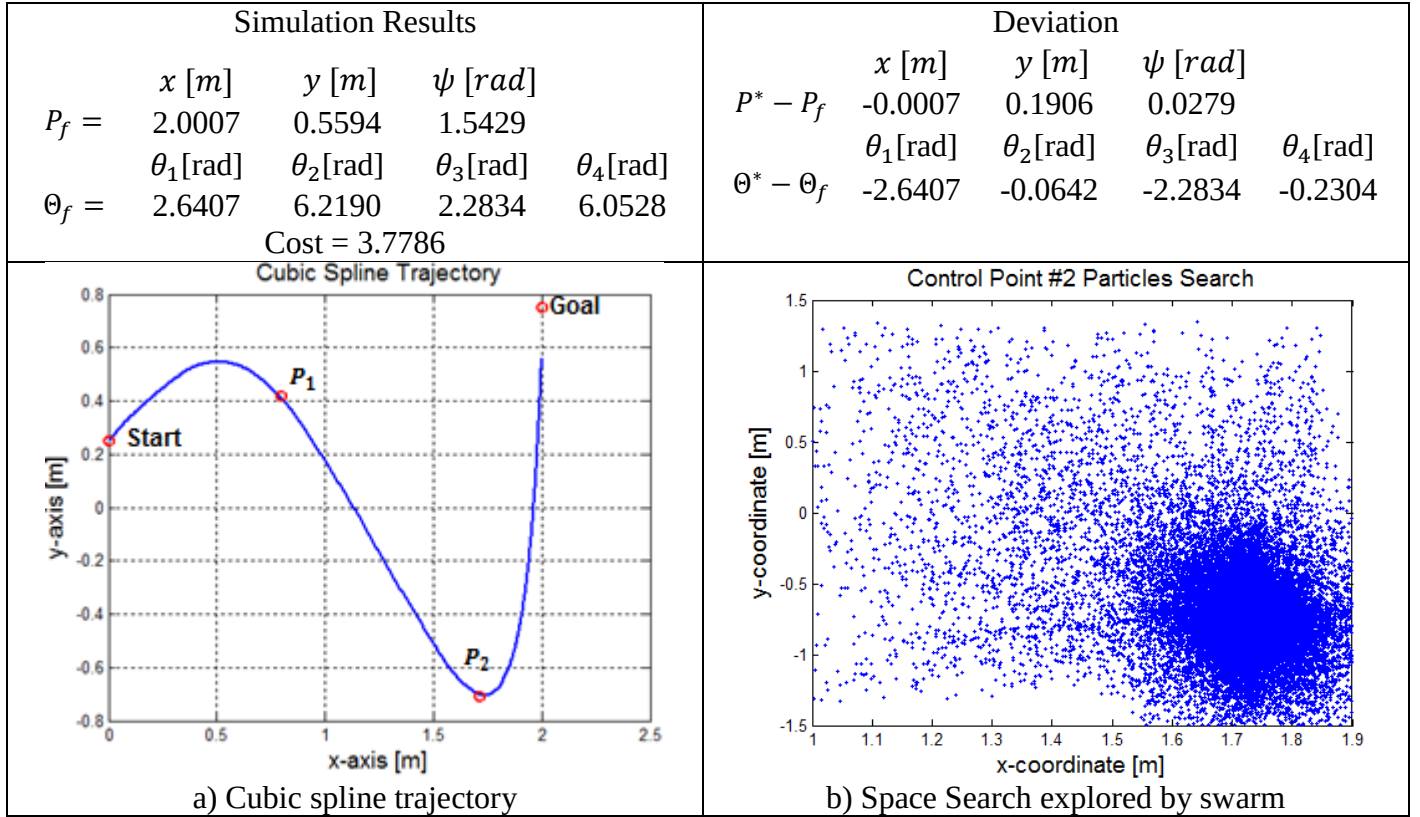


Figure 5.21 Enhanced PSO search algorithm simulation results #2

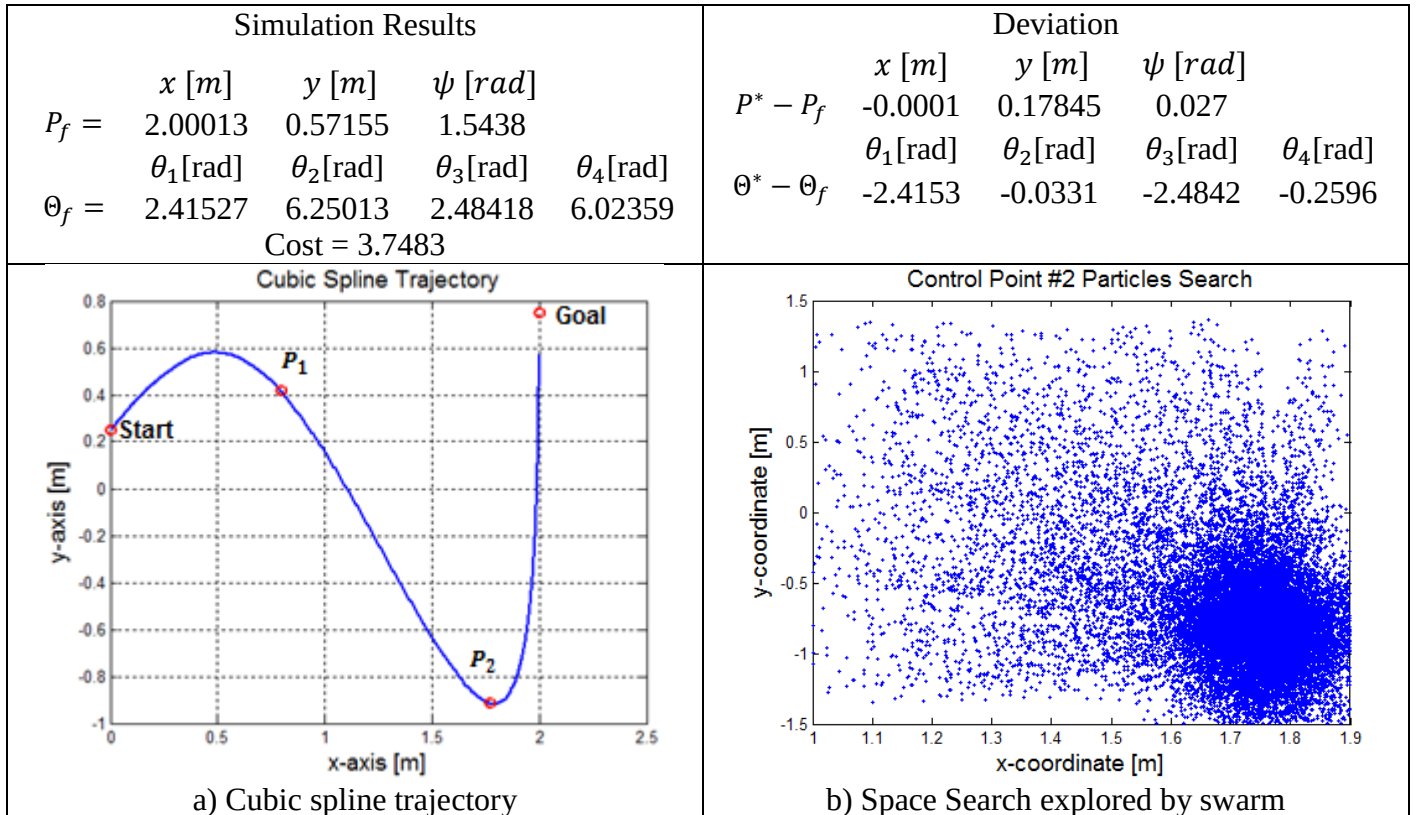


Figure 5.22 Enhanced PSO search algorithm simulation results #3

The capability of explore with more detail over the navigation space is a key factor for the improvements. The incorporation of inertia weight and velocity clamping parameters to the PSO algorithm were analyzed. However the test results showed that none of them improved the search. Based on the previous analysis of the different PSO search proposals, the enhanced PSO algorithm with particle rescent and re-defined trajectory was chosen to be used for the maneuvering methodology. This PSO algorithm is used to explore different control point locations in order to provide a trajectory suitable for the obstacle overcoming operation.

5.4.3 Trajectory time impact in PSO Search Algorithm

The final step of the PSO search module development is the selection of the time period used for the generation of the spline trajectory. As explained in previous sections the maneuvering methodology considers the use of piece-wise polynomials ($[x(s), y(s), \psi(s)]$) parameterized as function of time ($s = \frac{t}{3}$). Since the interpolation process and the fitness function evaluation are time dependent; the time interval to be selected, $t_0 \leq t \leq t_f$, will have a direct impact in the performance of the PSO module.

In order to analyze and illustrate the impact of the time interval selection a set of simulations was performed using the same test conditions (initial conditions and target states) for different time span values. Then we compared and evaluated the changes in the trajectory, deviation and fitness cost between the time intervals. The tests conditions are once again taken from Table 5.3 using 14 different time intervals evenly separated by 0.5 sec within the range $0.5 \leq t \leq 7 \text{ sec}$. The produced trajectories and the results of the simulation are presented in Figure 5.23 and Table 5.5.

The results show that the time interval selected can have a very noticeable impact on the shape of the produced trajectory (location of P_2) by the PSO module (Figure 5.23). The lower values of time (e.g. 0.5 seconds) produces high peaks of velocity which leads the robot to operate under saturation. When operating under saturation the robot did not reach the required velocities. The result will be that a considerable deviation will be present on the final states of the vehicle.

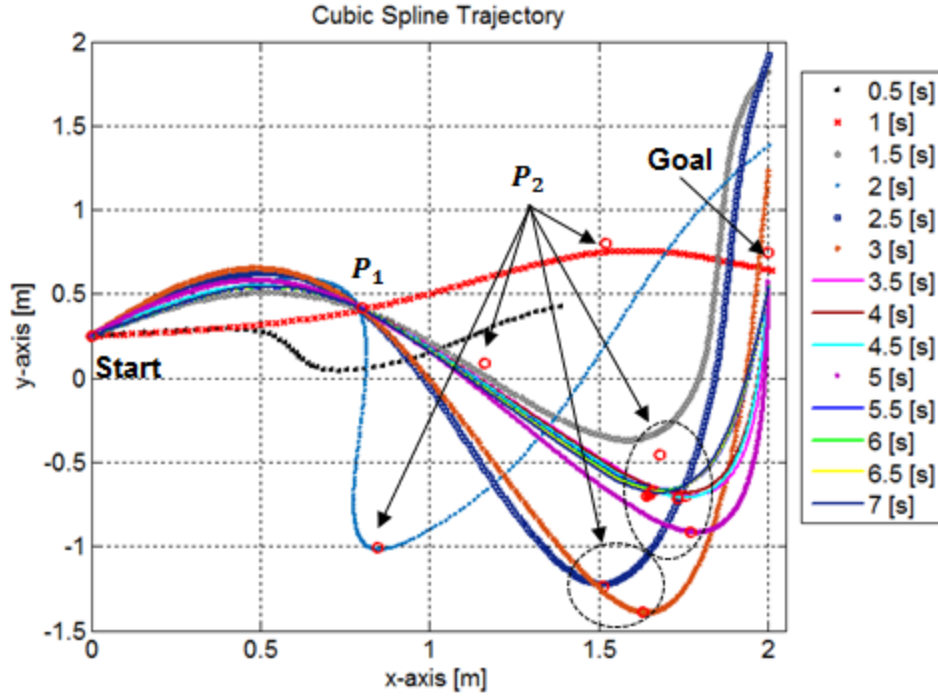


Figure 5.23 Time interval effects on PSO search module

Conversely larger values of the time interval can lead the robot to operate at a low velocity regime. Operating at low velocity can increase the slippage experienced by the vehicle. The main reason of the slippage is that the angular velocities of the actuators can fall within the dead-zone. At the dead-zone region the desired torque is lower than the one required by the motor to overcome the friction and inertia preventing the motion. Under this condition the open loop controller in the maneuvering methodology is not capable to correct such situation and deviations can appear on the actual motion of the robot.

As can be seen from the results for the different values listed in Table 5.5, the time value chosen can lead to a smaller deviation values and a reduced cost for the overall trajectory. When the trajectory shape and deviation are considered together it is observed that an appropriate selection of the time interval can lead to more refined trajectories. From these results the time interval of 1 second is the one at which the PSO algorithm exhibits a noticeable improvement in terms of trajectory overall distance, trajectory fitness and deviation compared with the other intervals tested.

Table 5.5 Time interval effects on PSO search module results

Time [s]	Position Deviation			Wheels Deviation $\theta^* - \theta_f$				Cost
	x [m]	y [m]	ψ [rad]	θ_1 [rad]	θ_2 [rad]	θ_3 [rad]	θ_4 [rad]	
0.5	0.6134	0.3240	0.4739	-0.9115	-0.0148	-0.1482	-0.0085	3.3571
1	-0.0095	0.1082	0.0199	-0.9369	-0.0055	-2.2332	-0.0011	0.1746
1.5	0.0006	-1.0727	0.0054	-2.2544	-0.0025	-2.3247	-0.0050	0.2432
2	-0.0017	-0.6303	0.0081	-1.4286	-0.0004	-0.1907	-0.0071	0.1988
2.5	-0.0033	-1.1698	0.0108	-0.3852	-0.0018	-1.9336	-0.0025	0.2377
3	-0.0005	-0.4778	-0.0106	-1.5601	-0.2019	-2.8154	-0.0429	3.1483
3.5	-0.0003	0.1779	0.0267	-2.4345	-0.0018	-2.4650	-0.2868	3.6984
4	-0.0007	0.1843	0.0247	-2.5328	-0.0012	-2.3459	-0.2486	3.2092
4.5	-0.0004	0.1837	0.0262	-2.5204	-0.0205	-2.3736	-0.2542	3.5230
5	-0.0001	0.1784	0.0270	-2.4147	-0.0332	-2.4842	-0.2601	3.7562
5.5	-0.0010	0.2005	0.0277	-2.8203	-0.0778	-2.1070	-0.2060	3.6451
6	-0.0008	0.2039	0.0241	-2.8417	-0.0738	-2.0291	-0.1650	3.0726
6.5	-0.0009	0.2064	0.0210	-2.8568	-0.0668	-1.9730	-0.1264	2.4958
7	-0.0009	0.2084	0.0184	-2.8673	-0.0591	-1.9254	-0.0975	2.0317

To better observe the time interval impact on the PSO performance, a set of 10 additional simulations were run with a time interval of 1 second. The trajectories produced and the deviation values for these simulation are presented in Figure 5.24 and Table 5.6 respectively.

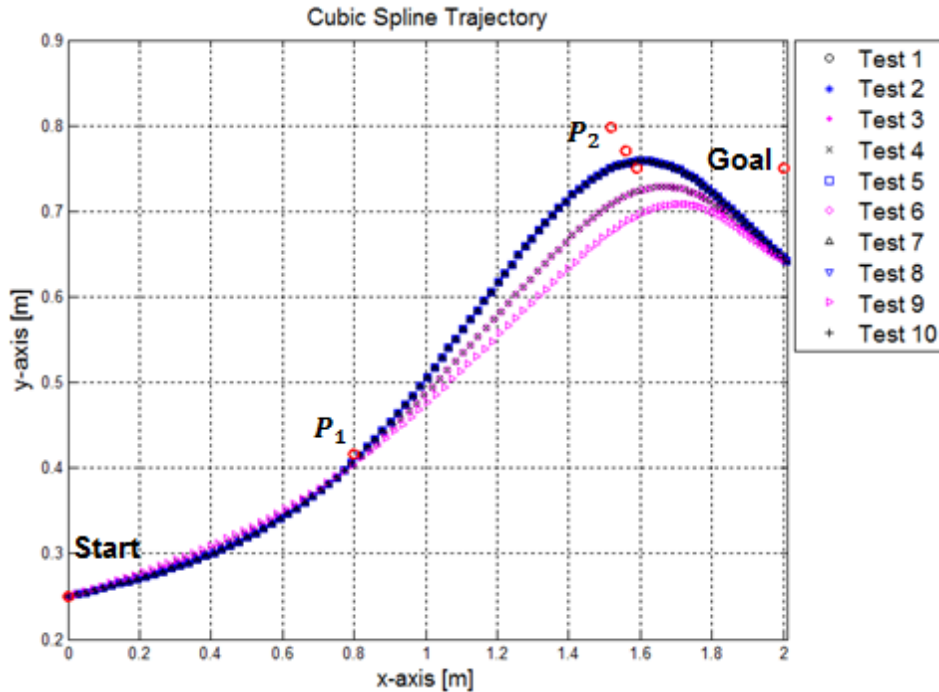
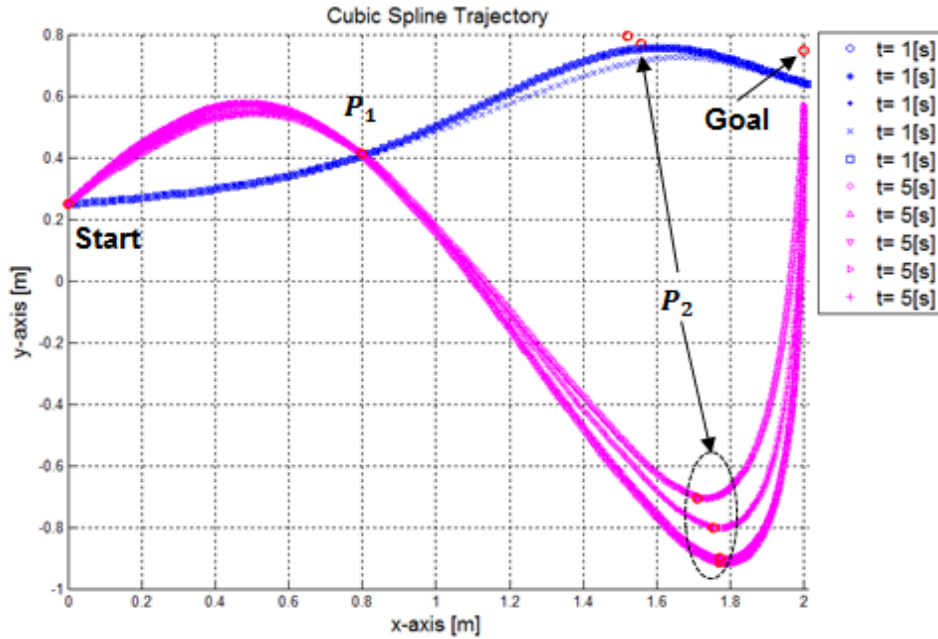
**Figure 5.24 PSO search for 1 sec time interval**

Table 5.6 PSO Search results for 1 sec time interval

	Position Deviation			Wheels Deviation $\theta^* - \theta_f$				Cost
	x [m]	y [m]	ψ [rad]	θ_1 [rad]	θ_2 [rad]	θ_3 [rad]	θ_4 [rad]	
Test 1	-0.0095	0.1080	0.0196	-0.9341	0.0000	-2.2355	0.0000	0.0909
Test 2	-0.0095	0.1080	0.0196	-0.9341	0.0000	-2.2355	0.0000	0.0909
Test 3	-0.0095	0.1080	0.0196	-0.9341	0.0000	-2.2355	0.0000	0.0909
Test 4	-0.0064	0.1086	0.0152	-0.8840	0.0000	-2.3082	0.0000	0.0709
Test 5	-0.0095	0.1080	0.0196	-0.9341	0.0000	-2.2355	0.0000	0.0909
Test 6	-0.0064	0.1086	0.0152	-0.8840	0.0000	-2.3082	0.0000	0.0709
Test 7	-0.0095	0.1080	0.0196	-0.9341	0.0000	-2.2355	0.0000	0.0909
Test 8	-0.0095	0.1080	0.0196	-0.9341	0.0000	-2.2355	0.0000	0.0909
Test 9	-0.0003	0.1064	0.0065	-0.7177	0.0000	-2.3868	0.0000	0.0305
Test 10	-0.0095	0.1080	0.0196	-0.9341	0.0000	-2.2355	0.0000	0.0909

The detailed simulation results (Figure 5.24 and Table 5.6) confirmed that for a 1 second interval the trajectories do provide a significant improvement. For example, compared to the trajectories produced for a 5 seconds interval (Figure 5.25) the new trajectories exhibited a shorter distance between the initial position and the target states.

**Figure 5.25 Trajectories comparison for 1 sec and 5 sec intervals**

The simulations results also showed a clear relationship between the selected time interval for the spline interpolation and the quality of the trajectories produced by the PSO module. However such relationship is complex and multidimensional. As the results showed, not

all the shorter path trajectories produced a reasonable fitness cost. Likewise in most of the cases a low fitness cost was not necessarily associated to a short distance path.

Since the selection of an optimal time interval is not within the goals of this thesis; an effective policy for time interval selection was defined based on the simulation results during the tuning of the PSO algorithm. The policy is intended to provide a starting point for an operational region that allows the production of trajectories feasible for the obstacle maneuvering. However, the path the length improvement was not considered for the policy selection. The time interval was defined as follow:

Table 5.7 Time Intervals for PSO Search Algorithm

Time	Navigation Space Size
1.5 sec	$(x_d - x_0) \leq 1 \text{ and } y_d - y_0 \leq 1$
2.5 sec	$1 < (x_d - x_0) \leq 2 \text{ and } 1 < y_d - y_0 \leq 2$
3.5 sec	$2 < (x_d - x_0) \text{ and } 2 < y_d - y_0 $

The intention of the intervals established above is to be a reference, and they can be modified in case that the trajectories produced exhibited a high deviation. Nevertheless in order to keep consistency and allow comparison; for the different study cases presented Chapter 6 the time intervals in Table 5.7 were kept.

The analysis of the time interval impact concluded the tuning and validation activities performed for the PSO search algorithm. After the different test performed it was decided to use the implementation with particle reset and a 2D space search. The convergence criterion was kept at 500 iterations. The incorporation of the inertia weight and velocity clamping parameters was discarded since they did not improve the behaviour of the quality of the produced trajectories. The parameters chosen for the final PSO algorithm used by the maneuvering methodology proposed in this thesis are summarized in Table 5.8.

Table 5.8 PSO search algorithm final implementation parameters

Acceleration Coefficients $c_1 = c_2 = 0.15$	Time Interval $\Delta t = \begin{cases} 1.5 \text{ s}, & (x_d - x_0) \leq 1 \text{ and } y_d - y_0 \leq 1 \\ 2.5 \text{ s}, & 1 < (x_d - x_0) \leq 2 \text{ and } 1 < y_d - y_0 \leq 2 \\ 3.5 \text{ s}, & 2 < (x_d - x_0) \text{ and } 2 < y_d - y_0 \end{cases}$	Convergence Criterion 500 iterations
PSO Particles Equations $\begin{bmatrix} x(k+1) \\ y(k+1) \end{bmatrix} = \begin{bmatrix} x(k) + v_x(k+1) \\ y(k) + v_y(k+1) \end{bmatrix} \quad c_1 = c_2 = 0.15$ $\begin{bmatrix} v_x(k+1) \\ v_y(k+1) \end{bmatrix} = \begin{bmatrix} v_x(k) + c_1 \cdot r_{x1}(k) \cdot (x_b(k) - x(k)) + c_2 \cdot r_{x2}(k) \cdot (\hat{x}_b(k) - x(k)) \\ v_y(k) + c_1 \cdot r_{y1}(k) \cdot (y_b(k) - y(k)) + c_2 \cdot r_{y2}(k) \cdot (\hat{y}_b(k) - y(k)) \end{bmatrix}$		
Control Point P_1 Location $P_1 = \left[\left(x_{min} + \frac{x_{max} - x_{min}}{2.5} \right) \quad \left(y_0 + \frac{y_f - y_0}{3} \right) \right] \quad \text{Where:} \quad x_{min} = x_0 \quad x_{max} = x_f$		
Particle Initialization $P_{ij} = \left[\left(x_b + \frac{x_{max} - x_b}{6} \right) \quad \left(y_{min} + j \cdot \frac{y_{max} - y_{min}}{11} \right) \right]$ Where: $i = \{1, 2 \dots 5\} \quad j = \{1, 2 \dots 10\} \quad x_{max} = x_f \quad x_b = \frac{x_f - x_0}{2}$		
Particle Location Reset If $\left((x_i > x_{max}) \text{ or } (x_i < x_{min}) \text{ or } (y_j > y_{max}) \text{ or } (y_j < y_{min}) \right)$ $P_{ij} = \left[\left(x_b + 0.9 \cdot r_i \cdot (x_{max} - x_b) \right) \quad \left(y_{min} + \frac{y_{max} - y_{min}}{2} + 0.9 \cdot (r_j - 0.5) \cdot (y_{max} - y_{min}) \right) \right]$		
Fitness Function Gains $\gamma = [3 \quad 0.125 \quad 2.5] \quad \beta = \begin{cases} [12.5 \quad 0 \quad 12.5 \quad 0], & \text{if wh1 and wh3 are closer to obstacle} \\ [0 \quad 12.5 \quad 0 \quad 12.5], & \text{if wh2 and wh4 are closer to obstacle} \\ [12.5 \quad 12.5 \quad 0 \quad 0], & \text{if wh1 and wh2 are closer to obstacle} \end{cases}$		

Chapter 6: Experimental Results

In Chapter 5 the V-REP model and the PSO search algorithm were refined and tested individually. The test results validate those components as reliable and accurate tools to be used for the overall performance evaluation of the proposed maneuvering methodology. In this chapter series of tests using all available tools in a single maneuvering simulation are performed such tests consider 2 main scenarios with different conditions for: robot position, wheels angular position, obstacle position, and height and target obstacle encountering position. Each one of these scenarios consider 3 possible configurations to perform the obstacle overcoming action.

Each experimental test performs three main tasks as part of the maneuvering methodology. It first estimates the required clearance for the given obstacle height for each configuration. The target position and wheels states are then calculated for each configuration based on: the required clearance, obstacle location and the desired encountering position along the obstacle. After the target positions are determined a set of 10 PSO searches were performed. Finally the obtained trajectories were executed in the V-REP simulator and the results analyzed.

The trajectory evaluation on the physical simulator was divided into 2 sub-tests. In the first sub-test (Figure 6.1) the trajectory execution was tested using an open loop trajectory controller and the ODE (Open Dynamics Engine). The robot executes the spline until the either it reaches the desired position on the x -axis or when a timeout equivalent to 1.5 times the time interval is reached. The main goal of this process is to evaluate whether or not the obtained trajectory is suitable for an open loop execution.

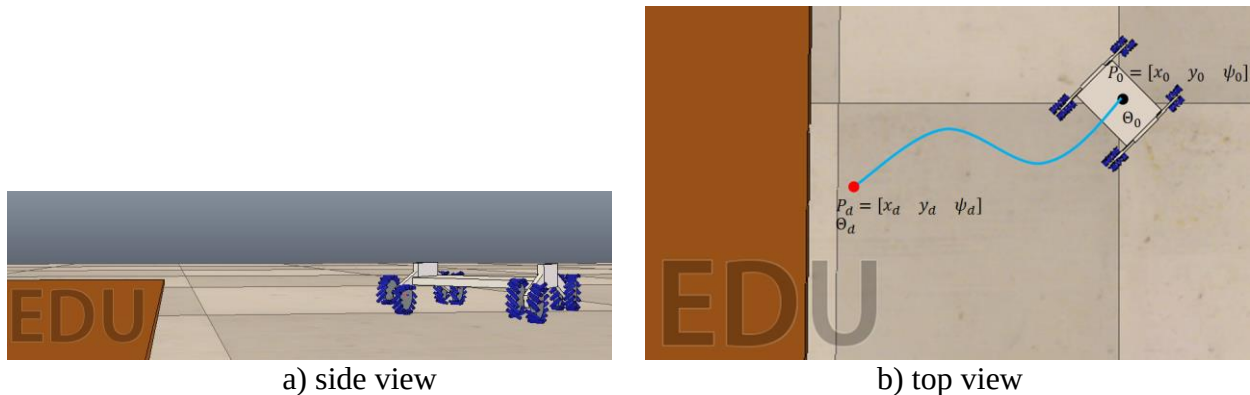


Figure 6.1 Trajectory evaluation sub-test #1

The second sub-test (Figure 6.2) assesses the OGV's obstacle climbing abilities. For this sub-test the Bullet physical engine was used. The robot was placed at the final position of the

trajectory and then commanded to run either a forward or lateral motion (depending on the configuration tested) for a fixed period of 2 seconds. The positions of the robot, joints and SMDWs' velocity data generated during the run were collected and the results analyzed to determine if the robot was able to overcome the obstacle at the given configuration.

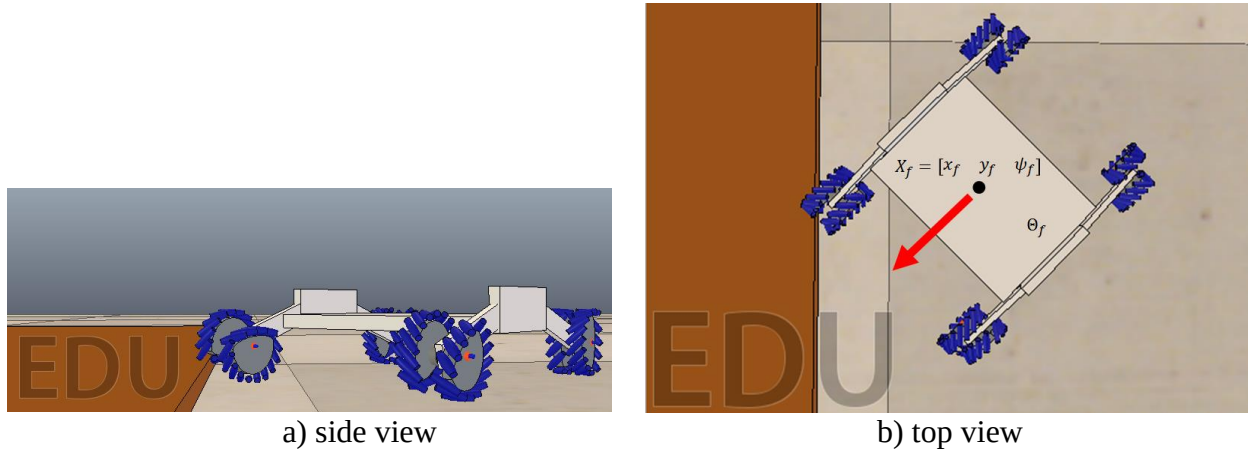


Figure 6.2 Trajectory evaluation sub-test #2

The results were evaluated in terms of the success at overcoming the obstacle. Furthermore, a comparison was performed between the expected overcoming capabilities and the capabilities provided by the final position of the physical simulation results. Under such terms it was possible to evaluate the effectiveness of the proposed methodology and also to establish a range of conditions at which the methodology can provide the best performances.

6.1 Test Scenario #1

The initial conditions for the first scenario are provided in Figure 6.3. This first scenario aims to test the performance of the methodology for a space having a range of approximately 3 meters where the robot is allowed to move. For this test an obstacle height of 5 cm, the maximum obstacle height the simulated SMDWs can overcome was used assuming maximum robot frame rotation of quarter turn.

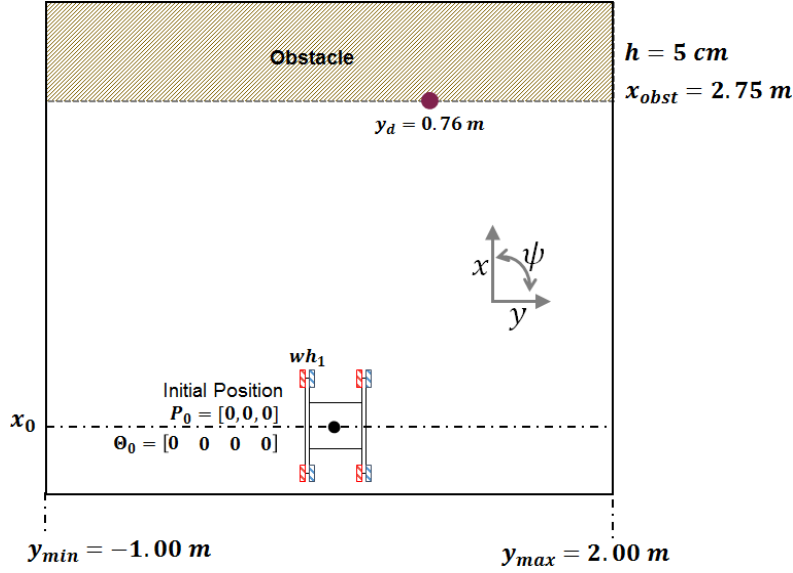


Figure 6.3 Scenario #1 Initial Conditions and Test Parameters

Using the geometric model presented in this thesis the necessary angular position for a 5 cm obstacle was calculated for the three selected configurations (Table 6.1). The target position of the robot's frame of reference was computed. Therefore, in order to find the desired position the fixed distance between the robot's frame center and the wheel in contact with the obstacle is subtracted.

For the desired wheels states only one or two critical wheels having contact with the obstacle were observed. However, since the PSO search requires a target state arbitrary values were chosen for the other wheels. The fitness function however only considers the critical wheels. The ground clearance estimation process determined that wheel should have an angular position, θ , smaller than 0.26 rad in order to climb the obstacle in forward motion. Instead, the most stringent criterion was applied and the required angular position was set as $\theta = 0^\circ$. The estimated desired states for each configuration are provided in Table 6.1.

Table 6.1 Scenario #1 desired robot position and wheel configurations

Configuration	Desired Robot Final Position [m, rad]	Desired Wheel's Final Position [rad]
Lateral motion, $\alpha = 0^\circ$	$X_d = \begin{bmatrix} 2.46 & 0.76 & -\frac{\pi}{2} \end{bmatrix}$	$\Theta_d = [0 \quad 0 \quad 0 \quad 0]$
Forward motion, $\alpha = -30^\circ$	$X_d = [2.2 \quad 0.76 \quad -0.611]$	$\Theta_d = [0 \quad 0 \quad 0 \quad 0]$
Forward motion, $\alpha = -45^\circ$	$X_d = \begin{bmatrix} 2.2 & 0.76 & -\frac{\pi}{4} \end{bmatrix}$	$\Theta_d = [0 \quad 0 \quad 0 \quad 0]$

Using the PSO search algorithm a set of trajectories was generated for each one of the different desired configurations. The trajectories were executed on the physical simulator of the LOC8 platform and a comparison between the simulation results and the obtained trajectories was performed (Figure 6.4). The results show that the PSO search algorithm produces consistent trajectories. In all cases the search technique provided the same control points (P_2) for the given conditions.

In all the configurations analyzed it is possible to observe a lag in the trajectory obtained with the physical simulation, which is expected for an open loop controller. It was observed that the magnitude of such lag varies over the time and distance traveled. For roughly two thirds of the overall trajectory the robot followed very closely the trajectories provided by the PSO search. However at the end of the trajectory the deviation with the expected trajectory became noticeable particularly in the cases a) and b) shown in Figure 6.4. While in all the cases the position along the obstacle (y -axis) deviated significantly from the desired location, it remained within the navigation space limits (y_{max}, y_{min}).

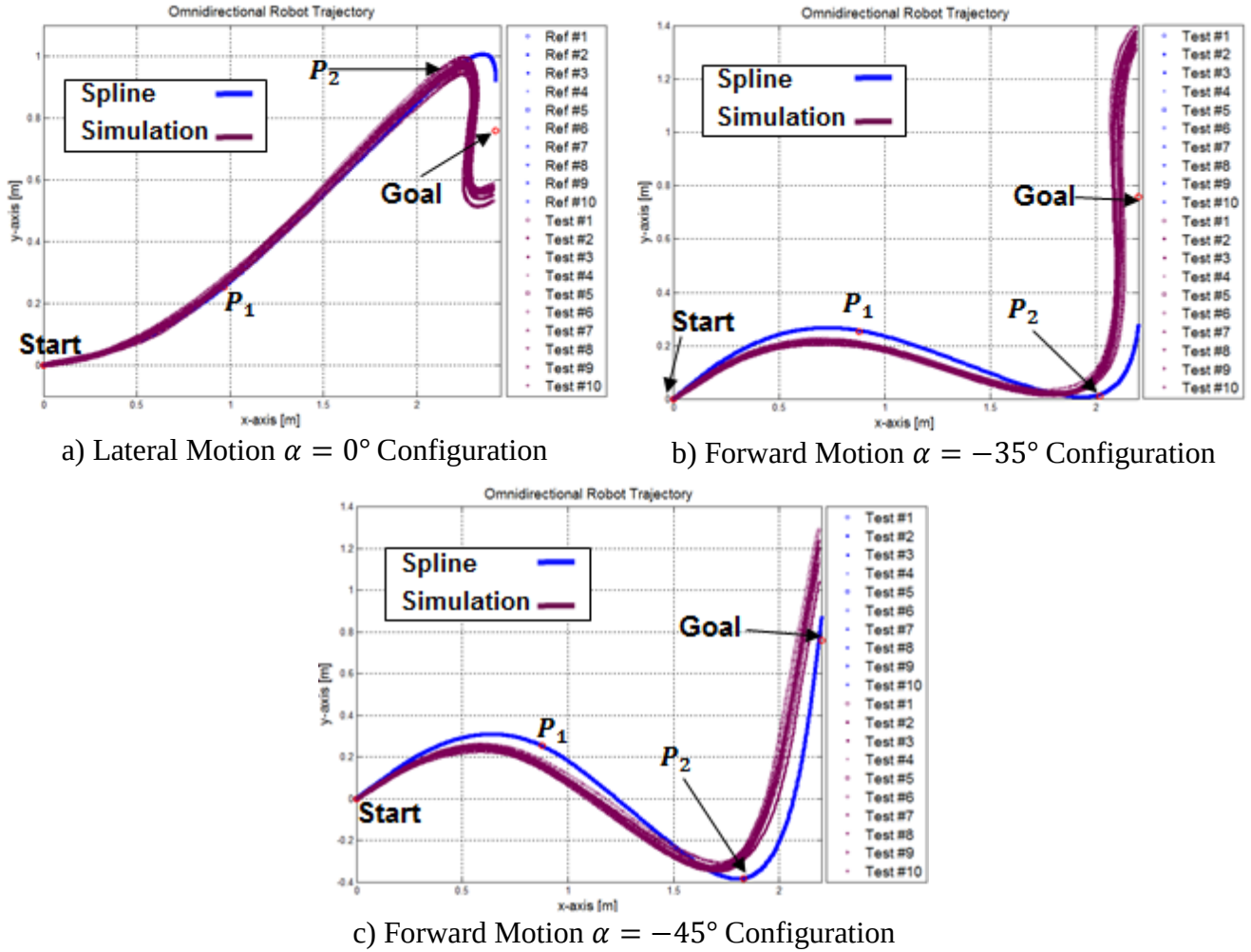


Figure 6.4 Scenario #1 first sub-test robot trajectories

The results regarding the trajectory following indicate the presence of shortcomings produced by the use of an open loop controller. While those shortcomings impact on the robot's final position along the y -axis, the robot's orientation and final position on x -axis were achieved within a minor deviation. The results for the angular position of the wheels observed a similar trend with slightly larger differences between the desired position and the final one.

A) *Lateral Motion* ($\alpha = 0^\circ$). The results for this configuration are displayed in Table 6.2. From such results in 4 out of 10 cases the deviation of the final angular position of the critical wheels (wheels #1 and #3) was minimal (not greater than 0.2 rad). In the remaining cases the observed deviation was significantly larger (up to 1 rad), having a significant impact on the maximum ground clearance achieved at such wheel positions. Due to the fact that the intended obstacle height of 5 cm requires of the maximum overcoming capability of the SMDW device

the acceptable deviation range is very limited. Based on these results it is expected to observe several negative results for the obstacle handling test in scenario #1.

Table 6.2 Scenario #1 lateral motion $\alpha = 0^\circ$ configuration simulation results

Test #	Final Position X_f			Final Wheels State Θ_f			
	$x_f [m]$	$y_f [m]$	$\psi_f [rad]$	$\theta_{1f}[rad]$	$\theta_{2f}[rad]$	$\theta_{3f}[rad]$	$\theta_{4f}[rad]$
1	2.4113	0.5327	-1.5152	5.0285	4.7747	1.1989	2.4351
2	2.4132	0.5685	-1.4954	6.0152	3.7757	6.2513	3.4277
3	2.4119	0.5813	-1.5035	6.2782	3.6950	6.1709	3.5666
4	2.4101	0.5797	-1.5076	5.5314	4.3442	0.5579	3.2759
5	2.4091	0.5684	-1.5010	5.6833	3.9065	6.2591	3.1933
6	2.4128	0.5897	-1.4608	6.0957	4.1823	0.1830	3.7616
7	2.4091	0.5542	-1.4534	0.1299	3.7104	6.1328	3.9984
8	2.4109	0.5695	-1.5076	5.8757	3.9722	0.1734	3.3366
9	2.4093	0.5490	-1.5116	6.0953	3.5796	5.8658	3.6101
10	2.4108	0.5704	-1.4326	5.7644	4.4297	0.5680	3.4007

B) Forward Motion ($\alpha = -35^\circ$). The obtained results for this configuration (Table 6.3) show a significant deviation for the critical SMDW (wheel #1) in 5 out of 10 simulations. Nevertheless for this configuration the strict criterion of $\theta = 0^\circ$ for the target position was adopted. Therefore, the obtained deviation is considered smaller since the ground clearance required allows some deviation on the wheel's angular position. These results indicate that for this second configuration a mixture of positive and negative results at the obstacle handling test is expected.

Table 6.3 Scenario #1 forward motion $\alpha = -35^\circ$ configuration simulation results

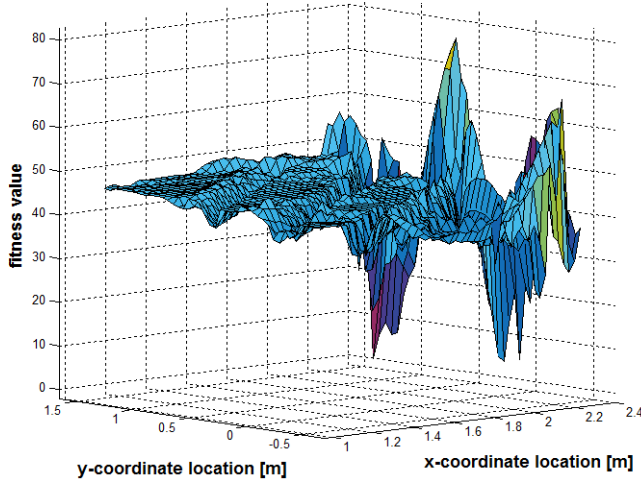
Test #	Final Position X_f			Final Wheels State Θ_f			
	$x_f [m]$	$y_f [m]$	$\psi_f [rad]$	$\theta_{1f}[rad]$	$\theta_{2f}[rad]$	$\theta_{3f}[rad]$	$\theta_{4f}[rad]$
1	2.1863	1.3138	-0.8528	0.6086	1.7219	2.0687	6.1932
2	2.1864	1.3319	-0.8802	0.8929	1.8828	2.1805	0.1384
3	2.1856	1.3793	-0.9222	6.1896	4.1434	4.5685	5.7776
4	2.1853	1.3786	-0.8952	0.3275	2.9940	3.1070	6.0941
5	2.1879	1.3780	-0.9047	0.4298	2.7583	3.1969	6.1248
6	2.1859	1.3994	-0.9098	0.8262	3.4088	3.7391	0.0809
7	2.1870	1.3577	-0.9098	0.3234	2.7592	3.1613	0.0837
8	2.1864	1.3631	-0.9481	0.5026	2.9966	3.5615	6.0133
9	2.1881	1.3480	-0.9120	0.6984	3.0575	3.4502	0.0303
10	2.1850	1.3218	-0.9173	6.1533	2.7018	3.0249	5.7448

C) *Forward Motion* ($\alpha = -45^\circ$). The simulation results for this configuration are contained in Table 6.4. The results show that final position achieved for the critical wheel (wheel #1) was within the range for obstacle overcoming in 5 out of 10 simulations. Based on these results it is expected that in the following sub-test nearly half of the trajectories simulated will allow the OGV to climb the given obstacle.

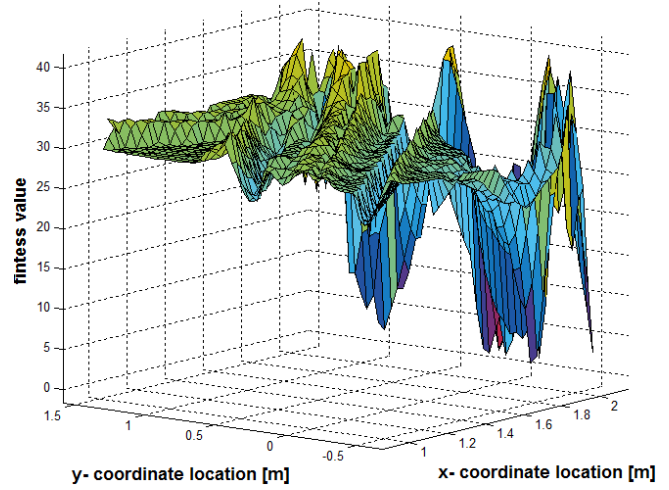
Table 6.4 Scenario #1 forward motion $\alpha = -45^\circ$ configuration simulation results

Test #	Final Position X_f			Final Wheels State θ_f			
	$x_f [m]$	$y_f [m]$	$\psi_f [rad]$	$\theta_{1f}[rad]$	$\theta_{2f}[rad]$	$\theta_{3f}[rad]$	$\theta_{4f}[rad]$
1	2.1864	1.2892	-0.8105	0.8589	3.2813	1.5928	2.7677
2	2.1853	1.1249	-0.8103	5.0074	3.3230	1.3811	0.2516
3	2.1859	1.0414	-0.7235	4.9006	1.6205	5.8297	0.4647
4	2.1850	1.0890	-0.7269	4.9006	1.0889	5.3903	0.4973
5	2.1851	1.1489	-0.7192	5.7782	1.5464	5.9523	1.1434
6	2.1866	1.1956	-0.7989	5.5954	3.1553	1.3356	1.2730
7	2.1874	1.2064	-0.7515	0.4508	1.6564	6.1595	2.2761
8	2.1856	1.1913	-0.7330	0.6106	1.7281	5.6993	2.5371
9	2.1866	1.2405	-0.7603	0.7951	2.4421	0.5501	2.7277
10	2.1873	1.2338	-0.7500	0.2255	2.4256	0.4720	2.3248

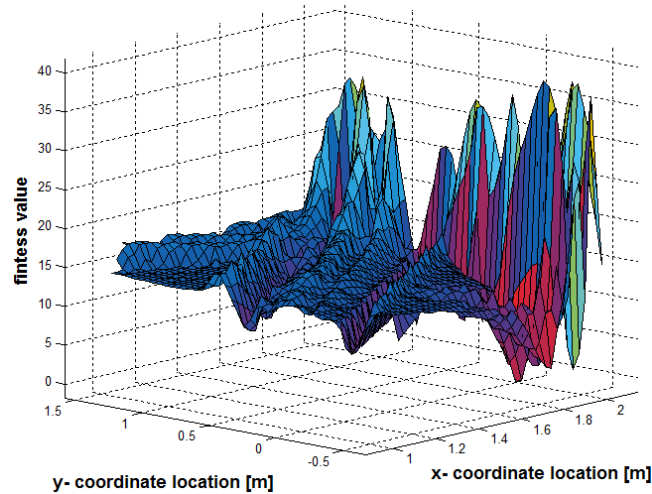
The fitness function cost for the location of P_2 among the space search is provided in Figure 6.5 for each one of the three selected configurations. As expected for the spline characterization problem, the fitness function value observed a different behavior for each configuration. For the conditions of Scenario #1, one or more local minima regions were present in each configuration. The evolution on the fitness cost observed for Scenario #1 presented a case in which the PSO search technique provides appropriate features for the handling of local minima regions [47].



a) Lateral Motion $\alpha = 0^\circ$ Configuration



b) Forward Motion $\alpha = -35^\circ$ Configuration



c) Forward Motion $\alpha = -45^\circ$ Configuration

Figure 6.5 Scenario #1 fitness function cost

The last section of the overall test, is intended to analyze the obstacle climbing capability for each one of the final configurations obtained by the trajectory physical simulation (Table 6.2 to Table 6.4.). For this sub-test a constant velocity input was considered for the obstacle climbing operation, the input was applied once the robot was located at the final states of the trajectory (X_f, θ_f) . The input for each configuration is shown in Table 6.5.

Table 6.5 Scenario #1 second sub-test velocity input

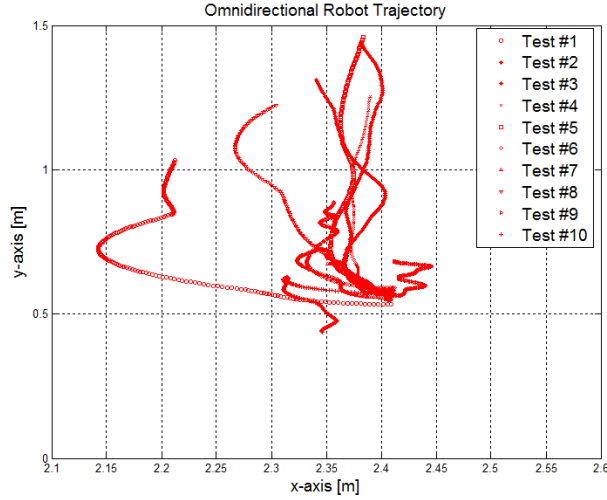
Configuration	Velocity Input $\left[\frac{km}{hr}\right], \left[\frac{rad}{s}\right]$
Lateral motion, $\alpha = 0^\circ$	$\dot{X} = [0 \quad 3 \quad 0]$, $\omega_1 = \omega_4 = 8.5$, $\omega_2 = \omega_3 = -8.5$
Forward motion, $\alpha = -30^\circ$ or $\alpha = -45^\circ$	$\dot{X} = [3 \quad 0 \quad 0]$, $\omega_1 = \omega_2 = \omega_3 = \omega_4 = 8.5$

A set of 10 simulations per configuration were performed using the V-REP physical simulator and final the configurations provided previously. The test results are provided in Figure 6.6 and then discussed. The lateral motion case exhibited unexpected results. In all simulations the robot with SMDWs was unable to climb on the obstacle. As can be seen in Figure 6.6a in each simulation the wheels #1 or #3 of the vehicle collided and bounced with the obstacle.

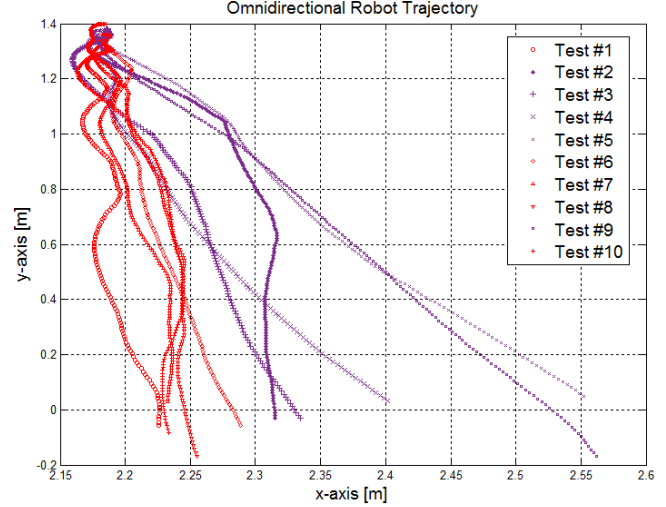
The trajectory plots and the physical simulator renders showed that the wheels were continuously bouncing on the edge of the obstacle, while the vehicle slid along the obstacle without being able to climb on it. The behavior described was expected for some of the final positions (e.g. Tests #1 and Test #9) at which the final states achieved exhibited a significant deviation. Nevertheless, it was expected that some of the configurations (e.g Test #2 and Test #3) would allow the robot to climb the obstacle in more than one case. The simulation results indicate that the maximum climbing capabilities at lateral motion are very sensitive to deviations on the wheel's angular position θ .

The forward motion configurations exhibited the expected obstacle climbing capabilities. The results for $\alpha = -35^\circ$ configuration (Figure 6.6b) show that in half of the simulations (marked purple in Figure 6.6b) the OGV was able to overcome the obstacle. However, the robot did not achieve climbing at the first contact, instead the SMDW slipped on the edge and at the next revolution, the SMDW engaged on the obstacle edge and allowed the climbing. For the remaining trials the OGV's wheels bounced against the obstacle edged and obstacle climbing was not achieved.

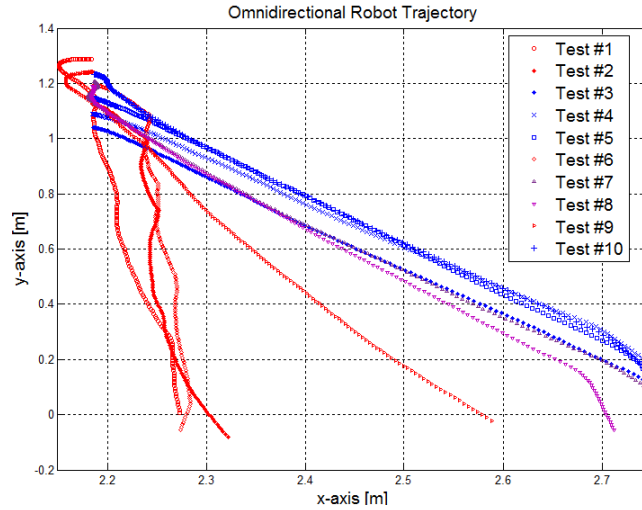
The results for the configuration $\alpha = -45^\circ$ (Figure 6.6c) observed a performance slightly better than expected. In 4 out of 10 cases (marked blue in Figure 6.6c) the robot was able to climb over the obstacle without any complication. In other 2 cases the robot's wheel remained on the obstacle edge sliding until the next revolution at which it engaged on the obstacle and the OGV climbed. For the remaining 5 cases (marked red in Figure 6.6c) the robot's wheel bounced against the obstacle edge and did not allow the OGV to climb it.



a) Lateral Motion $\alpha = 0^\circ$ Configuration



b) Forward Motion $\alpha = -35^\circ$ Configuration



c) Forward Motion $\alpha = -45^\circ$ Configuration

Figure 6.6 Scenario #1 second sub-test robot trajectories

In forward motion the SMDW positions that allowed a successful climbing were those with an angular position closer to the neutral position ($\theta = 0^\circ$). However some positions with a deviation inside the acceptable range did not allow to climb on the obstacle. Such cases of unsuccessful climbing were expected since the height of the obstacle used is within the maximum limit of the SMDW capabilities and the clearance estimation model used does not consider the rolling and slippage of the rollers.

The first obstacle handling tests showed that the maneuvering technique provides limited results when the obstacle height is at the boundary of the SMDW capabilities ($h = 5\text{ cm}$). However, it provided evidence of the utility and validity of the proposed ground clearance

estimation models. Despite the big simplifications made the proposed approach, provided a good reference for obstacle climbing suitable SMDW configurations.

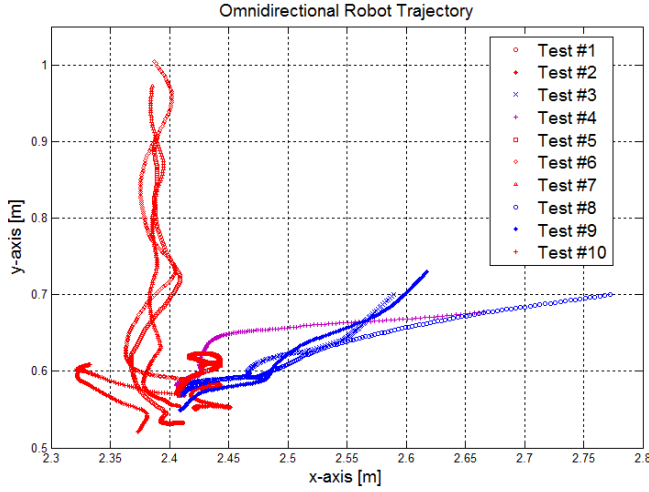
6.1.1 Test Scenario #1 Reduced Height Obstacle

The obstacle handling sub-test was repeated considering an obstacle of a height below the limits of the SMDW capabilities. The simulations were repeated for the previous three configurations considering an obstacle with $h = 3.5 \text{ cm}$. The tests were performed under the same conditions changing only the obstacle height. The robot trajectories for this second set of simulations are depicted in Figure 6.7.

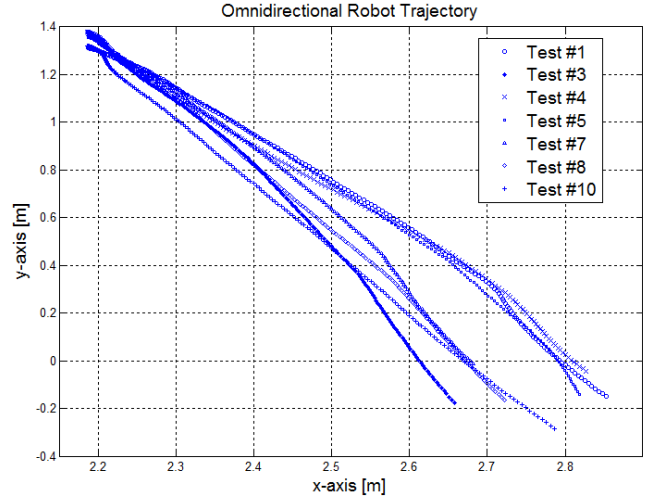
The results for lateral motion (Figure 6.7a) show that the OGV was able to climb the obstacle in 4 cases. In such cases the robot's trajectory observed a noticeable deviation and some bouncing prior the obstacle climbing. The robot was not able to overcome the obstacle in the remaining cases, in many of them the robot's wheels remained bouncing on the obstacle and the robot sliding around the y-axis. These results were unexpected in several cases in which the OGV's wheels had an adequate ground clearance for the given obstacle (e.g. Tests #6 and #7).

The simulation results for the forward motion configurations (Figure 6.7b and Figure 6.7c) exhibited the expected behavior. For most of the configurations the robot was able to climb over an obstacle of smaller height. The configurations that did not achieve to overcome the obstacle had in common that their angular position θ_{1f} had a value greater than 0.65 rad , having a reduced ground clearance at such configuration. For several of the successful cases especially in the $\alpha = -45^\circ$ configuration, the SMDW θ_{1f} was below the 2π value.

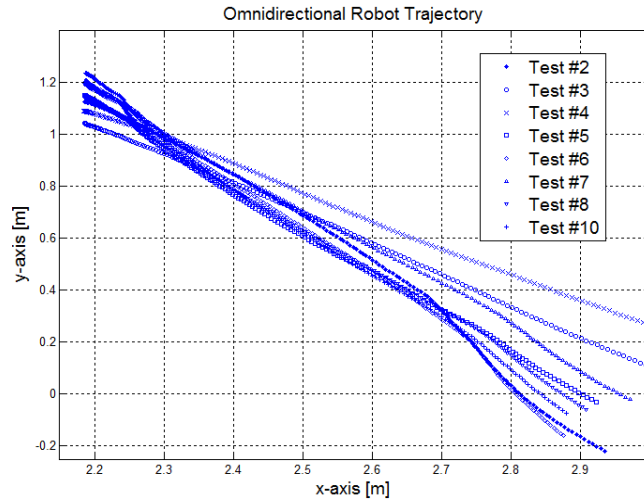
In both forward motion configurations cases the robot's trajectories exhibited some deviation over the time, such effect was expected since the SMDW #1 was already over the obstacle while the rest of the vehicle was still placed on the ground. That condition has an impact on the overall motion of the vehicle, however such behavior is not within the scope of this work.



a) Lateral Motion $\alpha = 0^\circ$ Configuration



b) Forward Motion $\alpha = -35^\circ$ Configuration



c) Forward Motion $\alpha = -45^\circ$ Configuration

Figure 6.7 Scenario #1 second sub-test robot trajectories at $h = 3.5\text{ cm}$

The test for a smaller obstacle height provided evidence of the methodology's utility for obstacle climbing at conditions below the operational limits of the SMDW. Furthermore, it also validates the feasibility of more than a single valid solution for the obstacle climbing. However, the methodology shows a somewhat limited performance for the lateral motion configuration, which was prior this research the only feasible configuration.

Due to the unexpected behavior of the robot previously described during the lateral motion ($\alpha = 0^\circ$) configuration, it was determined to perform a deeper analysis of such configuration. The obstacle handling sub-test was repeated for the configurations that achieved to overcome the obstacle, a closer look of the V-REP renders showed evidence of a correlation between the angular orientation of the robot ψ and the deviation in the trajectories.

The renders revealed a direct influence of the robot's orientation, ψ , on its inability to climb the obstacle. In the renders was observed that due to the deviation between the ideal orientation ($\psi_d = -90^\circ$) and the final orientation (ψ_d) a misalignment occurred between wheel #1 and wheel #3. Such misalignment led to the situation in which wheel #1 was already located above the obstacle while wheel #3 is still few of centimeters away from the obstacle. Under such circumstance when the wheel #1 had successfully climbed over the obstacle, wheel's #3 angular position θ_3 that did not provide adequate ground clearance which caused a collision. The collision made the robot bounce and slide along the obstacle edge.

In order to verify the correlation between the inability to climb and the robot's orientation ψ deviation, the tests were repeated this time setting manually the final orientation to $\psi_d = -90^\circ$. The results presented in Figure 6.8 were dramatically different after the orientation modification. In 8 out of 10 the cases the robot was able to overcome the obstacle without any significant trajectory deviations. In the remaining 2 cases, the robot presented some sliding while handling the obstacle which produced a moderate deviation on the vehicle's trajectory and orientation. However, at the end the vehicle was able to climb the obstacle.

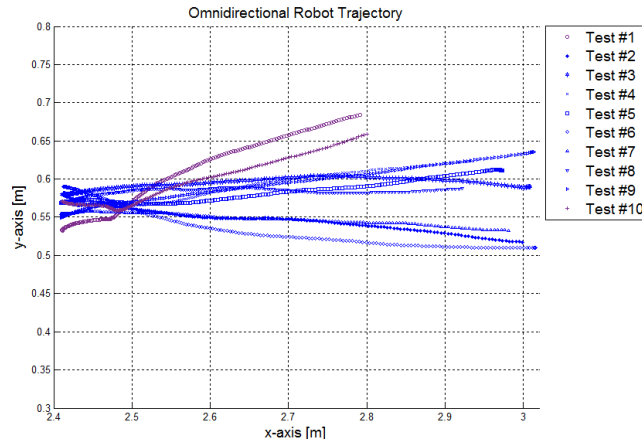


Figure 6.8 Robot Trajectories for Lateral Motion Configuration with $\psi_f = -90^\circ$

In order to verify the forward motion configurations where the robot was able to climb on the obstacle at $\theta_{1f} < 2\pi$ further analysis was performed. The geometric analysis for θ_{1f} between -90° and below 0° indicated that the SMDW possesses a considerable clearance for obstacle overcoming. Nevertheless, at the maximum clearance the point of contact with the obstacle does not allow the robot to successfully climb. But for the cases where the obstacle height is below the clearance the SMDW can engage with the obstacle by continuing the rotation and then

provide a suitable configuration for climbing. This situation is illustrated in Figure 6.9. It is thus concluded that several SMDW positions below the 2π range can allow the SMDW to overcome obstacles with a height lower than the maximum clearance.

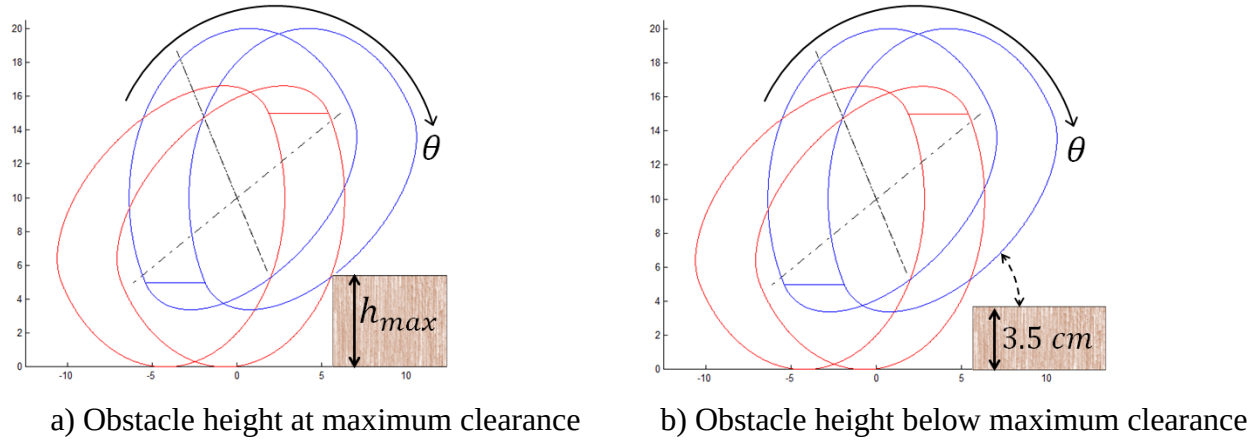


Figure 6.9 SMDW Clearance in forward motion at $\alpha = -35^\circ$ and $\theta = -30^\circ$

The test results provided evidence that indicates that vehicles with SMDWs are very sensitive to deviations on the robot's orientation, ψ , when lateral obstacle overcoming is desired. On the other hand the results also provided evidence that the methodology can generate suitable trajectories to overcome an obstacle in forward motion at orientations of $\alpha = -35^\circ$ and $\alpha = -45^\circ$.

6.2 Test Scenario #2

An additional set of tests were performed to analyze the methodology in confined spaces and determine if the maneuvering approach can generate paths under diverse constraints. For these tests a 2x2 meters area with an obstacle height $h = 3.5 \text{ cm}$ was used (Figure 6.10.).

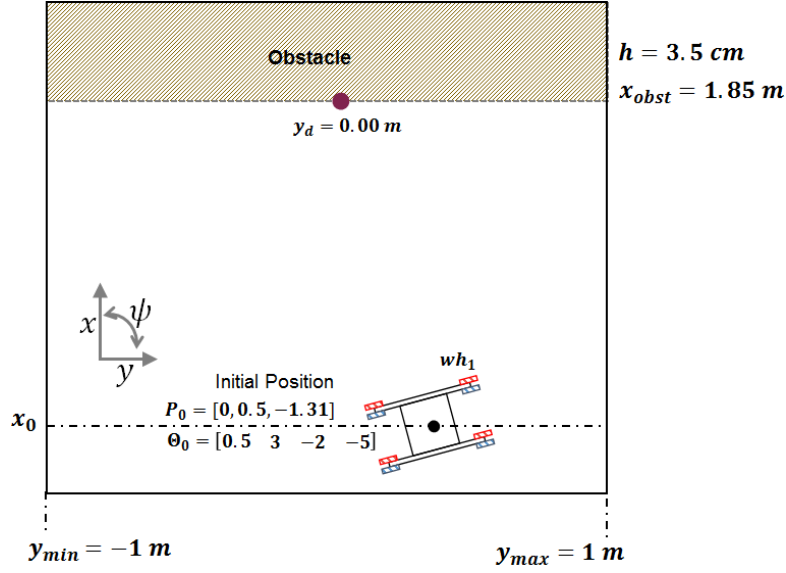


Figure 6.10 Scenario #2 Initial Conditions and Test Parameters

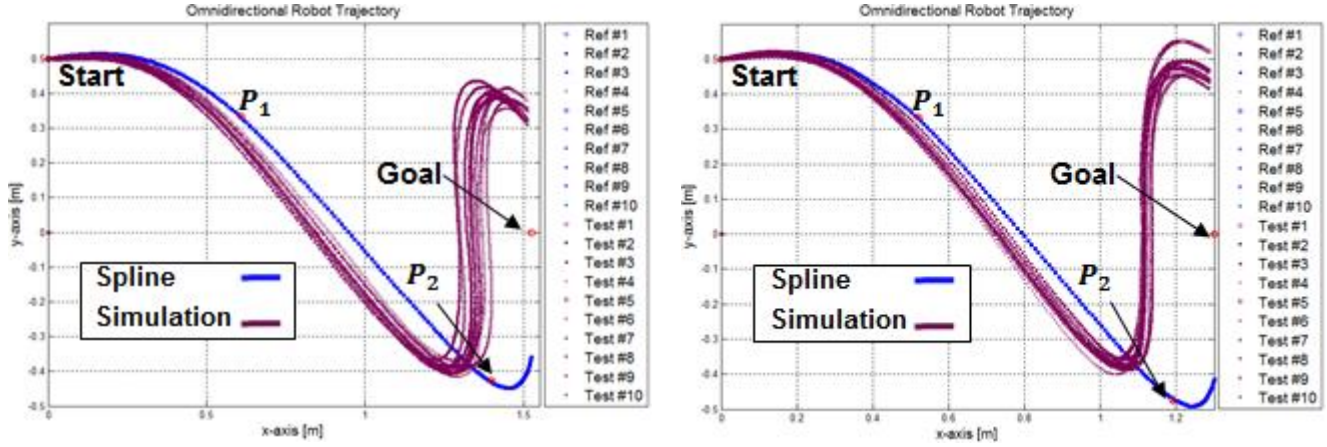
The robot's target position and SMDWs angles were determined using the geometrical model and the frame dimensions as describe in the Scenario #1 (Section 6.1). For these tests the results of the ground clearance estimation was used directly to determine the desired SMDWs position. The desired robot's position and SDMWs states for Scenario #2 are provided in Table 6.6.

Table 6.6 Scenario #2 Desired Robot Position and Wheel Configurations

Configuration	Desired Robot Final Position [m, rad]	Desired Wheel's Final Position [rad]
Lateral motion, $\alpha = 0^\circ$	$X_d = \left[1.524 \quad 0 \quad -\frac{\pi}{2} \right]$	$\Theta_d = [0.5 \quad \pi \quad -0.5 \quad \pi]$
Forward motion, $\alpha = -30^\circ$	$X_d = [1.3 \quad 0 \quad -0.611]$	$\Theta_d = \left[0.262 \quad 0.262 \quad \frac{\pi}{2} \quad \frac{\pi}{2} \right]$
Forward motion, $\alpha = -45^\circ$	$X_d = \left[1.3 \quad 0 \quad -\frac{\pi}{4} \right]$	$\Theta_d = [0.175 \quad 0.175 \quad 0 \quad 0]$

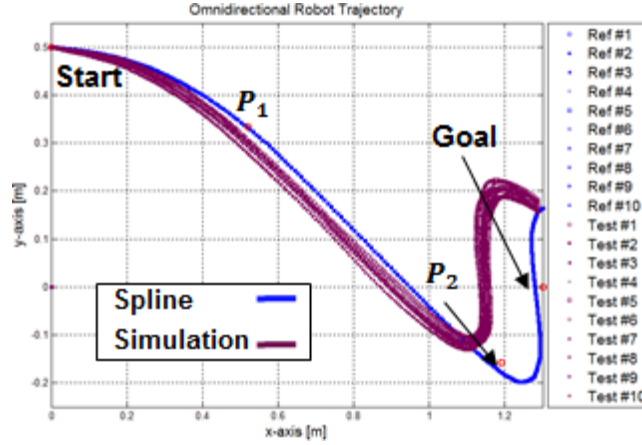
The PSO trajectory search was run for the three target destinations shown in Table 6.6. The resulting trajectories were fed to the V-REP simulator in order to perform the trajectory sub-test for the scenario #2. The comparison between the ideal trajectories and the ones produced by the physical simulator is shown in Figure 6.11 for each of the three configurations. Such results are as expected similar to the results presented in section 6.1 for the same test. The robot's trajectories observed a lag between the ideal trajectory and the simulation results. In all cases the OGV'S final position on the x-axis and orientation (ψ) observed minor deviations, the y-axis

position observed a larger, however, the positions achieved remained within the navigation space limits (y_{max}, y_{min}).



a) Lateral Motion $\alpha = 0^\circ$ Configuration

b) Forward Motion $\alpha = -35^\circ$ Configuration



c) Forward Motion $\alpha = -45^\circ$ Configuration

Figure 6.11 Scenario #2 first sub-test robot trajectories

The fitness function cost behaviour for the Scenario #2 is shown in Figure 6.12. Once again the fitness cost observed clearly differentiated trend in each one of the configurations analyzed in Scenario #2. In each case one or more local minima regions were observed. Such results are consistent with the ones obtained for Scenario #1 and provide further evidence of the suitability of the PSO technique for the spline characterization.

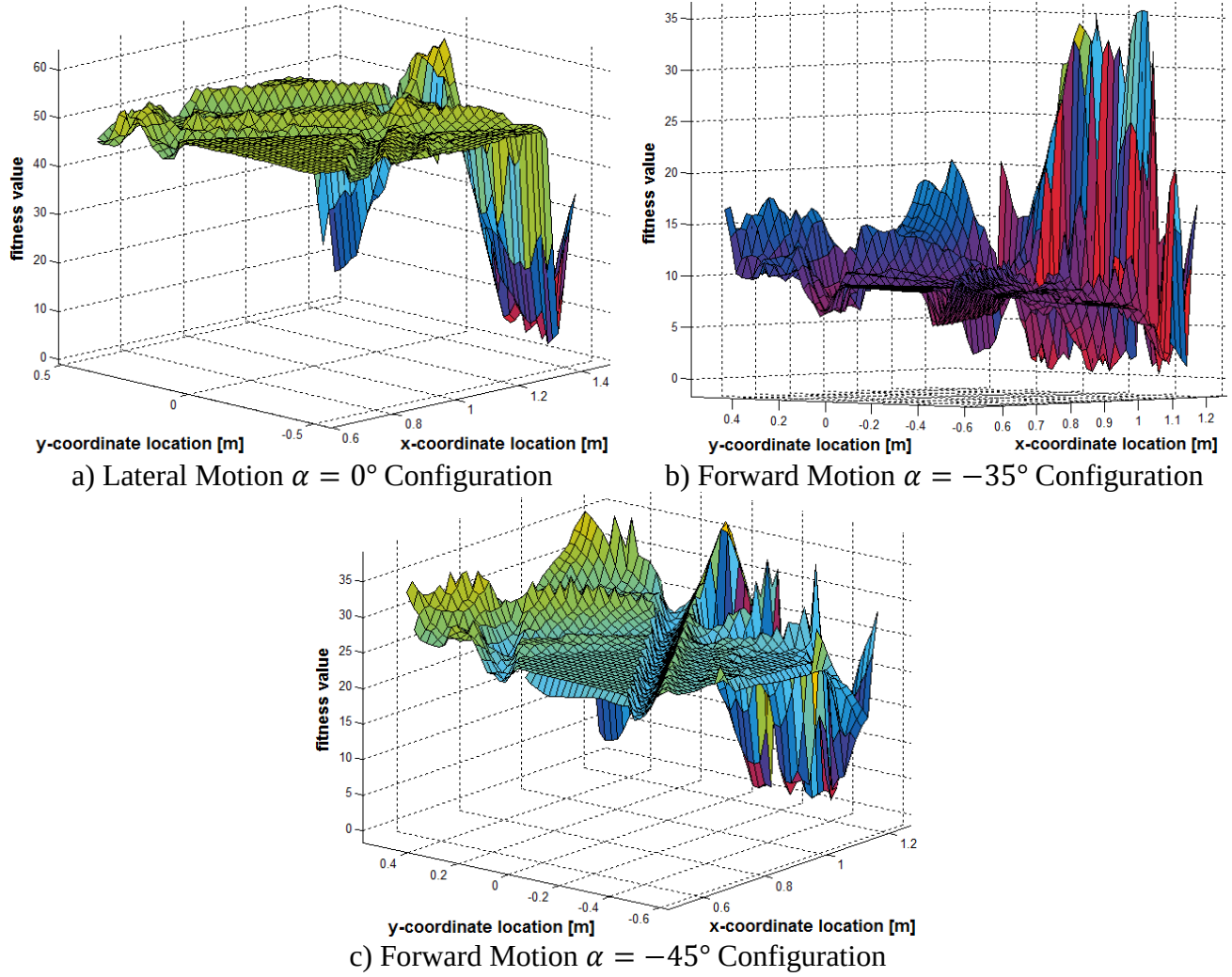


Figure 6.12 Scenario #2 fitness function cost

The robot's final position and wheels states (X_f , Θ_f) generated from the trajectories shown in Figure 6.11 were used to run the second sub-test to evaluate OGV's obstacle climbing capabilities. The input velocity used for this second subtest was the same input used in Scenario #1 tests (Table 6.5). A set of 10 simulations per configuration were run, the results are provided in Figure 6.13.

The forward motion configurations results (Figure 6.13b and Figure 6.13c) were as expected, the generated trajectories allowed the OGV to overcome the obstacle in all the cases. The reduction of the obstacle height increased the effectivity of the maneuvering technique. However, the lateral motion results (Figure 6.13a) showed an unexpected behaviour only in 2 cases the OGV was able to climb the obstacle. Further analysis showed that all the unsuccessful

climbing cases had a significant robot's orientation ψ deviation. Such results confirmed the conclusions presented in Section 6.1.1.

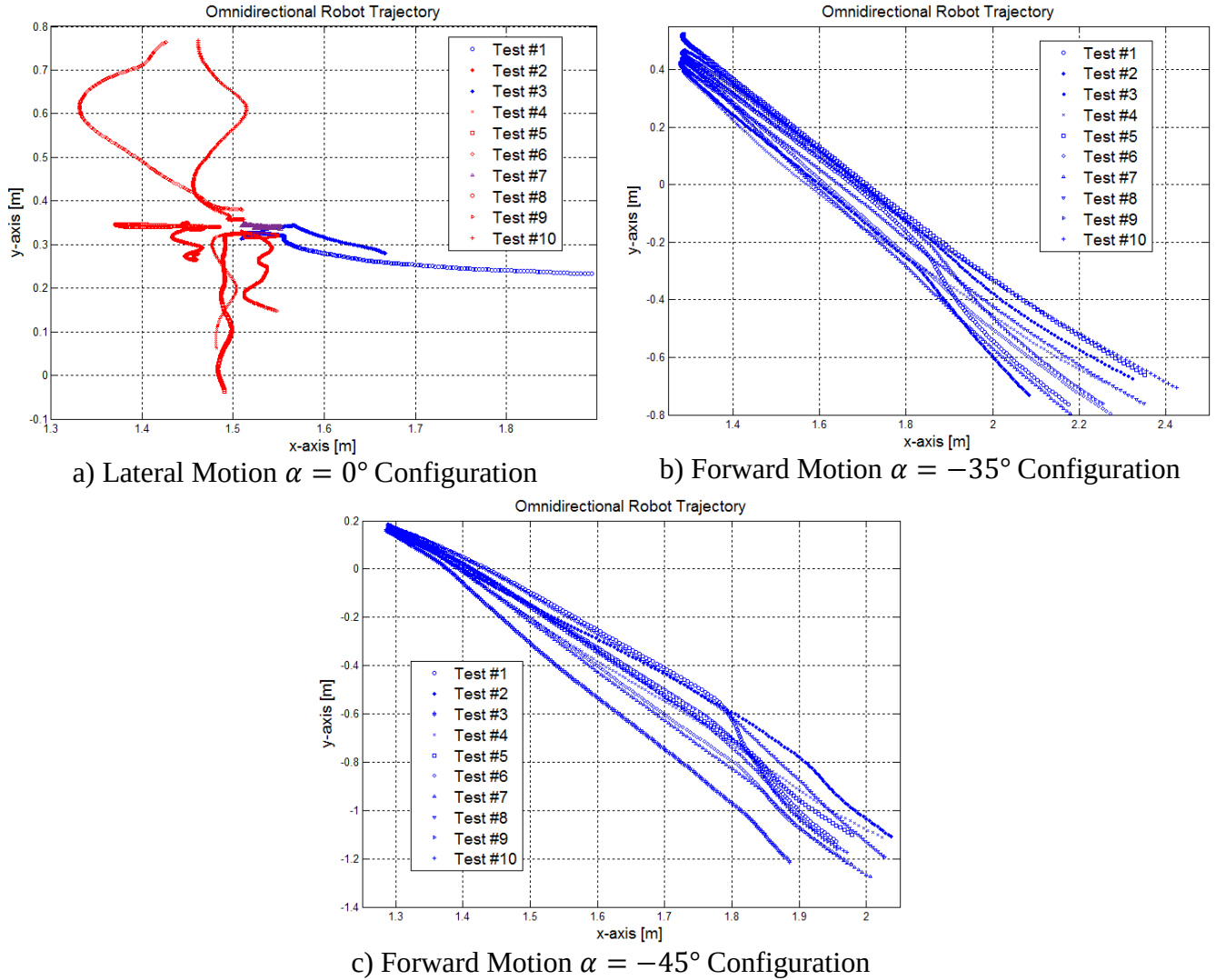


Figure 6.13 Scenario #2 second sub-test robot trajectories

The simulation results of this second scenario confirmed the conclusions previously presented in Section 6.1 and 6.1.1. The proposed maneuvering technique was able to generate trajectories that allowed an OGV equipped with SMDWs to overcome obstacles at diverse approaching configurations. Also the generated paths kept the OGV within the navigation space at all time in all the cases. Based in the results of both test scenarios it was concluded that the proposed methodology and its core components (PSO search, SMDW geometric model and physical simulator) achieved an enhancement in the omnidirectional capabilities of OGVs with SMDWs by allowing them to overcome obstacles in more than a single configuration.

Chapter 7: Conclusions

A maneuvering methodology that expands the capabilities of a robot to move in rough terrains was presented in this thesis. This maneuvering methodology provides an effective tool to enable robots equipped with SMDWs to move and navigate on complex outdoor rough terrains. The expansion of the holonomic capabilities is achieved by providing more than a single feasible solution (robot position and orientation) to climb over obstacles.

The maneuvering methodology was tested in different conditions. The tests proved that the methodology is capable to provide trajectories that allow robots with SMDWs to climb obstacles in more than a single robot orientation and motion direction.

The most significant contributions of these thesis include: the generation of effective tools for the development of applications with semi-circular Mecanum double wheels which expanded the ways in which obstacles can be climbed with SMDW. Prior to the work developed in this thesis the only explored and feasible solution considered to climb the obstacles was in lateral motion.

The work of this thesis provides a novel application for the PSO search techniques in the selection of control points for the generation of spline trajectories. The maneuvering control technique that existed prior this thesis defined the robot's trajectory by the numeric value of their spline's coefficients. The methodology proposed in this work defines the robot's trajectory by a set of points in the navigation space. The use of location points instead of equation's coefficients makes more accessible the use of the methodology since involves simpler mathematical definitions and provides a direct correlation with the physical world.

A novel and accurate physical simulator for the SMDW was generated for the testing of the methodology. Unlike most of the models for omnidirectional wheels, the developed simulator includes the peripheral rollers and its effects in a practical and accurate way. The approach used for the physical simulator has the potential to be extended to other omnidirectional wheel designs.

7.1 Future Work

Due to the complexity of the problem explored in this thesis some important details were not addressed in aims to simplify the endeavor. The following aspects are envisioned as the work that needs to be carried out:

- Include the trajectory length (distance traveled) as parameter of the fitness function. In order to obtain trajectories closer to near-optimal the overall distance factor shall be considered for the trajectory fitness evaluation. The redefinition of the fitness function to include the path length is a potential future improvement.
- Include the time factor as a search dimension of the PSO algorithm. In this research work was found that the time interval determined has an important impact on the trajectory qualities. The addition of the time interval parameter to the PSO search can potentially improve the quality of the trajectories generated.
- Substitute the open loop trajectory controller by a closed loop trajectory controller that can potentially reduce the deviations in the robot's final position, orientation and wheels state.
- Analyze the maneuvering once the vehicle is already on the obstacle. The stages that occur after the SMDW has climbed the obstacle successfully, are not included nor analyzed. In order to refine the methodology and expand the applications of the SMDW it would necessary to study the maneuvering of the robot on top of the obstacle.
- Modify the maneuvering methodology to include a hybrid Reactive-Planning navigation algorithm than can account to disturbances and the non-linearities of the robot operation. This hybrid approach can incorporate a closed loop trajectory tracking and a reactive component to correct the trajectory deviations and the wheel positions.
- Analyze the use of a navigation approach based on primitive motions. The study of a trajectory based in primitive motions is desirable in order to compare the performance of the spline based trajectories and also analyze potentially simpler paths.

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